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TARGET ARTICLE

≈ The VVIQ as a Psychometric Test of Individual Differences in Visual Imagery Vividness: A Critical Quantitative Review and Plea for Direction

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This paper assesses the construct validity of the Vividness of Visual Imagery Questionnaire (VVIQ) on the basis a quantitative review of content validity, reliability, and criterion-related validity. Consideration of the test instructions, items, rating scale, and distribution of scores revealed a number of flaws that challenged content validity. There was also evidence that VVIQ ratings could reflect confidence in criterion task performance if the latter was given first. However, although there was a positive correlation between reported vividness and social desirability, the degree of contamination was acceptable. Reliability was acceptable for internal consistency (.88), barely acceptable for test-retest (.74), and unacceptable for both immediate (.64) and delayed alternate-form (.54). However, it was argued that the last 3 values may be lower than is usually desirable because vividness is somewhat dynamic. The effects of social desirability and context also suggested that vividness resembles a typical performance trait. For 263 individual estimates across 71 criterion tasks, the validity coefficient for the VVIQ was estimated to be .269, which was deemed to be evidence of an acceptable relationship. However, it varied from self-report (.369), through cognitive/perception (.273), to memory (.137) tasks, with the last of these only marginally acceptable. It was concluded that the evidence was generally favorable to the construct validity of the VVIQ, and that the situation might improve if certain features of the test were altered. Modifications were suggested and recommendations were offered for future research with the VVIQ as an independent and dependent variable.

"In what ways does the behavior of a man who states that he has vivid visual imagery differ from that of another who says that his imagery is vague and dim?" (Marks, 1972, p. 83). To answer this question in the context of laboratory research on cognition (Marks, 1972, p.84; 1987), Marks (1972, 1973a) presented his Vividness of Visual Imagery Questionnaire (VVIQ) and related it to performance on visual memory and perception tasks. The

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VVIQ has enjoyed wide use in over 100 investigations (Marks, 1989a, 1989b).

Because Marks' initial question concerned the relationship between report and behavior, it was appropriate for him to adopt the questionnaire approach to measuring imagery. He also justifies it on the grounds that imagery is "a conscious, quasi-perceptual experience" (Marks, 1977, p. 280), for which verbal reports are a necessary source of evidence (Marks, 1983a). This applies particularly to vividness, which he defines as "a combination of clarity and liveliness. The more vivid an image, therefore, the closer it approximates the actual percept" (Marks, 1972, p. 83). Although verbal reports of experience are not infallible (Marks, 1983a), they can be empirically evaluated as predictors of other variables (Marks, 1983b, pp. 97, 106). These data address Marks' (1972, p. 83) purely behavioral question, but they also pertain to the validity of the VVIQ (Marks, 1972, p. 84; 1989b). The purpose of the present quantitative review of this evidence was to evaluate the validity of the VVIQ and to suggest directions for research.

Because there is no "real-world" (Ghiselli, Campbell, & Zedeck, 1981, p. 280) measure of imagery vividness, the validity of the VVIQ cannot be assessed simply by correlating it with a natural criterion. Under these circumstances, it is appropriate to examine construct validity (Ahsen, 1990, p. 5; Chara & Hamm, 1989; Cronbach & Meehl, 1955, p. 284; Ghiselli et al., 1981, p. 280; Marks, 1989b; McKelvie, 1990).

The Nature of Construct Validity

Cronbach and Meehl (1955, p. 282) state that "Construct validation is involved whenever a test is to be interpreted as a measure of some attribute or quality that is not 'operationally defined.'" According to Ghiselli et al. (1981, pp. 282-283), the procedure may be deductive or inductive. In the first case, where theory leads (Smith, 1992, p. 483), the investigator has a theoretical understanding of the construct and develops a test to measure it. If predictions are confirmed, the theory is supported and the validity of test is enhanced (Cronbach, 1990, p. 179). However, negative evidence may mean that the theory needs revision, or that the test is a poor measure, or even that the research was badly designed (Cohen, Montague, Nathan, & Swerdlik, 1988, pp. 139-140; Cronbach, 1990, p. 179; Cronbach & Meehl, 1955, p. 295). In the second case, the starting point is the test (Smith, 1992, p. 483), and research is conducted to establish what it measures (Anastasi, 1988, p. 162; Ghiselli et al., 1981, p. 282) so that it can be interpreted (Cronbach & Meehl, 1955, p. 283). Here, "any correlation of the test score with another variable is welcome" (Cronbach, 1988, p. 12),

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because all empirical relationships help to define what is being measured. For some tests, both approaches are involved. The construct may be loosely-defined (Cronbach, 1990, p. 159) or perhaps not defined at all, necessitating initial inductive "fishing" (Ghiselli et al., 1981, p. 284). These data suggest a theoretical construct to be tested deductively. Construct validation is a creative, never-ending process, in which interpretations (and, possibly, tests) are constantly being revised in the light of new evidence (Cronbach, 1990, p. 178; 1988, p. 5). If the construct is fairly well understood, negative evidence means that the test requires revision (Cronbach, 1990, p. 181). However, the complexity and fluidity of the construct validation process imply that it is impossible to state categorically whether or not a test "is valid" (Cronbach, 1989, p. 151; Cronbach & Meehl, 1955, p. 290).

Thus, construct validity includes both the nature of the trait and how successfully the test measures it, and all psychometric and experimental data feed into the judgments (Anastasi, 1988, p. 163). Citing the same references as Anastasi, Marks (1989b, p. 461) applies this point directly to the VVIQ. Because Marks (1972, p. 83; 1973a) did not initially make directional hypotheses and later observed that research conditions and criterion performance measures should be examined in order to "illuminate the processes involved" (1983a, p. 24), and because the concept of vividness has been debated, an inductive approach to construct validity might be appropriate here. Unfortunately, because the starting point is the test itself, this does not put the VVIQ at risk, and provides only a description of what it may and may not measure. Because the present review was conducted to evaluate the validity of the VVIQ as a measure of imagery vividness, the deductive approach was emphasized.

This enterprise involves a wide variety of evidence (Anastasi, 1988, pp. 153-161; Cohen et al., 1988, pp. 140-150; Cronbach & Meehl, 1955, pp. 287-289; Ghiselli et al., 1981, pp. 284-287). If the test measures a unitary trait, its content should be homogeneous and internally consistent. If the trait is static, delayed reliabilities should be high. There should be both convergent and discriminant validity (including a multitrait/multimethod matrix) where correlations with similar tests should be high (but not too high, otherwise the measures would be redundant), and correlations with measures of response bias and irrelevant variables should be low. Experimental studies with the test as a dependent variable should provide support for theoretically-derived hypotheses.

To assess the construct validity of the VVIQ, I examined the three major psychometric properties of content validity, reliability, and criterion validity.

ty. Estimates of relationships were obtained from individual correlation coefficients by aggregating them into 95% confidence intervals. To do this, investigators usually calculate a weighted mean of the coefficients (r_s) themselves, or of Fisher's transformation rr (Alexander, 1990). I converted the r_s to z_r s, then calculated a M and SD to give a 95% confidence interval for z_r , which was then reconverted back to give the confidence interval for r . Finally, these intervals were judged in accordance with recently developed standards and guidelines (McKelvie, 1994a), which are summarized at various points throughout this paper (Tables 3, 5, and 7).

Development and Description of the VVIQ

Marks introduced the VVIQ to overcome ambiguities in previous work that showed no relationship between individual differences in imagery vividness and visual recall (Sheehan & Neisser, 1969). Marks (1973a) argues that this association may have been obscured by the use of an imagery measure covering seven sensory modalities, so that its validity as a predictor in any one was questionable (Isaac, Marks, & Russell, 1986; Marks, 1987). The offending instrument was a shortened form (Sheehan, 1967) of Betts' Questionnaire Upon Mental Imagery (QMI; Betts, 1909), in which subjects rate 35 items (5 of which are visual) on a 7-point scale of vividness (see Table 1). Marks expanded the visual section of the QMI to 16 items and, because he thought that it required too fine a discrimination (Isaac, et al., 1986), he reduced the rating scale to 5 points. The original scale had 3 degrees of clarity and of vagueness, with a midpoint that was neither clear nor vague, whereas Marks' (1972, p. 91) VVIQ scale has 3 degrees of clarity and 2 of vagueness, with the midpoint at the lowest degree of clarity (Table 1). Although he developed the questionnaire in the context of memory research (Marks, 1972, 1973a), and selected items that were likely to be familiar (Marks, 1972, p. 91), Marks (1987, 1989b) states that they entail not only reproductive imagery but also the more dynamic qualities of imagination and creativity. In addition, he arranged them in sets of 4 (see Appendix) to stimulate "a natural development of interest, activity and absorption as different aspects of the scene unfold" (Marks, 1987, p. 4).

According to Marks' (1973a) standard instructions, each item is rated twice, once with the eyes open and once with the eyes closed. The person's score is the mean of the two ratings over the 16 items. Administration is by individual or group, but Marks (1983b, p. 108) recommends that people be tested twice over a few days or weeks in order to catch the rare subject who apparently reverses the polarity of the scale, giving vivid imagery a high instead of a low rating.

Logical Commentaries on the VVIQ: Content Validity

Content or logical validity refers to whether items represent the domain of interest. With only 16 questions, the VVIQ seems short, but the four blocks tap different experiences. However, logical concerns have been expressed about Marks' notion of vividness, the items, and the rating scale. These doubts will be discussed in light of the distribution of VVIQ scores and of the effects of extraneous variables.

Table 1
QMI and VVIQ Rating Scales and VVIQ Items

QMI Rating Scale (Sheehan & Neisser, 1969)	
The image aroused by an item of this test may be	Rating
Perfectly clear and vivid as the actual experience	1
Very clear and comparable in vividness to the actual experience	2
Moderately clear and vivid	3
Not clear or vivid, but recognizable	4
Vague and dim	5
So vague and dim as to be hardly discernable	6
No image present at all, you only "knowing" that you are thinking of the object	7
VVIQ Rating Scale (Marks, 1973)	
The image aroused by an item of this test may be	
Perfectly clear and as vivid as normal vision	1
Clear and reasonably vivid	2
Moderately clear and vivid	3
Vague and dim	4
No image at all, you only "know" that you are thinking of the object	5

Marks' Conception of Visual Imagery Vividness

Culling his overall position on the measurement of visual imagery vividness from numerous statements (Marks, 1972, pp. 83, 84; 1973a; 1977; 1983b, pp. 96, 97, 100; 1985, p. 1; 1989b, p. 462; see also Reisberg, Culver, Heuer, & Fischman, 1986, p. 54), I summarize it as follows. Visual imagery is a process that is encoded at a number of levels, the lowest of which contains concrete information such as color and shape. Activation of the lowest level produces a conscious, vivid, quasi-perceptual experience not based on current sensory input. Individual differences in the ability to activate this level create variations in the vividness of visual experience. Vividness means clarity and liveliness, and the more vivid the image, the closer the experience is to actual perception of sensory input. Such experience plays a functional role in cognition and verbal self-report is

required to study it. Assuming that reports reflect the experience, their relationship with performance is an empirical question decided by scientific investigation. Because the verbal reports are ratings on the VVIQ, the terms "good" and "poor" imagers (or visualizers) are convenient labels operationalized by lower and higher test scores, respectively. Although Marks assumes that vividness is functional, he did not initially specify whether vivid imagery would be more effective than dim imagery. Rather, as shown in the opening sentence to this paper, he posed a question, seeking information (Marks, 1972, p. 83; 1973a).

Whether critics are generally more (e.g., Katz, 1983, p. 44; A. Richardson, 1969, pp. 2-3, 1977, p. 31) or less (e.g. J.T.E. Richardson, 1980, p. 42) sympathetic to Marks' attempt to measure and investigate individual differences in visual imagery vividness, they agree that self-reports are required to investigate visual imagery as a phenomenal event. However, Marks' operational definition of vividness has been criticized as ambiguous, so that VVIQ reports may not adequately reflect it, and there is doubt that vividness reflects the "unique quality" (Dean & Morris, 1991, p. 332) or "essence" (Ahsen, 1986, p. 3; 1990, p. 2) of imagery as a functional experience. These two criticisms are directed mainly at the test and construct, respectively.

Ambiguity in the Operational Definition of Vividness

Writers observe "similarity to perceptual experience" in Marks' definition (Ahsen, 1990, p. 2; Dean & Morris, 1991, p. 332; Heuer, Fischman, & Reisberg, 1986, p. 162), but they disagree whether vividness has one or two components (Ahsen, 1986, p. 3; Chara & Verplanck, 1986; Heuer et al., 1986, p. 161; Kihlstrom, Glisky, Peterson, Harvey, & Rose, 1991, p. 140; Morris & Hampson, 1983, pp. 93-94). I think that Marks' (1972, p. 83) definition of vividness as "clarity and liveliness" and his scale descriptors as clear/vague and vivid/dim show that his conception has two components. However, he uses "vivid" in his rating scale, thereby partly defining the concept in terms of itself. If respondents also vary in their interpretation of what is required (Finke, 1980, p. 116; Slee, 1988, p. 126; Sommer, 1980, p. 117), scale ambiguity is a source of error that threatens validity.

Vividness as the Essence of Visual Imagery Experience

Pointing to variations in dictionary definitions and everyday use of vividness, Slee (1988, p. 128) argues that any measure of it is unlikely to capture the essence of imagery as a "perception-like memorial process." I disagree that Marks' interest was confined to memory, but ambiguity in his operational definition weakens VVIQ vividness as the essence of imagery experience.

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rience. Morris and Hampson (1983, p. 93) also object that the rating scale has "sullied" the accurate measurement of vividness by "lumping together" the two components without either a *priori* or empirical justification that they covary.

On the basis of his claim that VVIQ scores fluctuate radically according to certain instructions Ahsen (1985a, p. 14; 1986, pp. 2, 4; 1987, p. 14; 1990, pp. 11, 15; 1991, p. 24; 1993b, p. 34-35, 1993d, p. 156) also questions whether vividness is a stable maximum performance trait (an ability) where high vividness is more important than low vividness. In fact, he argues that "unvividness" is functional, giving it a paradoxical role (Ahsen, 1985a, 1986, 1987, 1988, 1990, 1993a, 1993b, 1993d). However, Marks (1972, p. 84) stated that the terms "poor" and "good" were labels ("mnemonics," "shorthand expressions") that represented operations, and he did not phrase his initial question directionally (Marks, 1972, p. 83; 1973a). This implies that good visualizers might not always perform better than poor visualizers (see also Marks, 1989b). On the other hand, Marks (1983b, pp. 109-123) presented predictions, most of which indicated better task performance by good visualizers. He also stated that his extreme group labels presume a quantifiable "ability" (Marks, 1972, pp. 84, 92; Marks, 1993), which implies that vividness is a stable maximum performance trait (see also Juhasz, 1977) and that higher levels of vividness are of greater value (Morris & Hampson, 1983, p. 93).

The Items

Although Chara and Hamm (1988) question whether subjects even consider their visual images when responding to the VVIQ (see also Chara, 1992a), this claim has been debated (Chara, 1989, 1992a; Marks, 1988; McKelvie, 1990, 1992d). Others argue that the VVIQ taps images of static visual memories (Chara & Verplanck, 1986; Slee, 1988, p. 128), but memories are not necessarily fixed or simply reproductive (Marks, 1986, p. 252; 1987; 1989b).

It is possible to answer all VVIQ questions without constructing novel images (Chara, 1992b; McKelvie & Demers, 1979), and Items 1 to 4 and 9 to 12 are preceded by an explicit instruction to think of something familiar. However, it does not appear before the other two blocks of items, implying that memory may or may not be used. Moreover, five of the eight items (numbers 5 to 8, and 16) in the latter two blocks demand a change in the picture that is before the mind's eye, and some items require control of imagery (Dean & Morris, 1991, p. 332; Gur & Hilgard, 1975; Kihlstrom, et al., 1991). Indeed, sequencing may move the test into the area of creative imagination (Hilgard, 1977, p. 103). Thus, while VVIQ items tap literal

memory images that preserve previous static visual input, they also involve dynamic processes, some of which will produce novelty. Together with Marks' (1987, p. 4) explanation that he adopted the blocked format to stimulate interest, these logical considerations suggest that scores on the VVIQ will be related to memory accuracy, but also to performance on tasks that reflect absorption, creativity, and control of imagery.

From the discussion of the meaning of vividness and of the items, and from further inspection of the questionnaire itself, I think that the following 12 (separately underlined) processes are logically involved when a person completes the VVIQ: from a (1) *verbal* stimulus, there is (2) *intentional* (or *cued*) (3) *generation* from (4) *memory* and (5) *imagination*, (6) *brief holding*, and some (7) *control/manipulation* of (8) *complex* (9) *visual* (10) *images* that preserve concrete details, whose (11) *vividness (similarity to perceptual experience)* is then (12) *judged and reported*. This list of processes will be used to develop predictions for VVIQ/task relationships.

The Rating Scale

Ahsen (1990, p. 13) criticizes the rating scale and instructions by pointing out that Rating 5 refers to "thinking of the object" without any imagery whereas, immediately afterwards, in the instruction for the first item, we are asked to "Think of some relative....and consider carefully the picture that comes before your mind's eye" (see Table 1). Ahsen detects a contradiction in the meaning of the word "think," because one implies an image and the other does not, so that "thinking" means both "seeing" and "not seeing." It is important that the test should distinguish vividness of knowing from vividness of picturing (A. Richardson, 1969, p. 47) but I disagree that there is a problem with inconsistent use of the word "think." In both cases, "think" means "Attend and try to retrieve an image." Initially, when we do this and experience an image, we rate it from 1 to 4. However, if we attend to the object but have no image, we rate it as 5, indicating that, although we are aware that tried to retrieve one, we only experience an abstraction. Evidence also supports Marks' use of "think." Reisberg and Leak (1987) instructed subjects to imagine faces of famous people then asked them whether their images matched the photographs of these people. Some subjects, whose VVIQ scores indicated that they were poor visualizers, could not answer because they had only been "thinking about" the faces, leading Reisberg and Leak (1987, p. 524) to state that "These responses provide striking confirmation for the VVIQ assessment."

Nevertheless, Ahsen is correct to observe that the instruction to "consider carefully the picture....," which follows "Think of some relative or friend....," contains the implication that an image will be forthcoming, par-

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ticularly since it is repeated before each block of items. This is not a problem with the word "think," but is a general demand characteristic that an image will be experienced, which may bias reports towards vividness.

Marks recommends two administrations of the VVIQ to catch the rare individual who misinterprets the direction of the rating scale. Because the VVIQ is presented as a test of imagery ability, and because it may be natural to give higher ratings to higher levels of skill, it is not surprising that some people reverse the scale. However, Marks' (1983b, p. 108) advice was published 10 years after the VVIQ was introduced and, as far as I can see from method sections, it is not followed (even by Marks). If it is important to readminister the VVIQ a second or even a third time (Marks, 1983b, p.108), in order to prevent "correlations from plummeting towards zero artifactually as a result of one or two careless subjects," any nonsignificant VVIQ/criterion task relationships will be difficult to evaluate. In addition, unless subjects are carefully questioned about their use of the rating scale, I think that it is hazardous to interpret a vastly different score on repeated testing as evidence of misunderstanding of scale direction. Even if the change is "far in excess of the standard error of measurement" (Marks, 1983b, p. 108), it may reflect unreliability. A simpler solution would be to reverse the numerical values on the scale, so that higher ratings represent greater vividness (e.g. Crawford, 1982; Lorenz & Neisser, 1985; Reisberg & Morris, 1985; Walczyk & Hall, 1988). This practice would lessen the problem of scale misunderstanding, and would also change the predicted direction of r_s , which are often expected to be negative. This can be a source of difficulty for data interpretation, because it is sometimes not clear (e.g. Childers, Houston & Heckler, 1985; Parrott & Strongman, 1985) whether r_s were positive (meaning that poorer vividness was associated with better performance) or negative (meaning that poorer vividness was associated with poorer performance), and whether signs were omitted to ease comprehension of the relationship (Parrott, 1986).

Others have criticized the rating task itself. By instruction, respondents must make an absolute judgment (J. T. E. Richardson, 1980, p. 124; 1988) in which they compare the clarity and vividness of their visual imagery experience with the standard of normal vision (perception of real-world visual stimuli). However, because respondents cannot know what response criterion is adopted by others (Kosslyn & Jolicoeur, 1980; Morris & Hampson, 1983, p. 95; Slee, 1988; Strosahl & Ascough, 1981), the meaning of scale values may vary across subjects (Chara, 1992b; Chara & Verplanck, 1986; Sommer, 1980, p. 117; White, Sheehan, & Ashton, 1977). Therefore, similar mental experiences among subjects may be given different ratings, and different mental experiences may be given the same ratings,

thereby introducing unsystematic error that may suppress between-subjects comparisons (Kaufmann, 1981, 1983). In contrast, if people are consistent with themselves, comparisons of the ratings of different stimuli can be made within-subjects (Kaufmann, 1981), because each person knows what the end-points of the scale signify in their own experience (J. T. E. Richardson, 1980, p. 124). Marks, 1972, p. 84) recognized this difficulty, but he responded (Marks, 1983a), not unreasonably, by appealing to empirical studies that compare the task behavior of good and poor imagers.

The subjectivity problem could be tackled by anchoring the rating scale with photographs that vary along a pertinent dimension such as clarity (Kaufmann, 1982). In fact, Chara and Verplanck (1986), operationalized vividness by associating each scale point with a version of a picture representing degree of focus (very sharp to very blurry) and of illumination (too bright through normal to very dark). Using Marks' VVIQ, Chara and Verplanck selected good and poor imagers, then administered a new 30-item questionnaire in which each step on the 5-point rating scale was accompanied by a slide that was chosen to represent it. For items common to each scale, Chara and Verplanck found that the mean imagery vividness scores for the poor imagers changed little, whereas those for the good imagers rose. Chara and Verplanck (1986, p. 918) characterize the latter difference as a "radical change," and Chara (1992a) argues that the change occurred as a consequence of initially lenient responding by good imagers. However, the increase in VVIQ scores, which resembles one reported by Marks (1973a) himself, could be a regression effect (McKelvie, 1990, 1992d). In addition, Marks (1973a) reports that VVIQ scores were significantly different on his second administration, showing that the good and poor imagers were clearly differentiated. Chara and Verplanck (1986) did not report the *t*-test for their corresponding ratings, but their scores for good imagers are absolutely lower than those for poor imagers on their new scale. Because the second questionnaire was not identical to the first and because there was a 10- to 14-day delay between administrations, Chara and Verplanck's data with the objectified scale may strengthen rather than weaken the authenticity of the original subjective one. Because variations in focus and in illumination of a photograph seem to capture important features of Marks' (1972, p. 84) notion of vividness [indeed, Marks (1988, 1989b) does not object], this approach deserves further investigation (McKelvie, 1990). Indeed, from reports of color and from assessments of sketches, Hishitani and Murakimi (1992) found more perceptual information in the images of good than poor imagers. This weakens the claim that the rating task is hopelessly subjective.

The problem of random error due to lack of objective scale anchors, like

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those of scale ambiguity and rating reversals, may be felt most acutely when discussing whether nonsignificant effects reflect random error in the VVIQ or zero-order relationships between the VVIQ and other variables. The Kaufmann-Richardson argument that the VVIQ rating scale is an inherently-flawed, ill-defined task and the concern over scale ambiguity also imply that VVIQ scores may be susceptible to demand characteristics, context effects, and response biases such as leniency and social desirability (J. T. E. Richardson, 1980, pp. 121-122). It has also been suggested (Chara & Verplanck, 1986; Cohen & Saslona, 1990; Finke, 1980) that scores on the VVIQ may reflect confidence (or overconfidence) in visual imagery, rather than experienced vividness per se. Of course, the subjectivity problem is not confined to the VVIQ scale, but applies to many forms of self-report (White, Sheehan, & Ashton, 1977), particularly typical performance tests (Anastasi, 1988, pp. 549, 553; Smith, 1992, p. 495). These matters are addressed by studies of score distribution and of the effects of confounding variables.

Distribution of VVIQ Scores and Effects of Extraneous Variables

If the instructional demand to experience an image induces subjects to respond leniently, that is, more vividly, the distribution of VVIQ scores will be positively skewed with a mean below the center (3) of the scale, violating the usual psychometric principles of normality and a distribution of item difficulties averaging out to 50% (Kihlstrom et al., 1991). The scores may then span a restricted range, adding another difficulty to the interpretation of insignificant correlation coefficients (Kihlstrom et al., 1991; Slusher & Anderson, 1987). In fact, investigators have remarked on overutilization of lower scale ratings (Chara, 1992b; Chara & Hamm, 1989; Chara & Verplanck, 1986; Kihlstrom et al., 1991; McKellar, 1972, p. 37; Reisberg, et al., 1986). To examine this issue, I searched empirical reports for the mean and standard deviation of VVIQ scores for all subjects in each the sample tested, and for means and standard deviations of "good" and "poor" visualizers.

Descriptive Statistics: Means and Standard Deviations

Total samples (norms). Thirty eight reports of means and 22 reports of standard deviations were found. The *M* of all 38 means, irrespective of the sample size, was $M = 2.307$ (see Table 2), which is below 3, $t(37) = 20.84$, $p < .01$. In one case, no sample size was given. This *M* of 2.32 was supplied to Irwin (1980) by A. Richardson as a population value for Australian university students. One of the others ($M = 2.08$, $SD = 0.74$; Kihlstrom et al., 1991) was also considered separately, since its sample size was much

larger than usual ($n = 730$). This left 36 means with a comparable sample size. The M of these means, weighted by their sample sizes, was 2.300 ($N = 1869$), almost identical to the unweighted value. The M of the 36 sample sizes was 51.92 (Table 2)

For the 22 reports with standard deviations, their M was 0.692. Because the corresponding mean sample size was 51.32, the standard error of the mean was estimated to be $0.692/(51.32^{0.5}) = 0.097$. However, a SE_m could also be calculated for each of the 22 samples. From Table 2, the M of these SE s was 0.123. Because these two estimates of SE_m were close (0.097, 0.123), they were averaged to give a final value of 0.11. When this was applied to the unweighted mean of the 38 mean scores, it resulted in a 95% confidence interval estimate for the VVIQ distribution of 2.091 to 2.523 (from $2.307 \pm 1.96 \times 0.11$), which is again below 3.

"Good" and "poor" visualizers. There were 33 reports of mean scores for these groups. In 15 of them, a median split was employed, but the M of the 33 percentage splits was 33.24. The corresponding value for the absolute sample sizes was 18.36 (Table 2).

The M s of these means for good and poor visualizers were 1.732 and 2.934, respectively. The latter was not significantly different from 3, $t(32) = 1.00$, $p > .10$. Although the SD of the means was slightly higher for the poor than good subjects, the difference was not statistically significant, $t(31) = 1.50$, $p > .10$ (test for correlated variances; Ferguson, 1981, p. 192).

From 14 separate reports, the M s of the standard deviations for good and poor visualizers were 0.361 and 0.565, respectively, showing that the variation in scores for poor visualizers was higher, $t(13) = 20.36$, $p < .01$. As before, SE_m s of each group could be calculated in two ways. From the two mean standard deviations just reported, and the corresponding mean sample sizes of 17.29, the SE_m s for good and poor visualizers were $(0.361/17.29^{0.5}) = 0.0868$ and $(0.565/17.29^{0.5}) = 0.1359$ respectively. However, from each of the 14 standard deviations and sample sizes, separate SE s were obtained. The M s of these SE s for good and poor visualizers were 0.0712 and 0.1567, respectively. Using the same averaging procedure as before gave estimates for the SE_m for good and poor visualizers of 0.079 (M of 0.0712 and 0.0868) and 0.146 (M of 0.1567 and 0.1359), respectively. Using these values, the 95% confidence intervals for good and poor visualizers were 1.58 to 1.89 ($1.73 \pm 1.96 \times 0.079$) and 2.65 to 3.22 ($2.93 \pm 1.96 \times 0.146$). Again, the latter includes 3.

Discussion of descriptive statistics. As predicted from considerations of scale ambiguity and demand characteristics, the central tendency of the VVIQ distribution was below the center of the scale. The general mean score was less than the midpoint of 3, and the mean score of "poor" visu-

Table 2
Estimates for Means, Standard Deviations and Standard Errors of the Mean for All VVIQ Scores and for those of Good and Poor Visualizers

	<i>M</i>	<i>SD</i>	<i>n</i>
All VVIQ Scores			
Means	2.307	0.205	38
Sample sizes	51.92	42.12	36
Standard deviations	0.692	0.215	22
Sample sizes	51.32	41.67	22
Standard errors of mean	0.123	0.067	22

Number of estimates and 95% confidence interval (midpoint):
 38, 2.091 (2.307) 2.523

Good VVIQ Scores			
Means	1.732	0.290	33
Sample sizes	18.36	14.49	33
Percentage splits	33.24	16.86	33
Standard deviations	0.361	0.267	14
Sample sizes	17.29	10.56	14
Standard errors of mean	0.0712	0.0529	14

Number of estimates and 95% confidence interval (midpoint):
 33, 1.58 (1.73) 1.89

Poor VVIQ Scores			
Means	2.934	0.378	33
Sample sizes	18.36	14.49	33
Percentage splits	33.24	16.86	33
Standard deviations	0.565	0.378	14
Sample sizes	17.29	10.56	14
Standard errors of mean	0.1567	0.1299	14

Number of estimates and 95% confidence interval (midpoint):
 33, 2.65 (2.93) 3.22

Note. The sources for means of all VVIQ scores were Alvarado (1984), Carroll, Baker, and Preston (1979), Chara and Hamm (1989), Cohen and Saslona (1990), Crawford (1982), Finke (1979), Finke and Kosslyn (1980), Finke and Schmidt (1978), Heuer, Fischman, and Reisberg (1986), Hishitani (1985), Irwin (1980), Kihlstrom et al. (1991), Kunzendorf (1982, 1984), Lorenz and Neisser (1985), MacLeod (1986), McKelvie (1979, 1986), Narchal and Broota (1988), Parrott (1986), Reisberg, et al. (1986), Reisberg and Leak (1987), Reisberg and Morris (1985), A. Richardson (1979), Rossi (1977), Sullivan and Macklin (1988), T Hoen (1978), Walczyk and Hall (1988). The sources for means of good and poor imagers were Berger and Gaunitz (1977, 1979), Campos (1988), Carroll et al. (1979), Chara and Verplanck (1986), Finke (1979), Dickel and Slak (1983), Dwyer (1988), Finke and Kosslyn (1980), Finke and Schmidt (1978), Gur and Hilgard (1975), Hishitani (1985a), Housner (1984), Housner and Hoffman (1978), Lash and Polyson (1987), Marks (1973a), Marks, Uemura, Tatsuno, and Imamura (1985), McKelvie and Demers (1979), McKelvie and Rohrberg (1978), O'Brien and Wolford (1982), Rossi and Fingeret (1977), T Hoen (1978), Wallace (1990a, 1991), Winczo-Kostecka and Cielecki (1985).

alizers was not different from 3 ("moderately clear and vivid"). This implies that responses on the VVIQ may be contaminated by a lenient response bias.

The overall mean of 2.3 and confidence interval of 2.1 to 2.5 was calculated from 38 samples, but it is based on more than 2600 cases, providing norms for the VVIQ. However, the vast majority of respondents were university students (see also Sommer, 1980). Although Gordon (1949) thought that such questionnaires demanded introspective abilities possessed by university (particularly psychology) students, investigators today assume that self-report tests should be constructed for everyone. The present population is not only restricted, but has also not been randomly sampled using formal stratification procedures. Although the data were gathered in a number of countries (mostly Australia, Canada, Great Britain, and the United States), gender is the only reported subject variable. From 4 studies (McKelvie, 1986, 1992a; Narchal & Broota, 1988; A. Richardson, 1979), it was possible to estimate the standardized effect size d (Cohen, 1977), giving a M and SD of 0.11 and 0.20, respectively. This yielded a 95% confidence interval of -0.09 to .31 ($0.11 \pm 1.96 \times .10$), which includes 0, showing no gender difference in VVIQ scores.

Overall, the present estimate of the VVIQ mean and standard deviation is based mostly on non-random samples of university students from a few countries. On the basis of Robinson, Shaver, and Wrightsman's (1991, p. 12) guidelines for the quality of norms, the representative value of these "norms" is minimal to moderate (1, 2) on a scale that extends from 0 (none) to 4 (exemplary). Of course, this problem will only arise if the VVIQ is used for individual assessment. In the experimental setting where it is usually employed, external validity, particularly random sampling, is less important than internal validity (Mook, 1983).

In addition to response leniency, there was concern that the distribution of scores was not normal and was restricted in range. Most authors do not report the shape of the distribution, but McKelvie (1992a), with 90 cases, and Kihlstrom et al. (1991) with 730 cases, both indicate that it was positively skewed. In addition, the overall mean of 2.307, together with the overall standard deviation of 0.692, implies that a normal distribution would have a range of 0.231 to 4.383. Because the minimum score is 1, this indicates that the VVIQ distribution is positively skewed. If the distribution was normal with a mean of 3, the range of 4 points on the 5-point scale implies that its standard deviation would be 0.667. Because the present value is 0.692, it does not appear to be low. However, the skewed distribution indicates that there are very few high scores, and that the effective

range is restricted. This poses a difficulty for researchers wishing to correlate VVIQ scores with other variables. The M (0.692) and SD (0.215) of the standard deviations also constitute further useful information. They show that the 95% confidence interval for the standard deviation is $0.692 \pm 1.96 \times (0.215/22)^{0.5}$, that is, 0.646 to 0.738. If a researcher finds that his/her value is below this, it is certainly low, implying that correlation coefficients may be underestimated due to a truncated range of scores.

The presence of skewness in VVIQ scores raised an additional difficulty for correlational studies. The shape of the distribution means that there will be a small number of individuals with high scores. If these outliers also have an extreme score on the second variable, they may have a disproportionate effect on r . Should the data of these individuals be kept or dropped? This places the investigator in a quandary because r may be significantly raised or lowered. Removing outliers has raised them from .72 to .89 (Rossi, 1977), from .62 to .79 (Marks, 1987), and from .71 to .83 (Marks, 1993). However, in a sample of 31 males (unpublished data) who completed VVIQ and A. Richardson's (1969) version of the Gordon Test of Visual Imagery Control (TVIC; Gordon, 1949), I found one who had a maximum score of 5 on the VVIQ (total score of 80), indicating no imagery, and a minimum score of 0 on the TVIC, indicating no control of imagery. Without that individual, the correlation between VVIQ and TVIC scores was -.080 (not significant); with him, it was -.487 ($p < .05$). Adding to the difficulty here is Marks' (1983b, p. 108) warning that r s may plummet because of a few subjects who reverse the VVIQ rating scale. Although this misunderstanding is apparently rare, it implies that some of the high scores in any skewed distribution should actually be low. Furthermore, Ahsen (1990, 1991) argues that idiosyncratic scores may represent important variations in imagery dynamics, which implies that outliers should be retained. Probably the best procedure, which has been followed (Marks, 1987; Rossi, 1977), is to report both r s, then exercise caution in judgment.

After discovering skewness in their large sample, Kihlstrom et al. (1991) also warn that VVIQ-criterion r s based on the typical small random sample are likely to be weak, and they suggest that studies might employ the extreme-groups technique. However, in their work with quartile (25%) splits, they found that "the cutpoint for the VVIQ 'low vividness' group was 39 on a 60-point scale" (p. 140) which implies that many "poor" visualizers were reporting fairly good images. Unfortunately, Kihlstrom et al.'s numbers are puzzling. On pages 135-136, they state that they used the standard VVIQ with 16 items and a 5-point scale, so that the range of total scores was 16 to 80 with a midpoint of 48 (3×16). Yet here (p. 140), they mention a 60-point scale, with a rating of "moderately clear and vivid"

(which is 3) on every item giving a total score of 36, and a rating of "vague and dim" on every item giving a total score of 48. Despite this ambiguity, Kihlstrom et al.'s general claim is supported by the present analysis, which showed that the mean score of "poor" visualizers was approximately 3 ("moderately clear and vivid"). This also implies that it is difficult to find people who report imagery that is "vague and dim" (a rating of 4) as demanded by Marks' (1972, p. 83; 1973a) initial question.

The argument that VVIQ scores are contaminated by response leniency also questions the wisdom that classifies subjects as good ("clear and vivid," Marks, 1972, p. 83; 1973a) if they report scores in the 1.6 to 1.9 range (better than "clear and reasonably vivid," a rating of 2). Their imagery may not be as vivid as it appears. However, this also implies that the imagery of poor visualizers may be less vivid than it appears. Thus, the surprising logical outcome of this reasoning is that, if extreme groups are selected, they may consist of people who actually possess the desired levels of vividness: good imagers may be close to 2 and poor imagers close to 4! Of course, this assumes that the leniency bias operates in a similar fashion throughout the scale. In addition, it is hazardous to recommend a scale that we mentally adjust for the contaminating effect of leniency.

If extreme groups are used, a 25% to 27% split will often provide optimal power to detect a relationship (Feldt, 1961). Since almost half of the extreme groups studies have employed a median split, with a mean percentage of 33.24, their overall power is slightly low. However, their mean sample size of 18.36 is close to the generally-recommended range of 15 to 20 for experimental groups (Cowles, 1974). It might be thought that the studies with median splits employed larger samples to compensate for loss of power. However, their mean sample size of 20.13 ($SD = 17.62$) was not significantly different from that of the others ($M = 16.89$, $SD = 11.60$), $t(31) = 0.63$, $p > .10$. It appears that the median split may usually be employed as a matter of convenience, because the mean total number of initial subjects required was 20.13×2 ($100\% = 2 \times 50\%$) = 40.26 or 41, whereas the corresponding value for the other studies was 18.89×5 ($100 = 5.19 \times 19.27\%$) = 94.45 or 95.

The logical analysis of scale ambiguity and of demands in the instructions implies that the low central tendency of VVIQ scores is due to biased responding. This interpretation is supported by reports that, on post-test questioning, subjects describe their imagery as less vivid than indicated by their ratings (Chara & Verplanck, 1986; Crawford & Allen, 1983; Sommer, 1980). However, these claims are anecdotal. It is also possible that the distribution of scores is not a reflection of biased or careless responding, but of the universality of imagery (Gordon, 1972, pp. 74, 79; McKellar, 1972,

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p. 35), and perhaps even of vivid experiencing. This assumes that the scale is accurate, but also implies that it will be very difficult to obtain subjects with poor imagery. These two interpretations were compared by examining whether VVIQ scores relate significantly to various extraneous variables: test format, social desirability, context, self-confidence, eyes open or closed, and "parental filter." If they do, the response bias alternative is supported.

Effects of Extraneous Variables

Questionnaire format. One factor that may encourage lenient responding on the VVIQ is the test format in which, following the structure of the QMI (Sheehan, 1967), the items are arranged in blocks of four. However, White, Ashton, and Law (1978) found that mean scores on the QMI increased, indicating less vividness when the items were reordered. For three studies with the VVIQ (McKelvie, 1979, 1986, 1992a), the differences between mean scores on the random and blocked versions were converted to effect sizes (d), whose M was 0.30, which is between small and medium in Cohen's (1977) terms. Because scores were lower with the blocked VVIQ, these studies suggest that this format encourages lenient responding.

However, in McKelvie's (1992a) study, it was also noted that the random distribution remained positively skewed and that its overall SD of 0.68 was similar to that of the blocked version. This suggests that, although the random VVIQ may be less contaminated by response leniency, the psychometric properties of its distribution remain less than ideal. At the same time, it was also reported that scores increased for both good and poor visualizers (27% split), and that the latter mean of 3.29, which was significantly higher than 3, may offer some hope to researchers seeking subjects whose imagery is reported to be vague and dim.

McKelvie's (1992a) claim that both blocked and random VVIQ distributions were skewed was based on a visual inspection of the scores in a frequency distribution table, but their skewness can be quantified. Using the third moment about the mean (g_1 , Ferguson, 1981, pp. 72-73), I calculated a value of 1.227 for the measure of skewness. Because $g_1 = 0$ for a symmetrical distribution, this indicates that there is a high positive skew. However, the corresponding value for the random distribution was 0.847 showing, as already indicated, substantial skew, but to a lesser degree than on the blocked version. To examine this issue further, I calculated g_1 for four samples reported in the 1986 paper. In the first two, with the blocked VVIQ, g_1 was 0.422 and 0.721, respectively, indicating that skew was present. However, in the second two, the corresponding values were -0.302 and 0.121, indicating negative skew and very little positive skew, respec-

tively. These data suggest that the random VVIQ reduces positive skew (overall change was from a mean g_1 of 0.790 to 0.222), and may remove it if it was not excessive in the blocked version.

Although these results favor the hypothesis that grouping of items is an instrument factor affecting VVIQ scores, thereby lowering validity of the VVIQ as a test of vividness, Marks adopted the blocked format because he thought that it would encourage natural development of interest and absorption in the rating task, and Hilgard (1977, pp. 103-104) recognized the sequencing as allowing the test to tap the creative imagination. Thus, the random VVIQ may not possess some of the original desirable properties and may be less valid as a predictor. Until the two versions have been directly compared, I cannot conclude whether the blocking of items undermines or enhances validity.

Social desirability, context, and confidence. Other studies have taken a different approach to response bias. Four (Childers, et al., 1985; Otteson, 1980; Parrott, 1986; A. Richardson, 1979) examined the relationship between VVIQ scores and social desirability as measured by Crowne and Marlow's (1964) Personal Reaction Inventory (PRI), yielding a total of 7 r s. I converted these to z s ($M = 0.226$, $SD = 0.150$) and found a 95% confidence interval 0.115 to 0.337 which, when reconverted back to r s, gave an interval estimate of .115 to .325 (with a midpoint of .222) (Table 4). This shows a significant relationship between social desirability and vivid imagery reports. From Table 3 (McKelvie, 1994a), the confidence interval includes the .25 standard, with its midpoint below. According to the interpretive guidelines, the standard is met, and the amount of contamination is acceptable.

If VVIQ task ambiguity invites extraneous effects on test responses, VVIQ scores may be a function of the context in which they are administered. However, because he expects that imagery vividness should be sensitive to certain contexts which, unfortunately, he does not specify, Marks (1989b) argues that these effects may enhance validity. I considered only context effects that I judged to be a threat to validity.

It has been found that VVIQ scores were lower when the test was administered following hypnosis than waking instructions (Crawford & Allen, 1983) and following a recognition test for upright (more difficult) than for inverted (easier) faces (McKelvie, 1993a). This suggest that subjects may respond more leniently on the VVIQ if they have received performance-enhancing cues. This possibility receives further support from three studies (McKelvie, 1993a; Reisberg et al., 1986; Reisberg & Leak, 1987) that yielded 5 r s between task confidence and VVIQ scores. Combining them in the

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same manner as with social desirability gave a confidence interval for r of .436 to .512, centered on .474 (Table 4), indicating that lower (more vivid) VVIQ scores were associated with greater confidence. This exceeds the acceptable standard (Table 3) and is evidence of an unacceptable relationship.

Table 3
Standards and Guidelines for Judging Content Validity Correlation Coefficients

VVIQ-Variable Relationship Hoped For	Judgment
Independence	
Nonsignificant correlation	No contamination
Correlation .15 or less	Inconsequential contamination
Correlation .16 to .25	Some contamination (acceptable)
Correlation above .25	Unacceptable contamination
Association	
Correlation .85 or more	No contamination
Correlation .75 to .84	Some contamination (acceptable)
Correlation below .75	Unacceptable contamination
Interpretation of 95% Confidence Intervals	
Interval lies above the standard (or below for low expected correlations)	Standard is exceeded
Interval includes standard with midpoint above (below for low expected correlations)	Standard is met
Interval includes standard with midpoint below (above for high expected correlations)	Standard is minimally met
Interval lies below the standard (or above for low expected correlations)	Standard is not met

Table 4
Content Validity: Confidence Intervals for the VVIQ and Social Desirability, Confidence, Eyes, and Parental Filter

Variables	<i>n</i>	Lower Bound	Mean	Upper Bound	Judgment
VVIQ/PRI correlation	7	.115	.222	.325	SC met: A
VVIQ/confidence correlation	5	.436	.474	.512	Exceeds SC: U
Eyes open and closed means (<i>d</i>)	5	-0.09	0.04	0.17	NC
Eyes open and closed correlation	5	.667	.759	.827	SC met: A
Parental filter means (<i>d</i>)	5	0.16	0.33	0.49	C ¹
Parental filters correlation	10	.776	.845	.894	Exceeds SC, NC met: A, i.e. NC

Notes. VVIQ = Vividness of Visual Imagery Questionnaire, PRI = Personal Reaction Inventory, SC = some contamination, A = acceptable, U = unacceptable, NC = no contamination, C = contamination; italics indicates my overall judgment of the relationship.

¹See text for further elaboration of this judgment.

Of course, the VVIQ is only likely to be contaminated by self-confidence if the VVIQ is given immediately after the experimental task. Moreover, VVIQ scores with neutral instructions were similar to those with difficult ones, and both were higher than with instructions emphasizing the ease of the task (McKelvie, 1979). These results demonstrate that VVIQ scores varied with instructions, but they do not warrant Kaufmann's (1983, p. 249) interpretation that the VVIQ is contaminated. Rather, they imply that subjects do not usually respond leniently because they assume the task to be easy.

Eyes open and closed instructions. Another variable that may affect test scores is the instruction to respond with eyes open or closed. Although Marks (1973a) did not explain why he introduced it (White et al., 1977), he points to research demonstrating interference effects between perception and imagery in the same modality (Marks, 1983b, p. 102). This suggests that imagery may be less vivid if there is concurrent visual input (eyes are open). Unfortunately, most investigators do not report if they followed Marks' instructions. Following Ahsen's (1985a, p. 5; 1990a, p. 32) line of reasoning on the effects of parental filter (see next section), it is even possible that subjects might be classified as good or poor when the distinction between them is whether they chose to respond with eyes closed or open, respectively!

Two kinds of data address this issue: comparisons of mean VVIQ scores and r_s of VVIQ scores under the two conditions. If the eyes open and closed variable does not threaten validity, M_s will not differ and r_s will be high. From four studies (Chara & Hamm, 1989; Dowling, 1973, cited in White, et al., 1977, p. 146; Narchal & Broota, 1988; A. Richardson, 1979), I calculated five effect sizes for differences whose M and SD were 0.04 and 0.29. A positive values of d indicate higher scores, that is, lower vividness, with open eyes. Because the 95% confidence interval of -0.09 to 0.17 includes 0 (Table 4), there is no significant effect of eyes. The absence of this effect is also notable in view of Marks' procedure, in which subjects complete all items on the VVIQ with eyes open, then return to the beginning and rate them with eyes closed (Ahsen, 1985a, p. 10). Because a practice effect has been found when the test is retaken, particularly very soon after (McKelvie, 1986), Marks' procedure does not control for the order effect. On the other hand, the instructions may offset this tendency, because they state: "Try and give your 'eyes closed' rating independently of the 'eyes open' rating. The two ratings for a given item *may not in all cases be the same*" (Ahsen, 1985a, p. 10) (emphasis mine). These words imply a demand that the ratings will (normally) be the same, somewhat undermining the obtained non-significant difference.

For the five r s between eyes open and eyes closed scores, the confidence interval was .667 to .827, centered on .759 (Table 4). Another value of .653 (Chara & Hamm, 1989) based on ρ rather than r is probably commensurate with these results, because it may slightly lower due to the loss of information from ranking. The present confidence interval contains the .75 standard, with the midpoint exceeding it, indicating that it has been met. Scores in the VVIQ are slightly contaminated by the choice of eyes open or closed, but to an acceptable degree.

These results alleviate the concern that a person's VVIQ score will be affected by choice of eyes open or closed, and they imply that Marks' (1973a) standard instruction has not been seriously breached by investigators who do not mention it. In fact, Ahsen argues that it is best to allow subjects to exercise their own preference, because he found that "most subjects compulsively return to their own natural way of functioning" (Ahsen, personal communication, December 31, 1990), no matter what the eyes instruction (see also Ahsen, 1987, p. 42, applying the same point to instructions in general). Indeed, he thinks that averaging the two sets of scores might conceal real imagery differences. This argument has some merit, but averaging also has the advantage that it would minimize the problem of misclassification. Some potential members of the extreme groups would be lost, but those that remain would have scores that were uncontaminated by the eyes variable. Overall, unless it is demonstrated that the VVIQ predicts differentially under the two conditions, I think that the investigator may exercise choice here, although the practice should be reported, particularly if Marks' recommendation is not followed (e.g. Alvarado, 1984; McKelvie, 1986). One point that might be considered is whether or not the criterion task will be accomplished with eyes open or closed. The VVIQ should then be administered in the same way.

Parental filter. In the first four items of the VVIQ, respondents think of a relative or friend who is frequently seen. Presumably, the closeness of the relationship and the high exposure rate is designed to increase the likelihood that the target is familiar and therefore easy to image. Indeed, Read and Peterson (1975) found that shorter image latency was associated with greater frequency of exposure for an imaged person for most subjects. Subjects also reported that their images were more vivid, clear and complete for people whom they had seen more recently and frequently and with whom they were emotionally involved. Although he cites a claim by Galton that some people have difficulty visualizing intimates, particularly if they have been lost, Yarmey (1975) also found that images were rated as most vivid for the person felt to be closest. These results are supported by evidence that the ease of forming an image was greatest for close friends and relatives, with parents best of all (Morris, 1992).

However, in his clinical experience, Ahsen (1991, p. 23) found that patients often had difficulty seeing vivid images of one parent. Thus, ratings on the first 4 items of the test (see Table 1) could be affected if one of the parents happened to be chosen as the relative or friend. Ahsen (1985a, 1990, 1991) then argued that selection of mother or father at the beginning of the VVIQ may influence responses on subsequent items. Consequently, if total test scores are radically affected by this choice, it could determine whether the individual is classified as a good or a poor visualizer (Ahsen, 1985a, p. 5; 1990, p. 32). The context effect of parental "filter" (Ahsen, 1990) as a threat to validity was addressed in the same manner as the eyes factor, by examining *Ms* and *rs* for each condition. Because Ahsen's claims have been debated (Ahsen, 1987, 1988; 1993d; Hilgard, 1987; Marks, 1987, 1993; McKelvie, 1993b; Reisberg & Heuer, 1988; A. Richardson, 1988; J. T. E. Richardson, 1988; Sheehan, 1987), these studies will be discussed in detail.

In Ahsen's (1985a) first investigation, 25 people rated the first four items of the VVIQ in turn for the father's image only and then repeated the procedure for each item for the mother's image. Ahsen states that ratings were clearly lower (i.e., more vivid) for father than for mother for 15 subjects, lower for mother than for father for 7 subjects, and similar for father and mother for 3 subjects, suggesting that a person's score on the VVIQ might be radically influenced if a parent was chosen at the beginning. In a second experiment, Ahsen modified Marks' VVIQ by including the instruction to keep each parent in mind before each block of items. Here, it is not clear if each block of items was rated first for father and then for mother, or whether the complete questionnaire was rated for father then for mother, or whether each item was rated for father then for mother. Moreover, Ahsen does not give his sample size, or distribution of scores, but only reports the complete data for a "typical" subject (Marks, 1987, p. 6). Here, ratings were generally lower for father than for mother. These two studies suggest that VVIQ ratings were *more vivid* with *father* than with *mother* in mind. However, Marks (1987) tested 22 students under the two parental filters by administering the father and mother versions of the VVIQ 10 days apart in a counterbalanced order. Later (Marks, 1993), he reported that the *Ms* did not differ.

Other studies have used Ahsen's AA-VVIQ, in which father or mother is kept in mind before each block of items and each item is rated twice (once for father and once for mother) before the next item is completed. Instructions do not specify in which order the parental conditions should be rated (only that they should be kept "alternately" in mind), but the place for the father rating appears to the left of the place for the mother rating.

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Reintroducing a classic practice that has been superseded by our conventions on descriptive and inferential statistics, Ahsen documented his results by providing complete raw data for two undergraduate samples (American Bent Hall Group, 16 males, 44 females, Ahsen, 1990a; Pakistani Women Group, 48 females, Ahsen, 1991). He summarized them by combining responses to all items from all subjects, and calculated that there was 238% (Ahsen, 1990; later corrected to 212%, Ahsen, 1991) and 190% (Ahsen, 1991a) less vivid imagery under the father than mother filter. Because only 56% of all ratings were duplicated across instruction (and 38% in the 1991a study by my calculation), Ahsen (1990) also argues that scores were inconsistent under the two conditions. For two additional American student samples (47 females; 13 males, 17 females), Sussman (1993) found 291% and 279% more dimming of vividness under the father than mother instruction. Ahsen (1990, 1991, 1993a) concludes that imagery is generally *dimmer* under the *father* than under the mother filter.

Although he argues that these data raise questions about the reliability of the VVIQ under the "special circumstances" (Ahsen, 1990, p. 15) of keeping parents alternately in mind (see also Ahsen, 1990, p. 19) and about the assumption that imagery is an ability (Ahsen, 1985a, p. 14; 1986, pp. 2, 4; 1987, p. 14; 1990, pp. 11, 15; 1991a, p. 24; 1993b, pp. 34-35; 1993d, p. 156), Ahsen (1986, p. 6; 1987, p. 13; 1990a, p. 15, 17; 1991a, p. 27; 1993a, p. 21) does not use them to spearhead an attack on the psychometric properties of the VVIQ. Instead, he emphasizes their implications for clinical and experimental studies of the nature and dynamics of vividness itself (Ahsen, 1985a, 1986, 1987, 1990, 1991, 1993a). Although he has both accepted (Ahsen, 1985a, p. 5; 1993c, p. 142; see also Marks, 1987, p. 3; J. T. E. Richardson, 1988, p. 117; Sussman, 1993, p. 326) and rejected (Ahsen, 1993d, p. 179) that his findings constitute a context effect on VVIQ scores, I think that they can be treated in this way from the perspective of assessing the psychometric properties of the VVIQ, and that his results are relevant to content validity.

Because Ahsen (1990, 1991) did not present his results in the usual form of *Ms*, *SDs*, and *rs*, I calculated them from his raw data (McKelvie, 1993b). Considering men and women separately, the three effect sizes (*d*) for father vs. mother scores from Ahsen's two main studies were 0.16 (men), 0.42 (women) (Bent Hall Group), and 0.64 (women) (Pakistani Women Group) indicating higher ratings (less vividness) under the father instruction. McKelvie (1993b) also conducted a new study of the AA-VVIQ with 23 male and 28 female Canadian undergraduates (Bishop's University Group or BUG). Ratings were higher under the father than mother instruction for women (*d* = 0.64) but were not significantly different for men (*d* = -0.23).

Together, the five samples from Ahsen and McKelvie give a mean effect size of 0.33 ($SD = 0.37$) and a 95% confidence interval of 0.16 to 0.49 (Table 4). This non-zero, small to medium effect (Cohen, 1977) shows an overall tendency for ratings to be higher (i.e., lower vividness) under the father than mother instruction, suggesting filter contamination. However, it conceals a gender difference. For males, mean $d = -0.04$, which is not different from 0, whereas for females mean $d = 0.57$ (a medium effect). Sussman's (1993) results were not included, because she did not provide raw data or M s, but they also suggest that the imagery dimming effect under father compared to mother is stronger for women than for men (Sussman, 1993, p. 323). In her second sample, where scores were 291% higher for father- than for mother-filtered imagery, the percentages were 386% and 173% for women and men, respectively.

Pointing to Ahsen's (1985a) report that imagery was dimmer under the mother than the father filter, and to his own (1987) finding of no significant difference, Marks (1993) argues that the father-dimming effect claimed by Ahsen is not robust. Unfortunately, I could not include the results from these two studies in my overall analysis, because the required information was not reported. However, in both cases, the two versions of the VVIQ were administered sequentially rather than simultaneously (i.e., side-by-side ratings of items under each filter). Differences may only emerge under the latter condition. Marks (1993) agrees, and argues that they may reflect a contrast effect which is eliminated by delayed sequential presentation. In fact, with order of presentation controlled, I recently compared AA-VVIQ scores in the side-by-side and immediate sequential conditions (McKelvie, 1994b) for undergraduates who were matched on the basis of their scores on the randomized-item version of Marks' VVIQ. With both side-by-side and sequential administration, there was no significant effect of parental filter. However, in the first case, imagery scores were higher (less vivid) for the father-first than for the mother-first order for women and for the mother-first than father-first order for men. Although scores on the two blocked parental versions of the VVIQ were generally lower than on the original random version, there was a notable exception: the score under father was higher than for the random version for women when father was rated before mother. Although these results do not demonstrate a contrast effect on the side-by-side version of the questionnaire, they suggest imagery dimming for the opposite-sex parent. This holds particularly for women in the side-by-side condition with father rated first, which is similar to the condition under which the previous imagery dimming effect for women was obtained. In fact, the effect size for dimming of father compared to mother imagery in the side-by-side father-first condition was $d = 0.363$ for women and $d =$

0.134 for men. Overall, I conclude that there is no consistent evidence across different administration conditions of dimmer imagery under the father than mother filter, although it may occur for women when ratings are made side-by-side for each item and given first for father.

However, I doubt that the arbitrary choice of a parent at the beginning of the VVIQ will contaminate the total score to an extent that compromises a person's classification as a good or poor imager. First, father dimming seems to be confined to a narrow set of conditions. Because any evidence for greater imagery vividness under the mother filter has occurred when the mother rating is placed to the right of the father rating, it could reflect the VVIQ order effect (McKelvie, 1986). Ahsen (1993d, p. 160) accepts both formats but objects to a delay in the sequential condition because it may interfere with "interhemispheric communication." I am not sure what he means by this, but if it implies that respondents can mentally compare the two images under each filter before recording either of their ratings, it may enhance accuracy. However, Marks' (1993, p. 341) objection to the side-by-side format is clear: It encourages an artifactual contrast/carry-over effect, particularly in view of the fact that the AA-VVIQ contains a demand that ratings not be the same (see also Marks, 1987, p. 6. In support of the latter claim, Marks quotes Ahsen [1990, p. 14] as follows: "*we tell the subjects in the way of normal routine that the scores when keeping mother in mind and when keeping father in mind may not be the same*" (Marks' italics). Notably, Marks fails to include the rest of the sentence, which states: "which preliminary instruction is not different from that of Marks' in the VVIQ where the subject is told that doing the test with open eyes or closed eyes may not give the same score." Of course, although this implies that Marks may have devised the instruction, it does not detract from the criticism that it contains a demand to respond differently. Indeed, Ahsen (1990, p. 10) recognizes this possibility, but questions it on the grounds that his results showed that ratings were clearly higher under father than under mother for some subjects and clearly higher under mother for others, and that there was a systematic effect in which ratings were generally higher under father. If there is a general demand to be different, I do not understand Ahsen's reply, which assumes that differences occurred. However, a closer examination of the AA-VVIQ itself shows that the exact instructions are as follows. In the introduction, it is stated that the image "may or may not change in vividness if one or other parent is kept in mind" (Ahsen, 1990, p. 22), which (to me) implies no expectation. Later, just prior to the ratings, it is stated: "*The two ratings for a given item may not in all cases be the same*" (Ahsen's 1990, p. 23). The critical difference in the latter case is the phrase "in all cases," which was also present in the instructions for eyes

open and eyes closed. As noted in that discussion, it implies (to me) that, normally, the ratings are likely to be the same, but that subjects may not always find this to be the case. Thus, if there is any demand here, it is to respond similarly under each parental filter. If there is an artifactual carry-over effect with the side-by-side format, I do not think that it results in differences due to demands. As recognized by both Ahsen (1993d, pp. 156, 160) and Marks (1993), more work on the conditions of administration (demand characteristics, order effect, carry-over effect, side-by-side versus sequential format) is needed.

A second reason for doubting that choice of parent will contaminate VVIQ scores is that Marks (1987) found that only 4 of his 22 subjects reported thinking of a parent at the beginning of the VVIQ when it was administered in the usual manner. In addition, Ahsen's AA-VVIQ instructs respondents to think of mother and father before each block of items. With the standard version of the VVIQ, any initial effect of choosing a parent on the first four items may not spontaneously affect subsequent responses (McKelvie, 1993b). I investigated this with a condition in which the instruction to think of parents was given only before the first block of items and was not repeated again (McKelvie, 1994b). Total VVIQ scores did not differ between parental filters, although there was a general order effect in which the second rating was more vivid than the first. I conclude that choice of a parent at the beginning of the VVIQ will not jeopardize a subject's classification as a good or poor imager.

There were 10 estimates of r between VVIQ scores under the two conditions (from Ahsen, 1990, 1991; Marks, 1987; McKelvie, 1993b, 1994b), giving a confidence interval of .776 to .894 (midpoint = .845) (Table 4) that exceeds the .75 standard and even overlaps with .85. Furthermore, in my condition in which the parental instruction was given only once and order was controlled, r between father and mother scores was .915 (McKelvie, 1994b). From the current guidelines, VVIQ scores are either somewhat contaminated (but to an acceptable degree) or inconsequentially contaminated by parental filter. Overall, and agreeing with Marks (1993, p. 343), I conclude that choice of parent at the beginning of the test is not a confounding variable. Therefore, I do not endorse Ahsen's (1986, p. 3; 1987, p. 14; personal communication, December 31, 1990) suggestion that the reference to "relative" in the first four items be dropped, in line with standard practice with ambiguous items. Ahsen argues that the term is a "relic" from the Betts (1909) questionnaire, which was written during an era when people had closer contact with members of their extended family than they do now. This may be true, but I do not think that content validity is seriously compromised by the reference to a relative.

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Content Validity: Conclusions

VVIQ scores are not unacceptably contaminated by either choice of eyes or choice of parent. In addition, subjects do not respond leniently because they assume the task to be easy. They may do so if they are approval seekers, but the contaminating effect is acceptable. The only factor that may seriously distort VVIQ scores is cueing from an immediately-preceding test. If subjects think that they have done well or if they are self-confident, they respond more leniently and report more vivid imagery. The obvious solution to this difficulty is to avoid administering the VVIQ soon after an experimental task, as Marks (1973a) originally recognized.

The status of the blocked format is unclear. Scores are lower than those of the random version, suggesting that it encourages response leniency, but its structure may tap imagery vividness more effectively. Although the random VVIQ gave a mean score for poor imagers that approaches Marks' (1973a) standard of "vague and dim," its distribution remained skewed and its variance was not greater than the blocked questionnaire. The two versions should be compared further.

The most serious implication of the present threats to content validity is that nonsignificant relationships with other variables are difficult to interpret. They could partly reflect the failure of vividness to capture the essence of experienced imagery, ambiguity in its definition, scale reversals, subjectivity of the rating task, or a truncated range of scores.

Reliability of the VVIQ

Reliability is part of construct validity, but is also a prerequisite for criterion validity (Marks, 1983, p. 106). I judged the reliability coefficients according to the guidelines in Table 5 (McKelvie, 1994a).

Studies of Reliability

Internal Consistency

The results of studies reporting split-half (Marks, 1973a, 1987; McKelvie, 1986; McKelvie & Gingras, 1974; Narchal & Broota, 1988; Rossi, 1977) and alpha (Childers et al., 1985; Dowling, 1973, cited in White et al., 1977; Rossi, 1977) estimates are shown in Table 6. It was not always stated whether split-half estimates had been corrected for length, implying that some may be low. Taken as they are, they gave a 95% confidence interval for r of .856 to .902 (midpoint = .875). Not included was an estimate of .82 (correction unknown) obtained by correlating the first and second halves of the test (Kosslyn, Brunn, Cave, & Wallach, 1984). For alpha, the confidence interval of six values was .870 to .906, centered on .890. Both intervals exceed the .85 standard.

Table 5
Standards for Judging VVIQ Reliability Coefficients

Coefficient	Judgment		
	Very Good	Acceptable	Unacceptable
Immediate test-retest	.95 or more	.85 to .94	Below .85
Internal consistency	.95 or more	.85 to .94	Below .85
Immediate alternate-form	.85 or more	.75 to .84	Below .75
Delayed test-retest	.85 or more	.75 to .84	Below .75
Delayed alternate-form	.75 or more	.65 to .74	Below .65

Interpretation of 95% Confidence Intervals

Interval lies above the standard	Standard is exceeded
Interval includes standard with midpoint above	Standard is met
Interval includes standard with midpoint below	Standard is minimally met
Interval lies below the standard	Standard is not met

Table 6
Confidence Intervals for Reliability Coefficients for the VVIQ

Coefficient	<i>n</i>	Lower Bound	Mean	Upper Bound	Judgment
Split-half	8	.856	.875	.902	Exceeds A: A
Alpha	6	.870	.890	.906	Exceeds A: A
Alternate-form: immediate	1		.64		U
Alternate-form: delayed	1		.54		U
Test-retest: Immediate	1		.940		Exceeds A: A
Test-retest: delayed	7	.722	.735	.782	A min met: A

Notes. A = acceptable, U = unacceptable, min = minimally; italics indicates my overall judgment of reliability.

Because Anastasi (1988, p. 124) points out that alpha will normally be lower than split-half reliability "unless the test items are highly homogeneous," these data suggest that the internal consistency of the VVIQ is acceptable, even good. The demonstration of a single factor for VVIQ items (Childers et al., 1985; Dowling, 1973, cited in White et al., 1977) reinforces this conclusion.

The only blemish to this rosy picture is a split-half estimate of .693 obtained with the random version of the VVIQ (McKelvie, 1986). This value is less than the unacceptable standard of .75. Although unreplicated, it suggests that the blocked format of the questionnaire is a factor in the previous estimates. In fact, McKelvie (1986) argued that the split-half estimate for the original VVIQ is likely to be high because of proximity error within each block of items. If subjects give similar responses within a block, the *r*

of scores from odd and even items will be elevated. Indeed, two other factor analyses (Dean & Morris, 1991; Kihlstrom et al., 1991) both produced four factors corresponding to the four blocks of items, and the another result led Kosslyn et al. (1984, p. 240) to state that "The VVIQ is clearly a more complex measure than is usually realized."

Of course, if the VVIQ taps a variety of processes, as I have suggested, it may not be unidimensional. It may also be premature to question internal consistency on the basis of the (unreplicated) low random split-half estimate, because it was noted earlier that randomization of the items may also disrupt some of the natural imaging processes tapped by the blocked version. The cautious conclusion is that internal consistency on the blocked VVIQ is acceptable.

Test-retest and Alternate-form

Results. The only estimate of immediate test-retest reliability is .94, (10-min delay, McKelvie, 1986). This exceeds the standard of .85 and the upper bound of the confidence intervals for internal consistency. However, the estimate could be artificially elevated by recall of previous responses. McKelvie's study supports this possibility, because a similar reliability coefficient of .92 was found when subjects on the second test were instructed to recall and duplicate initial responses. Moreover, he argued that this inflation due to carry-over was exacerbated by the blocked format of the VVIQ, because the corresponding test-retest and "memory" coefficients for a random version of the items were .85 and .98 respectively. The former value was consistent with the acceptable standard of .85 and was lower than the latter, indicating that carry-over was less of a factor for the random VVIQ.

As with immediate test-retest, there is only one estimate for each of immediate (10-minute) and delayed (3-week) alternate-form reliabilities. In the first case (McKelvie, 1993b), the value was .64, lower than the acceptable standard of .75. Similarly, the delayed value of .54 (McKelvie & Gingras, 1974) was lower than its .65 standard. Although unreplicated, these data suggest that the VVIQ is not consistent over forms. The poor immediate value is particularly worrisome in view of Buros' (1978) argument that the best type of reliability estimate is alternate-form.

Table 6 also shows the results for delayed (3 to 7 weeks) test-retest reliability (Isaac et al., 1985; Marks, 1973; McKelvie, 1986; McKelvie & Gingras, 1974; Parrott & Strongman, 1985; Rossi, 1977). For seven *rs*, the confidence interval was .722 to .782 (midpoint = .735). Because this interval includes .75, with its midpoint below, the acceptable standard for test-retest reliability is minimally met.

As a corollary to the investigation of carry-over effects on immediate test-

retest, McKelvie (1986) also included "memory" conditions at a 3-week delay for both blocked and random formats of the VVIQ. Although he concluded memory may play some role for the blocked but not for the random VVIQ, its effect was negative, suggesting that the true value for delayed blocked retest reliability might be higher than .735. Although conducted with another test, further work on this issue indicates that recall does not artificially increase the 3-week test retest estimate (McKelvie, 1992b).

Unfortunately, there is another reason to doubt whether the .735 value is acceptable, which was judged by a logical absolute standard. In practice, reliability coefficients usually exceed .80 (Anastasi, 1988, p. 115). More specifically, in a survey of studies of temporal stability for individual measures of intelligence, Schuerger and Witt (1989) found that the average value was .89 for delays up to 10 months. Moreover, in two further investigations of questionnaire measures of personality, Schuerger, Tait, and Tavernelli (1982) and Schuerger, Zarrella, and Hotz (1989) showed that the corresponding value for delays of up to 1 year was .73. In fact, basing his calculations on a number of large-scale studies with delays ranging from 1 to 40 years, Smith (1992, Tables 6 and 7) estimated that the annual stability coefficient was .99 for intelligence tests and .64 for personality inventories. If the VVIQ is a maximum performance test of ability, as Marks has claimed, then the present overall test-retest estimate falls short of the previous empirical standard for intelligence tests. On the other hand, I have argued above that VVIQ scores are slightly affected, although not to an intolerable degree, by contaminating variables such as social desirability, suggesting that it shares certain features with measures of typical performance. Although the present value of .735 is higher than Smith's overall estimate of .64 and similar to Schuerger et al.'s empirical average (.73) for these tests, these estimates include delays up to a year. However, because Smith (1992, Table 8) found a personality inventory test-retest value of approximately .76 up to 4 weeks, the present value may be taken to resemble the empirical standard for typical performance. Although he does not mention the possibility of a typical performance trait, Ahsen (1985a, p. 14; 1986, pp. 2, 4; 1987, p. 14; 1990, pp. 11, 15; 1991, p. 24; 1993b, pp. 34-35; 1993d, p. 156) also questions whether the VVIQ measures an ability.

Finally, although the immediate and delayed test-retest reliability estimates for the random VVIQ may be less contaminated by recall than those for the blocked version, this does not mean that the random VVIQ is preferable. First, as already noted, randomization may distort the natural process of assessment encouraged by the blocks. Second, McKelvie (1994) found that the correlation between the random VVIQ and the blocked AA-VVIQ given 3 weeks later ranged from .433 to .627 and was .520 and .540 when

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parents were mentioned only before the first block of items. The latter two estimates are similar to that for delayed alternate-form, which was judged to be low, and suggest again that the random and blocked VVIQs may tap different processes.

Filters, reliability, and duplication of responses. In developing the .75 standard for delayed test-retest and immediate alternate-form reliabilities, it was observed that the VVIQ was repeated under one variable for studies of filters and eyes. Thus, these coefficients can be viewed as forms of reliability, with the source of error as parent or eyes instruction rather than time or content. Indeed, although he did not set out to assess test-retest reliability of the VVIQ, Ahsen (1990, p. 17) presents calculations that seem to cast doubt on it. He begins by adopting .75 as a standard, since he thinks it is close to what Marks accepts. He then argues that it would be desirable, and in some sense "equivalent," if a subject were to repeat 12 out of 16 responses under the mother and father instructions. However, for his Bent Hall Group, over 960 responses (60 subjects on 16 items), only 56% were repeated, indicating that the VVIQ "flunked the most minimum standard of reliability" (Ahsen, 1990a, p. 18). Moreover, according to Ahsen (1990, pp. 18-19), "the VVIQ is designed to predict test-retest case by case (16 items per case) rather than over total number of items." He then shows that only 33% of the 60 subjects individually met the .75 criterion, a result that he characterizes as "shocking" for reliability. Furthermore, for the Pakistani Women Group (Ahsen, 1991a), the values are 38% and 8% by my calculation.

However, I do not agree with Ahsen that the VVIQ was designed to predict test-retest case by case. Marks (1987, p. 3) is clear that the VVIQ was designed to measure individual differences for research purposes, which involves relating scores to those on a criterion variable (Reisberg & Heuer, 1988). Here, the usual correlational estimates of reliability, not response repetition or "equivalence" (Marks, 1993, p. 342), are appropriate, because they indicate the degree to which scores remain relatively similar to other scores in the distribution. As Ebel (1965, p. 334) states: "Since a reliability coefficient looks like a proportion, some users are tempted to interpret it as the proportion of scores that agree. This is not correct." Of course, the reliability coefficient *is* a proportion, but is the proportion of obtained variance that is true variance, that is, not due to error. It is true that if all subjects duplicated their ratings on each item, total scores would be perfectly consistent and the reliability coefficient would be 1. However, it is also true that if all subjects showed a similar reactivity or practice effect, duplications would be 0 and the reliability coefficient would remain at 1. In fact, it was demonstrated above that the overall r across parental filter was .845, giving

a reliability estimate whose confidence interval exceeds my .75 standard.

Also notable is Marks' (1987) correlation between mother- and father-filtered VVIQ scores obtained 10 days apart. Because both delay and filter are possible sources of variable error across the two tests, the present logic governing standards for correlation coefficients dictates that a value of .65 would be acceptable and .75 very good (Table 5). Marks (1987, p. 6) initially reports a mother/father correlation of .62, which he characterizes as "quite consistent" but which, according to the present guidelines, indicates that both sources of variance combined to reduce reliability to an unacceptable level. However, Marks identifies a pair of data points as atypical. With the scores for this subject removed, the correlation became .79, which is very good. Care must be taken in the treatment of outliers, but Marks is justified in removing these scores, since the individual gave personal reasons for difficulty in imaging both parents and actually rated an aunt instead of mother. Logically, the value of .79 could be interpreted as an effect of delay or of parent. Since delayed test-retest reliability is .735, it is probably the former that was operating, implying that parental filter had no effect on VVIQ scores. Marks' results also raise the interesting possibility that previous estimates may also include subjects who did not follow instructions, implying that these estimates might be artificially low. Overall, his findings reinforce the earlier conclusion that VVIQ scores are little contaminated by parental filter. If so, delayed test-retest reliability for the mother and father AA-VVIQ should be commensurate with the .735 estimate. In fact, with a 19-day delay, McKelvie (1993b) found values of .839 and .790 respectively, both of which are acceptable and slightly higher than .735.

If Ahsen has fallen into Ebel's trap, and has taken a duplication approach to reliability, he would appear to be mistaken. However, this is unlikely for a person schooled in testing. Readers may think that Ahsen questions the psychometric properties of the VVIQ because he retains the term "reliability" in the discussion of his data (Ahsen, 1990, pp. 18-19). Indeed, that discussion appears in a section of the paper entitled "Test-retest Reliability Issue" (Ahsen, 1990, pp. 15-16) [see also Marks' (1993, pp. 341-342) reply under the same heading]. A clue to his motivation may be found in his numerous statements (Ahsen, 1986, p. 6; 1987, pp. 13; 1990, p. 15, 17; 1991a, p. 27; 1993a, p. 21) that he does not criticize the VVIQ qua test, and does not wish "to send the designers of these imagery tests back to the drawing table" (Ahsen, 1986, p. 6). Indeed, "The statistical success of the VVIQ is ample proof of its validity and importance" (Ahsen, 1987, p. 24). Rather, Ahsen is interested in reliability "under the special circumstances which have been spelled out by us regarding the adapted AA-VVIQ" (Ahsen, 1990, p. 15) and that "the reliability we are discussing is the one

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....in respect of our position (Ahsen, 1985) that the VVIQ did not show reliability when parents were kept in mind" (Ahsen, 1990, p. 19). Indeed, Ahsen (1993d, p. 161) contrasts this meaning of reliability with psychometric reliability, whose estimated value he accepts: "...it is clear that the test, no doubt, measures what it measures, and is reliable in that sense, as Marks points out." Ahsen's goal is to use the VVIQ as a vehicle for understanding visual imagery vividness as a dynamic process. From this point of view, variations over such factors as parental filter would not be surprising, and duplication rates would reflect them. As Sheehan (1987, p. 11) puts it: "The issue raised by Ahsen's results would appear to be not the inherent unreliability of items on the QMI and VVIQ, but the extent to which the choice of stimuli that are used to arouse imagery can affect subjects' vividness ratings and other aspects of their imagery performance." In fact, this issue is not confined to imagery tests. Radical fluctuations in performance on intelligence test items can occur as a function of variables such as motivation, yet this does not pose a serious problem for the reliability of the test or the stability of the assumed underlying ability. The crux of the matter may be the contrasting approaches between the psychometric perspective, which assumes that a test should be a valid and reliable measure of a relatively stable trait (Nunnally, 1978, p. 235), and which demands that reliability be treated correlationally, and Ahsen's structural perspective, which is directed toward understanding the dynamics of imagery (see also Ahsen [1993b, p. 35] and J. T. E. Richardson [1988, p. 117]).

Discussion of Reliability Data

Although accompanied by a caution about the blocked format and unidimensionality of the test, the split-half and alpha estimates, which exceeded .85, show that internal consistency is acceptable. In contrast, both immediate and delayed alternate-form reliabilities failed to meet the appropriate standards, and are unacceptable. This conclusion is reinforced by the claim (Garrett, 1958, p. 351) that most individual tests of intelligence possess alternate-form reliability coefficients above .90, the traditional standard described by Buros (1978).

In addition, delayed test-retest reliability was judged to be only minimally acceptable by the logical guidelines and lower than the empirical standards for maximum performance, although it matched the empirical standard for typical performance tests, with which the VVIQ may share some features. Practically, this implies the VVIQ should be administered fairly closely in time (perhaps within 3 weeks) to any criterion that is to be predicted. Here, the investigator is faced with a dilemma because, as was shown under content validity, if it is given soon afterward, responses may

be affected by perceived performance on the experimental task. Immediate prior administration runs a similar risk of contamination of the task by the VVIQ. The best solution may be to administer the VVIQ first, then the criterion task within 3 weeks, but to avoid contamination by dissociating the two activities.

In addition, the VVIQ might be lengthened to improve test-retest reliability. Sixteen items is short compared to most tests and, on the blocked version, is arguably only four. If the number of questions was doubled, perhaps by using the items for the alternate form (McKelvie, 1993b), test theory predicts that reliability would rise from .735 to .847 (Spearman-Brown formula), which is tantamount to "very good" by the present standards. Unfortunately, this may be easier said than done. For both immediate alternate-form and random split-half reliabilities, subjects were required to generate more images than on the usual VVIQ: 8 (2 x 4 major images) and 16 (since the items were unblocked), respectively. Yet, immediate alternate-form reliability was only .64 and the random split-half estimate was a low .693. These results suggest the counter-theoretical conclusion that increasing the number of items on the VVIQ may not improve reliability. Split-half and test-retest reliability estimates for an expanded VVIQ would help to settle this matter.

Is Visual Imagery Vividness a Stable Trait?

Rather than a weakness in the test, the marginally-acceptable test-retest reliability and the unacceptable alternate-form reliability may reflect inherent instability in the construct of vividness across time and circumstance (Reisberg, et al., 1986, p. 53; see also Morris & Hampson, 1983, p. 95). Indeed, Ahsen's major criticism of VVIQ work is that it does not sufficiently recognize that imagery vividness is dynamic, and he questions whether vividness is a stable trait (Ahsen, 1993d, p. 155). For dynamic traits, which "are presumed to be ever-changing as a function of situational and cognitive experiences" (Cohen, et al., 1988, p. 114), test-retest and alternate-form reliability will not be high and, if the dimension in question is purely a state, these estimates will be close to 0. Unfortunately, although Marks thinks that the VVIQ taps dynamic processes, and that VVIQ scores may vary with context (Marks, 1989b) and with mental states, some of which may be induced by VVIQ content (Marks, 1993, p. 344), he has never stated explicitly whether he thinks that the *trait* of imagery vividness is static or dynamic (McKelvie, 1990, p. 559). However, Marks' (1972, p.84; see also p. 92) assumes that imagery vividness is an ability, implying that it should be as stable as other maximum-performance traits.

Whether or not any trait is static or dynamic is part of the question of

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construct validity (Ghiselli, et al., 1981, p. 260), and the various reliability coefficients can offer a clue. According to Ghiselli et al. (1981, p. 260), high and low estimates from one and two administrations respectively suggest that the trait is dynamic. In the present case, the former coefficients exceed .85, and the latter are generally below .75, the lowest being .54 for delayed alternate-form. There is a difference here, but it is not clear whether Ghiselli et al. would consider the values to be high and low. However, in discussing the test-retest values, it was observed that they were commensurate with those obtained with tests of typical performance, suggesting that, although not a stable "ability," the VVIQ construct is somewhat dynamic and is a (reasonably) stable "trait." Nevertheless, it is dangerous to depend too heavily on this guideline, because it does not provide a clear standard by which to judge. If internal consistency is high and other estimates are low, how do we distinguish a dynamic trait from a test that may be unidimensional but inconsistent over time? This problem is endemic to construct validity, where it is often not clear when to attribute ambiguous data to a problematic construct or to a poor test.

However, Ghiselli et al.'s appeal to construct validity implies that there should be independent evidence other than reliability coefficients that could also bear on the nature of the measured dimension. Indeed, Cohen et al. (1988) invoked the criterion of sensitivity to situational and cognitive forces, and Ahsen (1991, p. 21) argues that his changes across parental filter indicate that imagery is a not a static process. Furthermore, although George (1982) found no change in VVIQ scores following imagery enhancement training, others report a decline (Parrott, 1986; Wallace, 1988) and removal of lower criterion test performance by poor than good imagers (Wallace, 1988). Marks (1989b, p. 462) has also noted the sensitivity of the VVIQ to surrounding conditions, including Ahsen's filter, which he describes as "surely a positive feature." Again, it is unclear where to draw the line between positive sensitivity and wayward variation but one possibility is to distinguish the effect of variables on *Ms* and on *rs*. If the latter show that test scores are seriously contaminated, the dimension in question may not be stable. In fact, although the present review of content validity showed that mean VVIQ scores could be altered via cueing from a prior task, it was concluded that the VVIQ was not seriously contaminated by the context variables that have been investigated. Therefore, this evidence does not support the hypothesis that imagery vividness is a dynamic trait.

On the other hand, Ahsen has presented cases showing how vividness of imagery and its rating can fluctuate. For example, one person who initially reported no imagery on the VVIQ scored above average after recalling a summer vacation (Ahsen, 1986, p. 6). Again, from a subject's descrip-

tion of the imagery content and of rating for a particular item (Ahsen, 1987, pp. 34-35), it is apparent that the rating was based on a part of the experienced image and that it changed when attention was shifted to another part. In addition, others have argued that images are dynamic and rapidly-changing (Bugelski, 1971, p. 58), and uneven (Reisberg & Heuer, 1988, p. 92) or incomplete (A Richardson, 1972, p. 121) in detail. I think that imagery vividness should contain these features, since it is defined by Marks as similarity to perceptual experience. Perception is not a fixed stimulus-driven automatic process, but is a joint function of sensory input and interpretation by higher-order factors such as stored knowledge (Anderson, 1990, pp. 47-84; Dodwell, 1970, pp. 205, 229; Zusne, 1970, p. 252). Signal detection theory (Green & Swets, 1966), which assumes that the stimulus itself has varied effects when repeated¹, points to variation in the first component, and evidence of set (Leeper, 1935) and perceptual regeneration from the stabilized image (Dodwell, 1970, pp. 193, 196) demonstrate variation in the second one. Of course, under normal conditions, phenomenal experience is not unstable (Dodwell, 1970, p. 188), because the continual presence of the stimulus permits integration of input and encourages a similar interpretation. In addition, just as Ahsen (1987, pp. 34-35) argues that imagery ratings change as attention is shifted to different parts of the image, so perceptual experience alters as a viewer shifts attention to different parts of the stimulus (e.g., eye movement studies of picture perception; Noton & Stark, 1971; Zusne, 1970, pp. 252-257). Notably, Howe (1985, p. 56) argues that ratings of vividness may partly reflect the subjective effort "to generate, integrate, and briefly maintain the coded spatial components of a designated image." This also implies that a vivid image, like normal perception, is experienced easily and continuously without effort, whereas a less vivid image is less stable, and perhaps more like perception under special circumstances (e.g., ambiguous figures, stabilized images). These considerations imply that imagery, like perception, is not simply a steady-state process.

The issue here is the *degree* of instability. If vividness fluctuates widely and randomly, there is little point in measuring it as a trait. However, although it must be handled with care, psychometric evidence can sometimes also provide insight into theory. Here, it may be unclear whether the the test-retest (.735) and alternate-form (.64, .54) reliability coefficients indicate weakness in the test or a natural property of the construct, but they are too high to support the radical conclusion that vividness is completely unstable, that is, that it is only a "state." In an interesting parallel to this argument, Smith (1992, p. 483) has proposed that the decline in longitudi-

¹ I am indebted to Lionel Standing for drawing my attention to this point.

nal stability of personality over time is due the "inherent indeterminacy of the global dynamics of the normal human brain." Although this randomizing agent operates also for intelligence, it is counteracted by forces that provide feedback on objective accuracy. However, with self-report personality inventories, responses cannot be scored in this fashion, and "there is no objective or consensual standard for the "correctness" of self opinions" (Smith, 1992, p. 495). I suggest that this reasoning also applies to the phenomenon of experienced vividness, which may be even more subject to randomizing influences than some personality traits. This is consistent with the suggestion that the VVIQ taps the creative process (Hilgard, 1977, p. 103; Marks, 1987; Marks, 1989b). According the Smith (1991), one of the features of such behavior is its novelty which, by definition, is unpredictable and therefore indeterminate. Indeed, the notion of a randomizing agent also applies to perceptual experience, to which vividness is supposedly allied, where it was argued that both sensory input and interpretation are variable factors. However, in a parallel to the case of intelligence, the randomizing influence of the brain is counteracted by the constant presence of the perceptual stimulus.

Overall, I think that the case data show that imagery vividness is not as stable as many other abilities, indicating that it will be difficult to create a test with optimal test-retest and alternate-form reliability properties. This does not mean that practical reliability standards, which were developed on logical grounds for research purposes, should be altered for the VVIQ if it is to be used as a psychometric device, but it does imply that every effort should be made to improve the test. In particular, it provides another reason for an empirical investigation of the effect of adding items, which is consistent with the general principle of aggregation, according to which random fluctuations can be averaged out (Reisberg & Heuer, 1988, p. 96). The test might also be improved in other ways. For example, the test-retest coefficients for the mother and father versions of the VVIQ were slightly higher than previous estimates, suggesting that reliability might increase if subjects adopt a focus when responding.

Conclusion on Reliability

In his review of the resurgence of interest in conscious mental imagery, Holt (1972, p. 9) states that "There is no better answer to the charge that a topic is too subjective than a set of stout reliability coefficients". I judged that internal consistency was acceptable, test-retest was minimally acceptable, and that alternate-form reliability was unacceptable. These results may add to the content validity difficulties that hamper the evaluation of nonsignificant VVIQ/criterion relationships, but they do not justify the conclusion that visual imagery vividness is too subjective.

Criterion Validity of the VVIQ

Establishing Predictions for VVIQ/Criterion Task Relationships

The most important evidence for construct validity is the extent to which a test is related as expected to criterion performance (Marks, 1983a, 1989b, 1993; Reisberg & Heuer, 1988). Following the first decade of work, Marks (1983b, pp. 109-123) gave a general account of the role of imagery and cognition followed by 14 predictions that he characterizes as theoretically meaningful (Marks, 1989b, p. 459-460). I also found three others in his work (Marks, 1972, p. 96; Marks, 1977; Marks, 1989b). Particular attention will be paid to these predictions, but others can also be formulated from the discussion of content validity, where I claimed that VVIQ scores involved 12 processes. Because self-report tests of imagery ability are more likely to be associated with criterion tasks whose requirements overlap with them (Cohen & Saslona, 1990; Dean & Morris, 1991, p. 332; Hall, Talukder, & Esposito, 1985; Heuer et al., 1986; Katz, 1983, pp. 44-46; Kaufmann, 1983; Reisberg, et al., 1986; A. Richardson, 1972, p. 120-121; Marks, 1977, 1983a, 1987; Slee, 1980; Strosahl & Ascough, 1981), I predict that *VVIQ scores should be related to scores on other tasks to the degree that they share these 12 components*, and will refer to it as the *overlap hypothesis*. At one extreme is the relationship between two administrations of the VVIQ (test-retest reliability), where all processes are shared, and at the other would be the relationship to performance on a task with no self-report or image production (e.g., abstract reasoning). On the basis of Marks' definition of vividness, and of the discussion of the content of the VVIQ items, the relationship with other tasks involving memory, perception, creativity, absorption, and control should fall in between. The overlap hypothesis is similar to the multitrait-multimethod matrix and the concepts of convergent and discriminant validation (Anastasi, 1988, pp. 156-158).

To evaluate their theory of imagery, Kosslyn et al. (1984) also assumed that test/task relationships would reflect shared processes. They postulated that imagery is composed of structures and of processing modules and that the latter are involved to different degrees in different tasks. Having specified these roles, they evaluated the model by examining how well it predicted the obtained similarity (measured by correlations among scores representing task performance) among tasks. The correlation between obtained and observed similarity was .56, which they regard as quite satisfactory. However, although the VVIQ was included in this analysis, Kosslyn et al. were unsure about its characterization. When they eliminated it and another test from the battery, the degree of agreement rose to .67, leading them to conclude that they had failed to adequately specify the processing modules in the VVIQ.

To tackle this problem, and to formulate specific predictions based on the degree of similarity of "trait" (i.e., number of shared task processes) and of method (i.e., self-report or not), I assigned weights to each of the 12 processes on the VVIQ according to my judgment of their importance from the VVIQ instructions and from my logical analysis of the VVIQ task. As recognized by Kosslyn et al., this procedure is speculative but, when it was also applied to the criterion tasks, it generated predictions from the overlap hypothesis. Details are given in the Appendix, but the weights assigned to the 12 processes in the VVIQ were as follows. Vividness (similarity to perceptual experience) was given the most weight (5), followed by cueing, vision, imagery, and self-report (4), verbal stimulus, generation, holding, and complexity (3), memory and imagination (2), and control (1). For purposes of comparing the VVIQ to other tasks, I summed the weights to give the similarity of each task to the VVIQ in terms of shared processes. Weights for the VVIQ and selected criterion tasks are shown in Table A of the Appendix, and the summed weights for each task are shown in Table 10.

Although the overlap argument implies that the VVIQ will predict more successfully if the criterion task involves imagery generated by instruction, Denis (1987, p. 207) claims that such instructions involve an expectancy effect that can be removed by studying imagery via individual differences. From this point of view, prediction will be more successful if the criterion task is incidental and does not involve instructed imagery, because the expectancy effect will be absent from both the VVIQ and task. Of course, this assumes that subjects are unaware of their VVIQ status and of its connection with the criterion task. In view of the debate over the importance of strongly-cued or instructed imagery generation, particular attention was paid to this factor.

The advantages and disadvantages of giving the VVIQ simultaneously or not with the criterion test were also noted, and taken into account when evaluating the research. Concurrent estimates avoid error due to delay, which was seen under test-retest reliability to reflect perhaps also instability in the vividness construct, but invite possible artificial dependence between test and criterion scores. It has been observed that an experimental task can cue VVIQ responses. If the criterion is administered first and subjects perceive their performance as good or poor, they may also rate themselves as good or poor visualizers. It is also possible that VVIQ responses may cue criterion task performance if subjects perceive a connection. Introduction of a delay (predictive or postdictive validity (Ghiselli et al., 1981), hereafter referred to together as delayed validity) is one way of minimizing such effects.

Predictions from the Overlap Hypothesis

Although the overlap hypothesis generated a model that quantified the similarity between the VVIQ and each criterion task, it was obtained by summing weights. This implies that predictions for validity coefficients can only be ordered in size, with higher ones expected with greater similarity of processes. However, more precise predictions can be obtained from the immediate and delayed alternate-form reliabilities, which set practical upper limits for concurrent and predictive validity coefficients (Kosslyn et al., 1984; McKelvie, 1994a). To derive them, I calculated the percentage of shared variance between the VVIQ and each criterion task by multiplying the upper limits of shared variance by the proportion of shared weights taken from the summed weights model. For delayed and concurrent validity, the values of .54 and .64 gave shared variances of .54² and .64², or 29.2% and 40.1%, rounded to 30% and 40%, respectively. Then, if a criterion task had a total weight of 38, which was the VVIQ total, the predicted percentage of shared variance was 100% of 30% and 40%, giving predicted delayed and concurrent validity coefficients of .548 and .632. Again, if the task had a weight of 19, the predicted shared variance was 50% (19/38) of 30% and 40%, or 15% and 20%, giving predicted validity coefficients of .387 and .447. Following this procedure, it was possible to calculate exact point predictions from the model for each task validity coefficient. Details are given later.

Review of Research Evidence for VVIQ/Criterion Relationships

Because this review was based on quantified VVIQ/task relationships, studies without the relevant information were not usually considered. Exceptions are justified. VVIQ scores have been related to those on 21 other self-report tasks involving a non-anchored subjective judgment (monomethod) and on 5 related objective performance tasks (heteromethod) (Tasks 1 to 26). Studies have also been conducted with 4 other cognitive and 5 perception tasks (Tasks 27 to 35), although quantified VVIQ relationships were available from only 4 of them (Tasks 26, 29, 30, and 35). Weights were also derived for a complete logical set of memory studies (Tasks 36 to 102), but only 40 quantified relationships were available. Again, *rs* were interpreted from my guidelines (McKelvie, 1994a; see Table 7).

In the first set of studies, the criterion was another self-report task, and the direction of the effect was predicted. It was expected that greater visual imagery vividness would be associated with better performance on the criterion.

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Table 7
Standards for Judging Criterion Validity Coefficients and Effect Sizes with Extreme Groups

Conditions	Relationship	<i>r</i>	<i>d</i>	Judgment
Unpredicted and self-report	Nonsignificant			No relationship
	2% to 5.9% of variance	.15 to .24	0.30 to 0.49	Inconsequential
	6% to 15%	.25 to .39	0.50 to 0.89	Acceptable
	16% or more	.40 or more	0.90 or more	Very good
Predicted or objective (not both)	Nonsignificant			No relationship
	Less than 2% of variance	Less than .15	Less than .30	Inconsequential
	2% to 5.9%	.15 to .24	0.30 to 0.49	Marginal
	6% to 15%	.25 to .39	0.50 to .89	Acceptable
	16% or more	.40 or more	0.90 or more	Very good
Predicted and objective	Nonsignificant			No relationship
	Less than 1% of variance	Less than .10	Less than .20	Inconsequential
	1% to 5.9%	.10 to .24	0.20 to 0.49	Marginal
	6% to 15%	.25 to .39	0.50 to .89	Acceptable
	16% or more	.40 or more	0.90 or more	Very good

Interpretation of 95% Confidence Intervals

Interval lies above the standard	Standard is exceeded
Interval includes standard with midpoint above	Standard is met
Interval includes standard with midpoint below	Standard is minimally met
Interval lies below the standard	Standard is not met

Predicted and Self-Report: $r = .15$ Marginal, $r = .25$ Acceptable

Questionnaires involving visual imagery or vividness. For Tasks 1 to 7, a judgment of the content of experienced visual imagery was required (Bowers, 1978; Chara & Verplanck, 1986; Crawford, 1982; Dean & Morris, 1991; Heuer et al., 1986; Marks, 1973a; McConkey & Nogrady, 1986; Phillips, 1978; Rossi & Fingeret, 1977; Sullivan & Macklin, 1988; Walczyk & Hall, 1988; Worthington, 1978). Weights for Tasks 1 and 6 are shown in Table A. In the first three cases, where vividness was judged directly, the confidence interval for the VVIQ/task r was .356 to .656 (midpoint = .522) (Table 8), which met the standard of a very good relationship (Table 7). In

Table 8
Correlation Coefficients (Criterion Validity) between the VVIQ and Self-report Tasks

Task	n	Lower Bound	Mean	Upper Bound	Judgment
1. Question Visual Vividness	9	.387	.591	.739	Exceeds A, VG met: VG
2. Complex Picture Visual Vividness	1		.296		A met: A
3. Simple Picture Visual Vividness	4	.063	.399	.654	A met, VG min met: A
1. to 3.	14	.356	.522	.656	Exceeds A, VG met: VG
4. Static VVIQ Components	8	.220	.435	.608	A, VG met: A
5. Controlled VVIQ Components	6	-.031	.142	.306	ns: NR
4. and 5.	14	.152	.316	.464	A met, VG min met: A
6. VES	2		.410		Exceeds A, VG met: A
7. Visual Elaboration	1		.310		A met: A
6. and 7.	3		.378		Exceeds A, VG min met: A
4. to 7.	17	.193	.327	.449	A met, VG min met: A
8. QMI	5	.383	.626	.787	Exceeds A, VG met: VG
9. VAIQ/VMIQ	5	.545	.677	.776	Exceeds VG: VG
8. and 9.	10	.531	.662	.747	Exceeds VG: VG
10. TVIC	18	.292	.377	.455	Exceeds A, VG min met: A
11. EI	1		.25		A met: A
12. Absorption	6	.362	.515	.640	Exceeds A, VG met: VG
13. Habitual Use	9	.049	.142	.233	MA min met: MA
14. Task Use	7	.227	.337	.437	A met, VG min met: A
13. and 14.	16	.144	.229	.311	MA, A min met: MA
15. Day/Night Dream Frequency	9	.045	.153	.256	MA met: MA
16. Aesthetic Judgement	3		.254		A met: A
17. Stereotype Maintenance	1		-.16		ns: NR
18. Configuration Recovery	1		.172		MA met: MA
19. Threat to Self-differentiation	6	.139	.238	.333	MA, A met: A
20. Pleasantness	1		.30		A met: A
21. Word Emotionality	2		.236		A met: A
All self-report by estimate	105	.328	.369	.446	Exceeds A, VG min met: A
All self-report by task	21	.244	.333	.426	A met, VG min met: A

Notes. (a) Italics indicate my judgment of the relationship: VG (very good), A (acceptable), MA (marginally acceptable), or NR (no relationship; min = minimally; (b) ns = not significant

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the second four cases, various aspects of visual imagery experience related to vividness were judged. Here, the confidence interval (Table 8) met the acceptable standard. However, for other vividness questionnaires that were only partly visual (Tasks 8 [weights in Table A] and 9; Hatakeyama, 1984; Isaac et al., 1985; Kunzendorf, 1982; Lorenz & Neisser, 1985; McConkey & Nogrady, 1986; Rossi & Fingeret, 1977), the relationship with the VVIQ was greater, and was classified as very good (Table 8).

Test of Visual Imagery Control (TVIC, Tables A and 8). On the TVIC (Gordon, 1949, 1950; revised version, Richardson, 1969, pp. 155-156), which has been frequently given in conjunction with the VVIQ, subjects indicate whether or not they can control the visual image of a car. Although it was argued that theoretical maxima are unlikely to be reached, they can be calculated here. The alternate-form reliability coefficient for the TVIC is .517 (McKelvie, 1992c), giving a theoretical maximum of .575 ($.64 \times .517$)^{0.5} for the validity coefficient (Ghiselli et al., 1981, p. 242). It is not clear whether two other alternate-form estimates of .73 (Lorenz & Neisser, 1985; McKelvie & Gingras, 1974) should be included in the calculation, because the alterations to the content of the test were minor. If they are, the combined alternate-form estimate becomes .670 (reconverted z_r mean of .811), giving a new upper bound for concurrent validity of .655 ($.64 \times .67$)^{0.5}.

For the 18 estimates of VVIQ/TVIC r (Task 10; Carroll, Baker & Preston, 1979; Childers et al., 1985; Dean & Morris, 1991; Hatakeyama, 1984; Kihlstrom et al., 1991; Kosslyn & Jolicoeur, 1980; Lorenz & Neisser, 1985; McKelvie, 1993b; McKelvie & Gingras, 1974; McKelvie & Rohrberg, 1978; Parrott, 1986; Parrott & Strongman, 1985; Phillips, 1978), the confidence interval exceeded the .25 standard (Table 8) and is an acceptable relationship. However, it also lies below both the theoretical maximum and the value for vividness questionnaires.

Eidetic imagery (EI, Task 11, Tables A and 8). Although it is often equated with "photographic memory," scientific investigators doubt that EI enhances recall and do not define it simply in terms of fidelity (Gray & Gummerman, 1975; Haber, 1969, p. 85; A. Richardson, 1969, p. 41, and 1983, p. 22). Rather, and whether or not the criteria are broad (structural eidetic imagery) or narrow (typographic eidetic imagery), the classification is made on the basis of self-report (Ahlsen, 1977; Marks & McKellar, 1982; A. Richardson, 1983, pp. 22-27). From the two brief reports (Marks & McKellar, 1982) relating EI to the VVIQ, I calculated the contingency coefficient to be .284 (maximum value .707), and Cohen's (1977, p. 222) effect size of w to be .296. This is almost "medium" (.30, Cohen, 1977, p. 225).

which, in terms of r , would be .30 and would meet the standard of acceptability. For present purposes, it was considered to be $r = .25$. Although the data base is weak, this indicates a marginal to acceptable relationship between the VVIQ and EI.

Absorption, imagery use, dream frequency, and aesthetic preference. The confidence interval for measures of absorption, effortless experiencing, and involuntariness (Task 12, Tables A and 8; Bowers, 1982; Crawford, 1982; McConkey & Nogrady, 1986) was .362 to .640, and met the very good .40 standard. This relationship is consistent with the proposal that the VVIQ taps the process of imaginative involvement.

The VVIQ has been related to a number of measures of imagery use (Tasks 13 [Table A] and 14; Bowers, 1978; Childers, et al., 1985; Cohen & Saslona, 1990; Hatakeyama, 1984; Heuer et al., 1986; Kosslyn & Jolicoeur, 1980; Kunzendorf, 1982; Lorenz & Neisser, 1985; Parrott, 1986; Priebe & Wallace, 1986; A. Richardson, 1984, p. 21; Wallace, 1988; Worthington, 1978). Although utilizing self-report, they reflect the functional aspect of imagery, which is an important question for the vividness construct. As Table 8 shows, the overall confidence interval for 16 estimates met the marginal standard (.15) and minimally met the acceptable standard (.25).

The relationship between the VVIQ and frequency of day or night dreaming (Task 15; Belicki & Belicki, 1986; Cathala, Laffont, Siksou, Esnault, Gilbert, Minz, Morethalmin, Buzare, & Waisbord, 1983; Crawford, 1982; A. Richardson, 1979) is of interest, particularly because Marks (1983b, p. 122) made a positive prediction for the latter case. Table 8 shows that the confidence interval was marginally acceptable. Finally, for aesthetic judgment (Task 16; Kunzendorf, 1982), which for Kunzendorf (1982) represents the imagery/creativity relationship, the mean correlation just meets the .25 standard.

Other self-report studies. In Tasks 17 to 19, the authors (Malhotra, Jain, & Pinson, 1988; Slusher & Anderson, 1987; Winczo-Kostecka & Cielecki, 1985) argued that visual imagery or vividness was involved. Table 8 shows that, for two of them, the relationship to the VVIQ was acceptable or minimally acceptable. Finally, two studies investigated the relationship between VVIQ scores and rated pleasantness of images generated to deal with pain from an ice-water test (Worthington, 1978) and rated emotionality of both pleasant and unpleasant words (Campos, 1988) (Tasks 20 and 21). Although the dependent variables were not visual, some relationship to the VVIQ might be expected, since Marks (1972, p. 91) suggests that vividness will be associated with "interest, affect, meaning, and the overall level of arousal evoked by a stimulus." Also, because many of the images generated by the VVIQ are likely to be from memory and therefore famil-

iar, they may be emotionally-toned. Table 8 shows an acceptable relationship between vividness and affect.

Commentary on Self-report Data

General. For all 105 estimates, the 95% confidence interval for the validity coefficient was .328 to .446, with a midpoint of .369. This exceeds and minimally meets the .25 and .40 standards respectively, indicating an acceptable and perhaps very good relationship between the VVIQ and other self-report tasks.

When investigating relationships between psychological tests and criterion tasks, experimenters should be unaware of the subject's test scores and subjects should be unaware of a connection between the test and task. These factors have also been cited as possible contaminants in VVIQ research (Berger & Gaunitz, 1977; Kaufmann, 1983). Because most of the present self-report studies involved group testing, it is likely that experimenters were blind as to subjects' VVIQ status. However, a potential problem is that, in 72 out of the 105 predicted cases, the VVIQ and other self-report test were given concurrently, with order of administration either not balanced or unclear for 70 out of the 72 concurrent estimates. This implies that carry-over effects may have been present, with subjects aware of their first set of responses when providing the others. Higher and lower correlations may, therefore, not only reflect more or fewer shared test processes respectively, but also demand characteristics to respond in a more or less consistent manner to a similar or less similar test, respectively. Indeed, Marks (1983b, p. 108) recognized this problem and recommended that testing conditions be controlled to avoid it. A delay between the two tests is one way to accomplish this.

Although easy to document as a possibility, it is difficult to judge the extent to which this confound actually operates to contaminate empirical relationships. For example, for the 18 estimates of the correlation between the VVIQ and TVIC, 16 were based on concurrent administrations of the tests (with the other two unclear), and the VVIQ was given first in 11 cases and only once with order controlled (the other 6 were unclear). Interpretation of the mean validity coefficient of .377 is therefore muddled by the possibility that it is inflated by positive cueing, that is, subjects responded similarly because they perceived the tests as similar (both involve visual imagery) or that it is lowered by a perceived differential emphasis on vividness and control, cueing subjects to respond differently. Authors did not state whether they minimized the effect of such demand characteristics by instructing subjects to answer the second test independently of the first.

To examine more systematically the possibility that this factor was associated with the significant predicted VVIQ/test relationship, I classified the studies and conducted a chi-square test on the following frequencies: significant and concurrent (45), significant and delayed (11), not significant and concurrent (24), and not significant and delayed (8). The total does not sum to 105 because some cases could not be clearly classified. This association was not significant, $2 \times 2 \chi^2(1, N = 88) = 0.34$ (without Yates correction for continuity), indicating that significant results were not more likely to occur with concurrent than with delayed testing. Furthermore, the obtained z 's for concurrent ($M = 0.421$, $SD = 0.315$, $n = 74$) and delayed testing ($M = 0.381$, $SD = 0.324$, $n = 18$) did not differ significantly, $t(90) = 0.48$, $p > .20$. Although the number of delayed studies was smaller, these results undermine the objection that the present data are contaminated by a subject cueing effect. Of course, if delay means that the VVIQ was given earlier so that extreme groups could be chosen to take part in subsequent testing, the subject effect is not ruled out if subjects know why they have been invited back. No criterion study of the VVIQ has provided enough methodological detail to decide whether this problem was dealt with. However, for the 10 self-report cases that fell into this category, only two involved extreme groups, so the difficulty does not arise. This fact is also contrary to an experimenter effect, which is likely to occur when the experimenter knows the VVIQ status of the subjects. Overall, I conclude that there is no evidence that the VVIQ/self-report relationship is compromised by a subject or an experimenter effect.

A related problem is the social desirability response bias which, although present in the VVIQ, was judged to be acceptable. However, it is likely that a similar level exists in many of the other self-report criterion tests, implying that it is a shared process. This may elevate the criterion validity coefficients. Indeed, it could be argued that a conservative estimate of the VVIQ/self-report task relationship is .147 (.369-.222), making it just marginally acceptable.

Predicted effects of variables from the overlap hypothesis. The division of self-report studies into those that did and did not require a direct judgment of visual imagery content implies that validity coefficients would be higher in the first case. The overlap hypothesis also predicts this, because the mean value for the summed weights over all 12 processes was higher for the 11 content ($M = 29.4$, $SD = 5.2$) than for the 10 non-content tasks ($M = 22.8$, $SD = 6.1$), $F(1, 19) = 7.24$, $p < .01$. I then calculated a mean expected z score for each task by classifying each of the obtained validity coefficients as concurrent or delayed (predictive or postdictive) (treating unclear cases as concurrent), and by averaging the corresponding expected

ed z_r values. (see Table 9). For example, of the four obtained validity coefficients for Task 3, two were concurrent and two involved a delay between the VVIQ and criterion task. Since the sum of all the weights for Task 3 was 26, the predicted concurrent and delayed shared variances were $26/38 \times 40\%$ and $26/38 \times 30\%$ or 27.37% and 20.52%, respectively, giving predicted validity coefficients of .523 and .453 or z_r values of 0.579 and 0.488, respectively. This yielded a mean expected z_r score for the two concurrent and two delayed estimates of 0.534 (mean of 0.523, 0.523, 0.453, 0.453) for Task 3. The mean expected z_r s for each task are shown in Table 10 (predicted z_r).

Using these derived z_r predictions for each task, the overall mean predicted value for the 105 z_r s was 0.576 ($SD = 0.097$). The mean for the 60 predicted z_r s from the 10 content tasks was higher than that for the 45 z_r s from the 11 non-content tasks ($M_s = 0.608, 0.535$; Table 9), $F(1, 103) = 16.89, p < .01$. This indicates that the obtained validity coefficients should be higher for the tasks requiring that imagery content be judged compared to the others.

The 11 content tasks were then grouped into 6 levels (see Table 9): (a) visual vividness: Tasks 1 to 3 all reflected visual vividness; (b) visual vividness related: Tasks 4 (VVIQ components) and Task 11 (eidetic imagery), reflecting tasks that were close to the VVIQ; (c) visual elaboration: Tasks 6 and 7; (d) vividness: Tasks 8 and 9 reflected vividness in mainly non-visual modalities; (e) control: Task 10 measured control; and (f) control related: Task 5. The 60 expected z_r s (again adjusted to represent concurrent or predictive estimates) corresponding to these 60 obtained validity coefficients were included in a one-way analysis of variance, which showed that they differed, $F(5, 54) = 38.17, p < .01$. As Table 9 (predicted z_r) shows, the highest predicted value occurred with vividness (0.682), followed by visual elaboration (0.667) and visual vividness (0.666) and control (0.636), then by visual vividness related (0.475) and control related (0.432).

Because some of the self-report tasks correspond to cognitive objective performance ones (control, hypnotic susceptibility, creativity) they were also considered separately: (a) control (Task 10), (b) absorption (Task 12), and (c) aesthetic judgment (Task 16). No analysis of variance was conducted, since there was no variation within each of the 3 levels (all cases were classified as concurrent), but the predicted z_r mean scores were 0.636, 0.587, and 0.299, respectively (Table 9).

Obtained effects. Each of the 105 obtained validity coefficients was transformed to z_r , and a mean value calculated for each of the 21 tasks (see Table 10). The M and SD for these 105 z_r s over the 21 tasks were 0.387 and 0.310, respectively, giving a 95% z_r confidence interval of .328 to 0.446, centered on .369. This was greater than 0, $t(104) = 12.77, p < .01$, but also lower than the

Table 9
*The Overlap Model: Predicted and Obtained Effects of Variables
 for z_T Scores (VVIQ/Task Relationships)*

Task	n	Predicted z_T		Obtained z_T	
		M	SD	M	SD
Self-report					
Content	60	0.608	0.100	0.484	0.338
No content	45	0.535	0.074	0.258	0.214
Self-report content					
VV	14	0.666	0.096	0.579	0.395
VV related	9	0.475	0.007	0.442	0.336
VE	3	0.667	0.031	0.398	0.075
Viv	10	0.682	0.068	0.779	0.302
on	18	0.636	0.000	0.396	0.205
Conrelated	6	0.432	0.000	0.143	0.217
Self-report: control, absorption, aesthetic judgment					
Con	18	0.636	0.000	0.396	0.205
Abs	6	0.587	0.000	0.597	0.237
Aesth	3	0.299	0.000	0.260	0.166
Cognitive: control, hypnotic susceptibility, creativity					
SpCon	15	0.397	0.033	0.064	0.162
Phys/MCon	11	0.567	0.000	0.275	0.183
HS	19	0.621	0.048	0.415	0.219
VisC	9	0.459	0.000	0.196	0.389
VerbC	6	0.473	0.000	0.206	0.212
Task					
SR	105	0.576	0.097	0.387	0.310
Cog/perc	70	0.507	0.100	0.279	0.257
Mem	82	0.466	0.057	0.159	0.358
Memory delay					
STM	67	0.461	0.046	0.168	0.355
LTM	15	0.489	0.090	0.113	0.388
Memory cueing					
Instruct	21	0.483	0.068	0.178	0.378
Pic	37	0.464	0.043	0.060	0.348
CW	15	0.471	0.067	0.227	0.337
AW	4	0.409	0.039	0.226	0.284
Memory task					
SS	30	0.486	0.064	0.072	0.405
PAL	12	0.448	0.040	0.166	0.292
FR	18	0.441	0.074	0.311	0.372
RN	22	0.471	0.153	0.146	0.289

Notes. (a) VV = visual vividness, VV related = visual vividness related, VE = visual elaboration, Viv = vividness, Con = control, Conrelated = control related; (b) Abs = absorption, Aesth = aesthetic judgement; (c) SpC = spatial control, Phys/MCon = physiological/motor control, HS = hypnotic susceptibility, VisC = visual creativity, VerC = verbal creativity; (d) SR = self-report, Cog/perc = cognitive/perception, Mem = memory; (e) Instruct = instructions to generate visual imagery, Pic = picture stimulus, CW = concrete word stimulus, AW = abstract word stimulus; (f) SS = single-picture, PAL = paired-associate learning, FR = free recall, RN = recognition memory.

Table 10
Predicted and Obtained Overlap of the VVIQ with Each Criterion Task

Task	Summed Weights	Predicted z_r	Obtained z_r
Self-report			
1	37	0.731	0.679
2	28	0.608	0.305
3	26	0.534	0.423
4	25	0.477	0.466
5	21	0.432	0.142
6	31	0.649	0.436
7	35	0.702	0.321
8	33	0.676	0.735
9	34	0.689	0.824
10	30	0.636	0.396
11	23	0.455	0.255
12	26.5	0.587	0.570
13	24.5	0.560	0.143
14	23.5	0.533	0.350
15	25.5	0.574	0.154
16	8	0.299	0.260
17	27	0.594	-0.161
18	24	0.553	0.174
19	24	0.510	0.242
20	28	0.608	0.310
21	16.5	0.446	0.240
Cognitive/Perception			
22	16	0.397	0.063
23	25	0.567	0.274
24	31	0.621	0.414
25	17	0.459	0.194
26	18.5	0.473	0.206
29	19	0.408	0.350
30	7	0.239	0.174
33	17	0.452	0.377
34	20	0.496	0.607
45	4	0.158	0.598
Memory: single-stimulus picture			
36	20	0.420	0.588
37	20.5	0.492	-0.034
38	21.5	0.518	0.045
39	20	0.490	-0.007
40	20	0.458	0.580
41	20	0.420	0.534
42	15	0.359	0.696
43	18	0.473	-0.060
44	10	0.336	-0.040
46	21	0.510	0.058
47	20	0.496	-0.598
48	29	0.621	-0.510
49	25.5	0.552	0.120
50	31	0.649	-0.513

Task	Summed Weights	Predicted z_r	Obtained z_r
Memory: paired-associate recall			
52	23.5	0.482	0.027
53	20	0.445	0.209
54	20	0.420	0.893
55	21	0.432	0.134
57	18.5	0.402	0.045
61	20	0.420	0.050
Memory: free recall			
67	16.5	0.378	0.125
68	18.5	0.473	-0.110
70	19	0.387	1.048
71	17.5	0.390	0.648
74	16.5	0.378	0.633
75	16.5	0.378	0.318
78	20	0.420	0.454
79	16.5	0.378	0.070
80	19.75	0.492	0.050
81	18.75	0.477	0.427
82	18.75	0.477	0.485
83	17.75	0.462	0.357
Memory: recognition			
87	19	0.481	0.027
88	18	0.466	0.510
89	16	0.436	0.192
90	18.5	0.473	0.352
92	19	0.481	-0.110
93	18	0.466	0.140
97	19	0.481	-0.472
98	18	0.466	0.400
<i>n</i>	71	71	71
<i>M</i>	21.10	0.484	0.259
<i>SD</i>	5.82	0.097	0.319

Notes. (a) Adjusted z_r s were derived by distinguishing concurrent and predictive estimates, with unclear cases considered concurrent; (c) For summed weights, maximum value was 38.

predicted z_r mean 0.576, $t(104) = 19.89$, $p < .01$ (single-sample t -tests using standard errors based on the obtained and predicted standard deviations, respectively). Thus, the relationship between the VVIQ and self-report tasks was acceptable, but the overlap hypothesis overpredicted the data.

However, 60 content z_r scores were higher than the 45 non-content z_r scores, $F(1, 103) = 15.78$, $p < .01$ (Table 9, $M_s = 0.484, 0.258$), confirming the general prediction that validity coefficients would be higher for criterion tasks that required a direct judgment of imagery experience compared to those that did not.

Within the six levels of content tasks, the obtained z_r scores differed, $F(5,$

54) = 3.70, $p < .01$. As Table 9 shows, the highest and lowest values occurred for vividness ($M = 0.779$) and control-related ($M = 0.143$) respectively, exactly as predicted from the overlap hypothesis. Relationships for the other four task groupings did not occur in precisely the expected order, but the Spearman rank-order correlation between predicted and obtained order over the six groups was .714, indicating good agreement.

For the three objective-related tasks, the analysis of variance of obtained z_r scores was almost significant, $F(2, 24) = 2.99$, $p < .07$. Inspection of the means (Table 9) shows that the VVIQ/absorption relationship was higher than the VVIQ/creativity relationship ($M_s = 0.597, 0.260$), as predicted, but that the VVIQ/control relationship was lower ($M = 0.396$) than the absorption one rather than higher as predicted from the overlap hypothesis.

Correlational evaluation of overlap models. Another way to gauge the success of the overlap model was to calculate a r between predicted and obtained validity coefficients. Here, cases that could not be unambiguously classified as concurrent or delayed were omitted, giving 98 pairs rather than 105. Although obtained z_r s were not a function of the delay between VVIQ and criterion task administration, this factor was retained and indeed sharpened by the exclusion of unclear cases, because it formed part of the overlap hypothesis. The r between predicted and obtained z_r scores was .392, $p < .01$.

Implications for construct validity of the VVIQ. Although Marks (1983b, p. 122) made only two specific predictions (for dream recall and hypnagogic imagery frequency) for the relationships between the VVIQ and other visual self-report tasks, I assume that he would expect all of them to be positive. The association between the VVIQ and dreaming was deemed to be marginal, and the acceptable validity coefficient of .369 from all 105 estimates confirms the general prediction. In addition, although the overlap hypothesis overpredicted the data in absolute terms, the r between predicted and obtained effects from the model was good, and the model was fairly successful in predicting the order of obtained effects for groups of tasks. Together, these results support both the vividness construct and the validity of the VVIQ.

Inductively, the ranking of the relationships for the six content tasks suggests that the most important VVIQ processes are general and visual vividness ($r_s = .652, .522$), followed by processes related to visual vividness ($r_s = .415, .378$), then by control ($r_s = .377, .142$). They are also consistent with the rank order of the VVIQ weights for vividness (5), vision and imagery (4), and control (1). However, the validity coefficient for absorption was .515, suggesting that imagination should have been given a higher weight than 2. The correlations for pleasantness and word emotionality

were acceptable, suggesting that affect is a component of the test. Although the relationship between the VVIQ and imagery use was minimally acceptable, the estimate may be high, because it did not include three nonsignificant relationships whose values were not reported. Therefore, the results support Cohen and Saslona's (1990) assertion that the VVIQ only taps natural deployment of imagery to a minor degree.

On a more cautionary note, the high validity coefficients for vividness in mainly non-visual modalities may undermine the VVIQ as a pure measure of visual imagery. Although the fault could lie in some of the other "non-visual" questionnaires [QMI, VMIQ (motor)], which have visual components, the individual VVIQ/task validity coefficient for the VAIQ (auditory) was .64, which is very close to the mean (.652) of all "non-visual" estimates. Furthermore, because the vast majority of studies clearly tapped imagery, it was not possible to compare the VVIQ as a predictor for imagery and non-imagery tasks to evaluate the claim that VVIQ ratings are not based on imagery (Chara, 1992a; Chara & Hamm, 1988). These concerns about vision and imagery reflect doubts about the discriminative validity of the VVIQ (Kaufmann, 1983).

Predicted and Objective Cognitive Performance Tasks:
 $r = .10$ Marginal, $r = .25$ Acceptable

Given that the processes of control, absorption, and creativity were initially identified as components of the VVIQ, the acceptably-high self-report validity coefficients of .377, .515, and .254 were not unexpected. However, if imagery vividness is functional, VVIQ scores should also be related positively to corresponding objective performance tasks that are less likely than the self-report tasks to be contaminated by extraneous variables such as demand characteristics or social desirability. The results of VVIQ studies with these tasks are shown in Table 11.

Control. The first of the two control tasks (Task 22: Spatial, Table A), which occurs in two forms ("Folding" and Spatial "Rotation") involves some direct manipulation of visual imagery itself. For "Folding" (Bethel-Fox, Lohman, & Snow, 1984; Dean & Morris, 1991; Hatakeyama, 1984; Lorenz & Neisser, 1985; McKelvie & Rohrberg, 1978; Parrott, 1986), the confidence interval overlapped with 0 (Table 8). Similarly, for Spatial "Rotation" (Dean & Morris, 1991; Hatakeyama, 1984; Kosslyn, et al., 1984; McKelvie & Rohrberg, 1978), and for all spatial tests combined, the confidence intervals contained 0 (Table 11), and indicate no relationship between the VVIQ and spatial performance. In the second kind of control

Table 11
Correlation

Variable

22. Spatial
"Folding"
Spatial
All Spatial
23. Physic
Physiol
Motor

All Physio
24. Hypno

25. Visual
Genera
High IC

All Visual
26. Verba
Genera
High IC

All Verbal
29. Objec
30. Proof
Visual Se
45. Gesta

All cognit
All cognit

33. Perce
34. Fields

All perce

All cognit
All cognit

Notes. (a) I
acceptable

Table 11
Correlation Coefficients (Criterion Validity) between the VVIQ and Cognitive and Perception Tasks

Variable	n	Lower Bound	Mean	Upper Bound	Judgment
<i>Cognitive</i>					
22. Spatial					
"Folding"	9	-.029	.089	.205	ns: NR
Spatial "Rotation"	6	-.082	.025	.131	ns: NR
All Spatial	15	-.018	.063	.145	ns: NR
23. Physiological/Motor Control					
Physiological	6	.056	.191	.318	MA met: MA
Motor	5	.230	.356	.470	A met, VG min met: A
All Physiological/Motor	11	.165	.268	.365	A met: A
24. Hypnotic Susceptibility	19	.306	.392	.473	Exceeds A, VG min met: A
25. Visual Creativity					
General IQ	6	-.235	.013	.260	ns: NR
High IQ	3		.510		Exceeds A, VG met: VG
All Visual Creativity	9	-.058	.192	.421	ns: NR
26. Verbal Creativity					
General IQ	3		.072		ns: NR
High IQ	3		.326		A met: A
All Verbal Creativity	6	.036	.203	.356	MA met: MA
29. Object Search	1		.338		A met: A
30. Proofreading	1		.172		MA met: MA
Visual Search (29, 30)	2		.256		A met: A
45. Gestalt Closure	4	.378	.536	.663	Exceeds A, VG met: VG
All cognitive by estimate	66	.201	.263	.322	A met: A
All cognitive by task	8	.167	.276	.380	A met: A
<i>Perception</i>					
33. Perceptual Aftereffects	3		.360		A met: A
34. Fields of Resolution	1		.542		Exceeds A, VG met: VG
All perception	4	.282	.409	.522	Exceeds A, VG met: VG
All cognitive/perceptual (estimate)	70	.212	.273	.330	A met: A
All cognitive/perceptual (task)	10	.211	.315	.412	A met, VG min met: A

Notes. (a) Italics indicate my judgment of the relationship: VG (very good), A (acceptable), MA (marginally acceptable), or NR (no relationship; min = minimally; (b) ns = not significant.

task (Task 23: Physiological/Motor Control, Table A), subjects were asked to alter physiological or motor processes, usually by utilizing visual imagery. For physiological control (Carroll, et al., 1979; Kunzendorf, 1981), the confidence interval met the .10 standard of marginal acceptability. For the five *rs* with control over a motor behavior (dance movements, trampolining; Housner, 1984; Isaac, 1985), the confidence interval overlapped with .25 and .40, indicating that the acceptable and very good levels were met and minimally met, respectively. For all 11 physiological/motor correlations, the interval of .165 to .365 exceeded the .10 marginal standard and met the acceptable standard (Table 11).

Hypnotic susceptibility. Measures of hypnotic susceptibility (Task 24, Table A) were included with performance tests, since they require an overt behavior from the respondent. Assuming that "once a good imager, always a good imager," Marks (1983b, p. 122-123) predicted that good visualizers should be more susceptible to hypnosis than poor ones. For the 19 estimates here (Table 11; Bowers, 1978, 1982; Coe, St. Jean, & Burger, 1980; Crawford, 1982; Crawford & Allen, 1983; Dwyer, 1988; McConkey & Nogrady, 1986; Priebe & Wallace, 1986; T Hoen, 1978; Wallace, 1988, 1990b, 1991), the confidence interval exceeded and minimally met the .25 and .40 standards respectively, giving a positive relationship between the VVIQ and hypnotic susceptibility.

Creativity. For visual creativity (Task 25; Parrott & Strongman, 1985; Shaw & Demers, 1986), the confidence interval showed no relationship. However, for high-intelligence subjects, the mean of three estimates was .510, indicating a very good relationship. For verbal creativity (Task 26; Parrott & Strongman, 1985; Shaw & Demers, 1986), the mean value of three *rs* for general intelligence subjects was .072 (no relationship). For the three values with high-intelligence subjects, the mean was .326 (acceptable).

Although the two confidence intervals for the visual (-.058 to .421) and verbal (.036 to .356) either overlapped with 0 (no relationship) or was close to it (marginal relationship), they concealed differences between the high- and average-intelligence subjects. For both tasks, the mean validity coefficient for the former met the .25 standard and was acceptable as evidence of a positive relationship between vividness and creativity. However, for the latter, neither the visual or verbal showed any relationship.

Other studies of objective cognitive performance. The mean of two *rs* between VVIQ scores and performance on two visual search tasks [Tasks 29 and 30 (weights in Table A), Table 11; Wallace, 1988, 1991] was .256, which represents an acceptable relationship. Similarly, Wallace (1990b) reported four estimates of the relationship between the VVIQ and performance on gestalt closure tasks (Task 45, Tables A and 11). This relationship was very good.

Commentary on Objective Cognitive Performance Data

General. For all 66 numerical estimates of the VVIQ/cognitive performance relationship, the 95% confidence interval was .201 to .322, with a midpoint of .263, indicating that the acceptable standard was met.

Of the 66 numerical relationships, 32 were obtained with concurrent administration of the VVIQ and criterion task, and in 31 of the 32 cases, order of testing was not balanced or unclear. However, the frequencies in the six significance/design categories were as follows: 17 (significant positive/concurrent), 12 (significant positive/delayed), 13 (not significant/concurrent), 10 (not significant/delayed), 2 (significant negative/concurrent), and 0 (significant negative/delayed). These show no significant association, $3 \times 2 \chi^2(2, N = 54) = 1.53, p > .30$, between significance and design, undermining any subject effect. In addition, a comparison of the mean z 's for the concurrent ($M = 0.277, SD = 0.211, n = 34$) and delayed ($M = 0.272, SD = 0.196, n = 20$) conditions was not significant, $t(52) = 0.08, p > .20$. It is also notable that the VVIQ was given first with a delay in 13 cases, in 10 of which extreme groups were chosen. Seven of these gave a significant positive VVIQ/task relationship and three gave no significance, suggesting that there may have been some subject cueing via knowledge about VVIQ status. However, four of these results came from Wallace (1990b), who took particular care to avoid subject and experimenter effects, leaving three positive and three nonsignificant results. Together, these data indicate that both subject and experimenter effects were unlikely. As with self-report studies, the vast majority of estimates were obtained with group testing, undermining further any experimenter effect.

The highest VVIQ/objective performance correlation was .536, for the Gestalt closure tasks. Although this was higher than the relationship with proofreading (.172), both surprised me, because I did not judge that imagery was involved (see Tasks 30 and 45, Table A). However, they support Wallace's (1990b, 1991) analysis, in which subjects perform the tasks using visual images from long-memory.

The next highest objective performance correlation of .392 between the VVIQ and hypnotic susceptibility is consistent with the self-report absorption validity coefficient of .515, and supports Marks' (1983b, p. 123) contention of a positive relationship. However, because measures of hypnotic susceptibility have themselves been characterized as showing the subject's tendency to respond to demands rather than to experience a special state (Predebon & Wenderoth, 1985; Spanos & Chaves, 1989), the present value cannot be taken as evidence that the VVIQ is demand-free. Indeed, the ability to experience vivid images, to become absorbed, to follow suggestions, and to respond to demands may all reflect a common dimension. The

cautious conclusion is that the absorption and hypnotic susceptibility coefficients indicate that "imaginative involvement" is a component of the VVIQ. It may partly reflect the global experience of vividness, in which the image resembles normal seeing if the person is imaginatively involved. If so, the weight for vividness in this task should be increased from 2 (Task 24, Table A).

Although it has been argued that creative processes are involved in the VVIQ (Hilgard, 1977, p. 103; Marks, 1987; 1989b), there was little evidence of this for most subjects. The validity coefficient for self-report aesthetic judgment is acceptable, but does not represent creative thinking and may be inflated by demand characteristics. For the creative production tasks, the only association with vividness occurred for subjects with relatively high intelligence test scores (Kunzendorf, 1982). Perhaps the imagination component of the VVIQ underlies this relationship. The differential relationship between the VVIQ and creativity for subjects of high and general intelligence may reflect the relationship between intelligence and creativity. It has been suggested (McNemar, 1964) that creativity requires a certain intelligence level, implying that creativity and intelligence will be positively correlated at this level and below but uncorrelated above it. In the latter case, creativity may also be correlated with imagery vividness.

The validity coefficients from the control tasks showed no relationship between the VVIQ and spatial alteration, a minimal one with physiological control, and an acceptable one with motor control. The first association has been in doubt for some time (A. Richardson, 1969, p. 53), and other research on the relationship between self-reported imagery and objective spatial tests has shown the same null result (Ernest, 1977; A. Richardson, 1972, 1977). One possibility is that the tests of spatial "folding" (the first of the two kinds of spatial test) do not involve visual imagery, but only general reasoning (Katz, 1983, p. 44; A. Richardson, 1969, p. 55; Reisberg et al., 1986; Yuille, 1986). However, the second kind of spatial test (spatial "rotation") almost certainly requires the formation and manipulation of a visual image (e.g., Cooper & Shepard, 1973), and the same lack of significance was obtained. This suggests that vividness is not an important process for spatial rotation (Horne, 1993), and perhaps also that the control or manipulation component in the VVIQ is smaller than hypothesized.

To argue that the element of imagery manipulation is essentially without function may seem to ignore the relationship between the VVIQ and tests of physiological/motor control. Overall, the association was judged to be just acceptable, but it is consistent with theories (e.g., Ahsen's Triple Code Model, 1982, 1984) suggesting that imagery can influence physiological function, and with evidence that subjects' reports of imagery accompany

physiological change (A. Richardson, 1984), e.g. suppression of the alpha rhythm (Kamiya, 1972; Nowlis & Kamiya, 1973) and motor performance (Suinn, 1983). However, these reports usually emphasize *having* the image rather than manipulating it: for example, "...all they would have to do was to conjure up an image of a person's face, hold it, and look at it very carefully, so much so that they could actually see the features of the person's face" (Kamiya, 1972, p. 522, emphasizes his). Perhaps generation, holding and vividness, not control, are the key elements of the image for such regulation.

There is evidence from the studies of visual scanning and visual search (Powell, 1986; Wallace, 1988, 1991) that the VVIQ taps visual processing. Wallace (1991) also makes this interpretation from its relationship with proofreading. However, he also found that that proofreading performance was related to hypnotic susceptibility (Wallace, 1987), suggesting that the holistic strategy reflects the ability to become absorbed. On the other hand, Wallace (1990b) found that vividness and hypnotic susceptibility made related but independent contributions to Gestalt closure performance. In discussing the evidence relating the VVIQ to self-reports, it was also observed that significant *rs* with non-visual tests undermined the discriminant validity of the VVIQ. Few VVIQ relationships with non-visual performance have been examined but it has been associated negatively to numerical reasoning (McKelvie & Rorhberg, 1978) and nonsignificantly with verbal and abstract reasoning (McKelvie & Rohrberg, 1978; Rossi & Fingeret, 1977). Hatakeyama's (1984) finding that size judgments were almost related to the QMI but not to the VVIQ is also somewhat puzzling, because the two questionnaires were themselves significantly associated. More work with non-visual controls is required. Finally, all but one of the studies clearly involved imagery, so that they cannot discriminate whether that process is important in the VVIQ.

Cueing. The objective performance test data can also be considered in light of the argument that the VVIQ is more likely to be associated with a task that explicitly instructs subjects to generate visual images. Of the five main tasks (Tasks 22 to 26), this requirement was met in two (hypnotic susceptibility, .392 and physiological/motor control, .268), in both of which the relationship was significant and acceptable. For the other three cases (spatial, visual, and verbal creativity), the associations were not significant. However, for the four additional performance tasks, the relationship was significant for the three cases (Tasks 27, 29, 30) with no instructed imagery, and not significant for the one (Task 28: Scan), where it did occur. Overall, these results indicate that instructed imagery generation does not predict a significant VVIQ/criterion relationship. This conclusion is supported by the

observation that the five significant relationships were distributed among all four levels (weights 4 to 0) of cueing.

Predicted and Objective Performance Tasks for Perception:
 $r = .10$ Marginal, $r = .25$ Acceptable

Theorists treat both perception (e.g., Anderson, 1990, pp. 47-84; Dodwell, 1970, p. 205, p. 229) and imagery (Finke, 1980) as a series of hierarchical levels, some of which may be functionally equivalent. In particular, it has been suggested that stimulus perception and imagery production both involve eye movements (see Marks, 1972, p. 99), which predicts an association between VVIQ scores and eye movement measures (Marks, 1983b, pp. 114-118). Although Marks (1983b) claims support for this, unfortunately, I found that his summaries of his VVIQ/eye movement studies (Marks, 1972, 1973b, 1977, 1983b) were confusing in places, and I could not meaningfully quantify his data. Other studies have compared visual illusions (Reisberg & Morris, 1985; Wallace, 1984a, 1984b) or subjective contours (Wallace, Persanyi, & Gerboc, 1989) in imagery for good and poor visualizers. However, the methodology of the first set has been criticized (Predebon & Wenderworth, 1985) and, from the information given, I could not estimate effect sizes in either case. Third, Finke and his colleagues (Finke, 1979, 1981; Finke & Kosslyn, 1980; Finke & Kurtzman, 1981a, 1981b; Finke & Schmidt, 1978) investigated the effects of substituting physical stimuli with visual images in various tasks. Although this methodology has also been debated (Broerse & Crassini, 1981; Harris, 1982), I was able to extract some VVIQ/task relationships (Task 33, Tables A and 11; Task 24, Table 11). The confidence interval meets the .40 standard (very good).

Discussing Finke's work, Kaufmann (1983) accepts that a relationship has been found between VVIQ scores and visual perception performance. However, Kaufmann doubts that the results indicate test validity, because, with such "pure" imagery tasks and careful experimental control, "it is possible that even a crude measure of imagery ability like the imagery questionnaire may provide a sufficient basis for predicting performance" (Kaufmann, 1983, pp. 247-248). Kaufmann then states that the VVIQ will be much less successful in predicting performance in "more complex criterion tasks, where imagery is only one of several determinants" (Kaufmann, 1983, p. 248), presumably demonstrating the weakness of the test. However, Kaufmann's argument places the VVIQ in a no-win situation. If it measures visual imagery vividness, then it is unlikely to predict more complex, and by implication, more important cognitive tasks. But its success with more purely visual tasks is only obtained in the rarefied atmos-

phere of tight experimental control, where any crude measure would work. In either case, the evidence does not demonstrate test validity. I agree that relationships between the VVIQ and criterion tasks may decline as the number of shared component processes decreases, but I would interpret this trend as enhancing validity, and that control of extraneous variables adds confidence to this conclusion.

Evaluation of overlap models for cognition and perception. The M and SD of the 70 obtained z_r s were 0.279 and 0.257, giving a validity coefficient 95% confidence interval of .212 to .330, centered on .273, meeting the acceptable .25 standard. However, this was also less than the expected mean z_r of 0.507 ($SD = 0.100$), $t(69) = 19.00$, $p < .01$, indicating that the overlap hypothesis overpredicted the data.

Corresponding to the self-report tasks of control, absorption, and aesthetic judgment, predictions were formulated for the cognitive/perceptual tasks of control (Tasks 22 and 23, spatial and physiological/motor control), hypnotic susceptibility (Task 24), and creativity (Tasks 25 and 26). Analysis of variance of the predicted z_r s for these five tasks was significant, $F(4, 55) = 79.93$, $p < .01$, with the order as follows (see Table 11): hypnotic susceptibility, physiological/motor control, verbal creativity, visual creativity, and spatial. Matching this perfectly, the obtained z_r s followed exactly the same order (Table 9), with $F(4, 55) = 3.08$, $p < .05$. Thus, the overlap hypothesis was quite successful in predicting self-report and objective performance for the effect of type of task (control, absorption and creativity), the only exception being self-reported control, which was overpredicted both in absolute and relative terms.

Finally, showing that the model predicted overall, r between 50 expected and obtained z_r s (20 were unclear as to delay), was .275 ($p < .05$).

Construct Validity of VVIQ on the Basis of Cognitive and Perception Studies

Marks (1983b, pp. 114-120) argues that his VVIQ/eye movement data are consistent with his predictions. In addition, Marks' (1983b, pp. 120-121) predictions for the imagery substitution effects were upheld by Finke's findings. Third, his prediction for hypnotic susceptibility was also supported. Together with the overall acceptable relationship between the VVIQ and objective (cognitive and perception) task performance, these findings support the construct of vividness and the validity of the VVIQ. However, the size of the association ($r = .273$) is slightly smaller than that for self-report tasks ($r = .369$).

The overlap hypothesis received further support. The overall r between predicted and obtained scores was slightly smaller than with self-report

tasks (.275 vs. .392), but was significant. As with self-report tasks, the absolute size of the obtained VVIQ/task relationship was overpredicted, but the order of the control, hypnosis, and creativity task relationships with the VVIQ was predicted perfectly. Generally, I conclude that the deductive evidence from cognitive and perception tasks supports the construct of vividness and the validity of the VVIQ, but less strongly than with self-report tasks. One reason may be that the cognitive tasks are themselves not well understood (Reisberg & Heuer, 1988). They may be insufficiently varied and may not sufficiently reflect the VVIQ process of vividness to provide a critical test (Horne, 1993). For example, size judgements, scanning, and search may only require imagery contours rather than rich detail.

Inductively, the cognitive data support the suggestion from self-report data that imaginative involvement is a more important VVIQ process than control or creativity. Finke's perception substitution data indicate that the VVIQ taps a veridical visual imagery process. Wallace's analysis of his proofreading and Gestalt closure work implies that the VVIQ taps accessibility to visual memory images. VVIQ effects in imagery but not in perception conditions (Finke, 1979; Finke & Kosslyn, 1980) contradict the claim that imagery is not a VVIQ process (Chara & Hamm, 1988; Chara, 1992a) and also demonstrate that vividness does not simply reflect sensitivity of the visual system (Finke, 1981).

Predicted and Objective Performance Tasks for Memory:
 $r = .10$ Marginal, $r = .25$ Acceptable

Researchers have frequently employed memory tasks as a criterion, yielding quantitative estimates from 40 tasks. Indeed, Marks (1972, p. 83; 1973a) phrased his initial question in terms of visual recall, distinguishing between the literal and associative function of imagery in memory (Marks, 1972, p. 85). In the first case, "a subject is asked to remember a stimulus pattern by forming an exact mental image of the pattern....during retrieval, the subject obtains at least some of the required information directly from the image, as if the original stimulus were still present" (Marks, 1972, p. 85; his emphasis). Second, "Imagery functions associatively when...a subject is required to form an association between a stimulus and some previously-presented stimulus by use of mental imagery...and retrieval occurs by decoding from this image" (Marks, 1972, p. 86; his emphasis). The first function has been investigated by assessing immediate memory for details from a single visual stimulus, and the second with paired-associate recall for a number of stimuli. In addition, the VVIQ/memory relationship has been studied with free recall and recognition for a number of stimuli.

Marks (1989b) also states that imagery is a construction from both mem-

ory and imagination. Furthermore (and following McKellar, 1987), he argues that even imagery of past experience ("memory imagery") is not purely reproductive (i.e., veridical), and that the imaginative component becomes increasingly important as the retention interval lengthens. Marks (1989b, p. 462) then goes on: "For these reasons, the validation experiments reported by Marks (1972, 1973a, 1973b) and many of the others that have followed, have been concerned with the role of imagery in immediate, or nearly immediate, memory when accuracy is used as the criterion..." This argument implies that the VVIQ is more likely to predict short-term rather than long-term memory performance. In fact, Marks has confined his attention to the short-term task, predicting that "Good visualizers are able to recall pictures more accurately than are poor visualizers" (Marks, 1983b, p. 109) and that "Good visualizers are able to detect a difference between a picture stored in memory and a new picture more quickly than poor visualizers" (Marks, 1983b, p. 111). He also predicted that good imagers should display superior associative recall using a visual imagery mnemonic (Marks, 1972, p. 96) and that they would recognize pictures more accurately than poor visualizers (Marks, 1983b, p. 113). Table 12 presents the confidence intervals for the VVIQ/memory task relationships.

Short-term recall of details from a picture. Although his two predictions were offered later (Marks, 1983b), Marks (1972, pp. 92-95; 1973a) began with a VVIQ investigation of the literal function of memory. It deserves scrutiny.

Subjects saw a picture of a scene or collection of objects, generated and held a visual image, and were then tested for their memory of detail. In his first study, 18 good and 18 poor visualizers, balanced for gender and selected from a pool of 74 students, viewed 15 photographs (1 per trial), knowing that an image would be formed and memory tested. During each trial, a picture was shown for 20 sec, after which subjects were instructed to "keep in mind a 'picture' of the displayed slide" (Marks, 1973a, p. 20) for 40 sec. To prevent verbal rehearsal, they also counted backwards during the first 30 sec of the delay. This was followed by a memory test with 3-alternative questions about details of the picture. Although characterized as recognition (Chara, 1989; Chara & Verplanck, 1986), this task involves retrieval of information from the held image (McKelvie, 1990) and is referred to by Marks (1988) as prompted recall. In addition, although Hishitani (1985a) thinks that the 40 sec delay period involves long-term memory, subjects were instructed to hold their image and to engage in a rehearsal-preventing task. Under such conditions, short-term memory lasts longer than the traditional 20 to 30 sec (Reitman, 1971). Process weights are shown in Table A (Task 36). Marks found that memory performance was

Table 12
Correlation Coefficients (Criterion Validity) between the VVIQ and Memory Tasks

Task	<i>n</i>	Lower Bound	Mean	Upper Bound	Judgment
<i>Single-stimulus</i>					
36. Single-stimulus Marks	3		.528		Exceeds A, VG met: VG
37. Single-stimulus Chara 1	7	-.165	-.034	.078	ns: NR
38. Single-stimulus Chara 2	5	-.021	.045	.111	ns: NR
36. to 38.	15	-.036	.136	.300	ns: NR
39. Single-stimulus CS	1		-.007		ns: NR
40. Motor Immediate	2		.523		Exceeds A, VG met: VG
41. Motor Imagery	1		.488		Exceeds A, VG met: VG
42. Motor Interference	1		.602		Exceeds A, VG met: VG
43. Visual Doodles	1		-.060		ns: NR
44. Verbal Doodles	1		-.040		ns: NR
45. Simultaneous Discrimination (RT)	1		.134		ns: NR
46. Successive Discrimination (RT)	2		.523		Exceeds A, VG met: VG
Accuracy	3		.058		ns: NR
47. Polygon/Color Recognition	2		-.536		Exceeds A, VG met: VG
All short-term single-stimulus recognition memory:					
37. (one), 46. and 47	6	-.423	-.157	.134	ns: NR
48. LTM Color Recognition	1		-.470		Exceeds A, VG met: VG
49. LTM Money Recognition	4	-.195	.119	.409	ns: NR
50. LTM Face Recognition	1		-.472		Exceeds A, VG met: VG
LTM Recognition (not two in 49.)	3		-.394		Exceeds A, VG min met: A
All single-stimulus	24	-.157	-.005	.146	ns: NR
<i>Paired-associate</i>					
52. PAL/STM/InstructedCW	4	-.211	.027	.265	ns: NR
53. PAL/STM/InstructedAW	3		.206		MA met: MA
54. PAL/STM/NotInstructedP	1		.713		Exceeds A, VG: VG
55. PAL/STM/NotInstructedCW	2		.133		ns: NR
57. PAL/STM/NotInstructedAW withCW	1		.045		ns: NR
All STM paired-associate recall	13	-.016	.141	.290	ns: NR
61. PAL/LTM/NotInstructedP	1		.050		ns: NR

Task	<i>n</i>	Lower Bound	Mean	Upper Bound	Judgment
<i>Free Recall</i>					
67. FR/STM/NotInstructedP	4	-.394	.124	.583	ns: NR
68. FR/STM/NotInstructedCW	1		-.110		ns: NR
70. FR/STM/NotInstructedCW withP	1		.782		Exceeds A, VG: VG
71. FR/STM/NotInstructedAW withP	1		.570		Exceeds A, VG: VG
74. FR/LTM/InstructedAW	1		.560		Exceeds A, VG: VG
75. FR/LTM/NotInstructedP	1		.308		A met: A
78. FR/LTM/NotInstructedCW withP	1		.425		Exceeds A, VG met: VG
79. FR/LTM/NotInstructedAW withP	1		.070		ns: NR
80. FR/STM/Imagery	1		.050		ns: NR
81. FR/LTM/Imagery	1		.402		Exceeds A, VG met: VG
82. FR/STM/Sentence	1		.450		Exceeds A, VG met: VG
83. FR/LTM/Sentence	1		.342		A met: A
All Free Recall (67. to 83., three from 49.)	18	.138	.301	.448	A met, VG min met: A
<i>Recognition Memory</i>					
87. RM/STM/NotInstructedAP	9	-.145	.027	.198	ns: NR
88. RM/STM/NotInstructedSP	2		.470		Exceeds A, VG met: VG
89. RM/STM/NotInstructedOP	3		.190		MA met: MA
90. RM/STM/NotInstructedCW	4	.156	.338	.498	A met, VG min met: A
92. RM/STM/NotInstructedCW withP	1		-.110		ns: NR
93. RM/STM/NotInstructedAW withP	1		.139		ns: NR
97. RM/LTM/NotInstructedAP	1		-.440		Exceeds A, VG met: VG
98. RM/LTM/NotInstructedSP	1		.380		A met: A
All face recognition memory	7	-.069	.089	.242	ns: NR
All recognition memory	26	-.015	.097	.206	ns: NR
All memory (estimate as unit)	88	.064	.137	.209	MA met: MA
All memory (task as unit)	40	.081	.196	.303	MA met, A min met: MA
All clearly visual memory	74	.020	.101	.179	MA met: MA

Task	<i>n</i>	Lower Bound	Mean	Upper Bound	Judgment
<i>Recognition Memory</i>					
Cueing: Instructed imagery	12	-.067	.194	.424	ns: NR
Picture stimulus	42	-.041	.058	.156	ns: NR
Concrete word stimulus	16	.012	.191	.351	MA met,
Abstract word stimulus	4	-.006	.226	.465	A min met: MA
Delay: STM	66	.075	.159	.241	ns: NR
LTM	17	-.141	.028	.194	MA met: MA
					ns: NR
<i>All Criterion Tasks</i>					
Task as unit	71	.184	.253	.231	A met: A
Individual estimate as unit	263	.252	.269	.310	Exceeds A: A

Notes. (a) Italics indicate my judgment of the relationship: VG (very good), A (acceptable), MA (marginally acceptable), or NR (no relationship; min = minimally); (b) ns = not significant; (c) RT = response time, PAL = paired-associate learning, FR = free recall, RM = recognition memory, STM = short-term memory, LTM = long-term memory, P = pictures, AP = art pictures, SP = scene pictures, OP = object pictures, CW = concrete words, AW = abstract words; (c) For criterion validity objective performance tasks (cognitive, perception, memory), all dependent variables reflected accuracy, with the exception of response time in Tasks 46 and 47.

higher for good than poor visualizers, with $d = 0.98$ (my calculation). Two additional studies replicated this result under slightly different conditions (e.g., 11 different pictures), with $d_s = 1.91$ and 0.91 , respectively (my calculations). Marks notes that the combined probability of all three results is less than .001, and I found that the mean value of the three effect sizes (converted to r s) was $r = .528$. In fact, although he did not initially mention it, Marks (1983b, p. 111) stated later that correlations between recall accuracy and the reciprocal of VVIQ scores ($1/V$) range from .50 to .65, with the highest value of .73 in his laboratory obtained by Hodgson (1977; cited in Marks, 1983b, p. 111).

Marks' original research has many commendable features. Although I argued that scores on the VVIQ are not radically affected by whether it is completed with eyes open or closed, Marks followed his own advice, giving it under both instructions. Second, the groups of good and poor visualizers in the three studies were matched for gender and represented 24%, 7%, and 24% of the initial pools. Two of these are close to Feldt's (1961) optimum of 27%. Third, great care was taken to prevent verbal rehearsal of pictorial content while the image was being held. These aspects of his design permit the inference that recall was based on an intentionally-generated visual image, and that the two groups of visualizers were clearly separated. On the other hand, the delay between VVIQ and experimental task is not stated, and it is not clear if subjects knew why they were selected. If

they did, it is possible that the performance differences between good and poor visualizers may have been at least partly due to this knowledge. For example, if good visualizers assumed that they would perform well, they may have tried harder than poor visualizers. Berger and Gaunitz (1977) agree, and also wonder if there might have been an experimenter effect, since Marks did not make clear whether or not a double-blind procedure was followed. However, Hishitani (1985a, p. 10) reports that Marks informed him in a personal communication that experimenters were blind as to the VVIQ status of subjects. On the other hand, Marks can be criticized for not employing controls either in the form of subject variables (e.g., intelligence) or of baseline task conditions (e.g., a non-visual memory task on which the two groups would not be expected to differ).

There are four other studies of this general kind (Chara & Hamm, 1989; Chara & Verplanck, 1986; Cohen & Saslona, 1990; Phillips, 1978), of which Chara's have been hotly debated (Chara, 1989, 1992a; Marks, 1988, 1989b; McKelvie, 1990, 1992d). I combined all the VVIQ/picture recall data in three ways: (a) all results together; (b) because some measures of recall were taken after others, only single measures or first measures were considered; and (c) because only Marks (1973a) exercised complete control over image holding and verbal rehearsal, his results were separated from the second set. The confidence interval of r for the 15 individual effect size estimates (Tasks 36 to 38; Table 12) was -0.036 to 0.300 , centered on $.136$, indicating no overall relationship between VVIQ scores and recall. This estimate does not include Hodgson's value of $.73$, because Marks (1983b, p. 111) only mentions it in passing while summarizing the work from his laboratory. However, even with it, the interval contains 0 ($-.002$ to $.361$). For the eight measures (three from Marks and five others but excluding Hodgson's) that were clearly not contaminated by earlier ones, the corresponding validity coefficient confidence interval was $-.021$ to $.494$, also overlapping with zero. For Marks' (1973a) work alone, the correlation was $.528$ (or $.656$ with Hodgson's estimate), and for the other five uncontaminated estimates, the confidence interval was $-.049$ to $.021$. Only Marks' results show an acceptable, even very good, association between VVIQ scores and recall.

Certain factors seem to differentiate Marks' work from the others. He gave each subject trials with either 11 or 15 different pictures, whereas the others gave only 1 (Chara & Hamm, 1989; Chara & Verplanck, 1986; Cohen & Saslona, 1990) or 2 (Phillips, 1978). Although two studies (Chara & Verplanck, 1986; Chara & Hamm, 1989) followed Marks in using interfering tasks to prevent verbal rehearsal, none of them retained both features of Marks' procedure. Although two estimates from slightly different tasks

(Tasks 43, 44; MacLeod, 1986) were not significant (Table 12), further support for the idea that the VVIQ predicts performance with generated visual imagery was obtained from studies of motor memory (Housner, 1984; Housner & Hoffman, 1978; Tasks 40 to 42; Table 12), where correlations were significant and greater than .25.

Short-term recognition memory of details from a picture. Marks (1983b) predicted that good visualizers would be faster to detect a difference between a memorized picture and a new one, but no mention was made of accuracy. Given that he also predicted that good visualizers would perform better than poor visualizers on recognition memory for a number of pictures (Marks, 1983b, p. 113), I was surprised that he did not also predict that good visualizers would be more accurate in detecting the difference.

The relationship between the VVIQ and response time has been investigated with simultaneous and delayed presentation of the two pictures to be discriminated. In the first case (Task 45, Table A), r was .134 (one estimate; Gur & Hilgard, 1975). In the second case, the mean of two estimates was .523 (Berger & Gaunitz, 1977; Gur & Hilgard, 1975), which was acceptable. There are six estimates of the VVIQ/accuracy relationship (Table 12, Tasks 37, 46, 47; Berger & Gaunitz, 1977; Cohen & Saslona, 1990; Gur & Hilgard, 1975; Heuer et al., 1986; Reisberg et al., 1986), giving a confidence interval that includes 0. However, the studies fall into two groups. Work with changed Meier Art Test pictures, where there were no instructions to utilize imagery or to prevent verbal rehearsal (Berger & Gaunitz, 1977; Gur & Hilgard, 1975) shows no significant relationship (overall $r = .058$). Second, work on recognition of color, where imagery was not generated and verbal rehearsal was not prevented, demonstrated either lower performance by good than poor visualizers (Heuer et al., 1986; Reisberg et al., 1986) or no significant relationship (Cohen & Saslona, 1990), with a combined correlation of -.358. However, for the two cases in which the relationship was negative, instructed rehearsal prevention seems redundant, because the task required discrimination of shades on the recognition test. These considerations suggest that there may be a positive association between the VVIQ and short-term recognition accuracy if an image is utilized without verbal processing, and that the association will be negative if the recognition test is extremely difficult. Indeed, from the three studies of long-term memory accuracy for details of a visual stimulus, all of which involved fine discrimination (Reisberg et al., 1986, Experiments 1 and 3; Reisberg & Leak, 1987), the mean r was -.394, an acceptable negative relationship (Tasks 48 to 50; Table 12). Unfortunately, I could not calculate a quantitative estimate from Cairns and Coll's (1977) study of polygon recognition, which probably involved imagery generation, no verbal processing,

and a difficult test (Task 47, Table 10). Because good visualizers performed better than poor visualizers, the study suggests that the first two factors are of primary importance in the VVIQ/memory relationship.

All single-stimulus studies. For all studies combined (24 estimates; Table 12), the confidence interval contained 0, indicating no relationship between VVIQ scores and memory performance. However, a significant association between the VVIQ and accuracy may occur if imagery processing is encouraged and verbal processing discouraged. In addition, for tests of moderate difficulty, the association was positive (Housner & Hoffman, 1978; Marks, 1973a; mean $r = .599$ without an estimate from Housner & Hoffman), but it was negative if the test response requires an extremely fine discrimination (Cohen & Saslona, 1990; Heuer et al., 1986; Reisberg & Leak, 1987; Reisberg et al., 1986; mean $r = -.460$).

Others also suggest that the negative VVIQ/performance relationship may appear when the test discriminations are extremely fine (Heuer et al., 1986; Reisberg & Leak, 1987; Reisberg et al., 1986) but, consistent with the hypothesis that imagination is a VVIQ process, they contend that good imagers may be more likely to confuse incorrect alternatives with true ones because they are more able to imagine them as correct. It might be argued that this suggestion would have received stronger support if the negative relationship had appeared only, or more strongly, on the long-term than the short-term memory task, because the former is more likely to involve imagination (Marks, 1989b). However, this objection overlooks that difficult discriminations were required in the present short-term tests showing a negative relationship. The interaction between VVIQ scores and short versus long-term memory might appear on easier tasks. This hypothesis should be investigated systematically by manipulating delay and task difficulty in a study of memory accuracy for good and poor imagers who are instructed to form images and prevented from verbalizing. If the present account is correct, good imagers will perform better than poor imagers with a short but not a long delay on a task of moderate difficulty. However, on a task requiring fine discriminations, good imagers will perform more poorly with both delays.

To account for the negative relationship between the VVIQ and memory performance, Cohen and Saslona (1990) contrast it with the finding that greater vividness was associated with greater confidence (Reisberg et al., 1986), and they suggest that low scores on the VVIQ may be inflated. If they represent overconfidence, it follows that they will be associated with poor performance, particularly in demanding tasks that require fine discriminations. This proposal has merit, particularly since it was concluded that VVIQ scores may be contaminated by self-confidence if the question-

naire follows a criterion task. However, I think that the negative relationship between the VVIQ and memory performance would have occurred more often if vivid reports simply represent confidence.

Paired-associate recall for a number of stimuli. In studies of Marks' (1972, p. 86) second, associative, function of imagery, subjects were usually shown word pairs, and were then asked to recall one member given the other. In some cases, they were cued to generate imagery by instruction or by an interactive picture, and in some cases they were told that a memory test would follow. No study has attempted to prevent verbal rehearsal.

Although he subsequently dropped it in view of his interpretation of the evidence (Marks, 1983b, p. 123), Marks (1972, p. 96) initially predicted that, if visual imagery can mediate association learning, then good visualizers will be able to utilize visual mnemonic devices better than poor visualizers. This is reasonable, because interactive images are likely to contain information about spatial relationships that can be used to form links between their elements. He then summarizes unpublished work in which good and poor visualizers did not differ in recall using a "mental walk" technique. Marks (1972, p. 98) accounts for this null finding by pointing out that the number of visual images reported generated was similar for subjects who were questioned about their actual use of imagery. Moreover, the correlation between imagery use and recall was significant ($r = .67$). Marks makes a distinction between the "ability to generate and employ imagery", which "varies across people," and "the *potential* to do so," which "is probably universal...given appropriate and optimal conditions" (Marks, 1972, p. 98; emphasizes his). Because Marks does not provide either descriptive or inferential statistics, I could not compute an estimate of the size of the nonsignificant VVIQ-performance relationship. However, Marks (1972, p. 98) contrasts its weakness with the significant effect that was obtained in his work on the literal function of imagery (Marks, 1973a; and partly reported in Marks, 1972, pp. 92-95), and he argues that the two sets of findings may be due to differences between the literal and associative tasks (Marks, 1972, p. 98). In particular, he argues that the latter, with its inclusion of the mental-walk mnemonic, is more likely to tap imagination and creativity.

There have been nine other studies of the relationship between the VVIQ and paired-associate learning (Tasks 52 to 57, Table 12; weights for Task 52, Table A; Childers & Houston, 1984; Childers et al., 1985; Dickel & Slak, 1983; Hall, Tulukder, & Esposito, 1989; Hishitani, 1985a; O'Brien & Wolford, 1982; Rossi & Fingeret, 1977; T Hoen, 1978; Wallace, 1990a). The validity coefficient confidence interval overlaps with 0 (no relationship).

In all three cases with a positive association (Childers & Houston, 1984;

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O'Brien & Wolford, 1983; Rossi & Fingeret, 1977), recall was incidental. In contrast, of the seven studies giving no significant effect, four employed an intentional test (Hall et al., 1989; McKellar et al., cited in Marks, 1972, p. 96; 'T Hoen, 1978; Wallace, 1990a). The remaining three tests were incidental (Childers et al., 1985; Dickel & Slak, 1983; Hishitani, 1985a), but the extreme groups were formed with a median split in the last two cases. Imagery-cueing by instruction does not appear to be a factor. When this occurred, results were significant in one case (Rossi & Fingeret, 1977), but not in five others (Dickel & Slak, 1983; Hall et al., 1989; Hishitani, 1985a; McKellar et al., cited in Marks, 1972, p. 96; 'T Hoen, 1978). However, pictures may promote the relationship, since they appeared in two cases that gave significant results (Childers & Houston, 1984; O'Brien & Wolford, 1983). Thus, although there is no overall relationship between VVIQ scores and paired-associate learning performance, it may occur if the good and poor visualizers are truly extreme groups, if pictures are provided, and if the test is incidental and immediate. However, intentional learning seems to inhibit the relationship, perhaps because it encourages verbal processing. These findings contrast with the results from studies of literal memory, where imagery cueing (via instructions or visual stimuli) was a factor and intentionality was not.

Free recall for a number of stimuli. Although Marks did not make a specific prediction for free recall, he states that good and poor visualizers should differ in their processing of stimuli that are encoded in a visual-analog manner and in a verbal manner (Marks, 1977). In particular, he argues that good visualizers will be at an advantage with pictures whereas poor visualizers will be at an advantage with words. Thus, Marks predicts a good visualizer superiority when visual-analog coding is invoked by stimuli, not just when imagery is generated by instruction (as was found in the literal studies). From Paivio's (1971) dual coding theory that higher recall for pictures than concrete words and for concrete words than abstract words is due to the superiority of two codes over one and of the visual over the verbal code, it can be predicted that good visualizers will perform better than poor visualizers on free recall of pictures and concrete words but not of abstract words, and that the superiority should be greater for pictures than for concrete words. However, the overlap argument makes a slightly different prediction. When no imagery instructions are given, the total weights over all 12 processes are more similar to the VVIQ for concrete than for abstract words (Tasks 68 and 69 respectively), but are also more similar for concrete words than for pictures as stimuli (Tasks 68 and 67 respectively) (see Table 10 and Table A for process weights for Task 68). Again, however, the

good visualizer superiority is predicted to be greater for pictures than for abstract words.

On the other hand, although visual imagery detail may provide retrieval cues, vividness may not be as important here as for literal recall or even associative recall, where spatial properties of the image have a function. Given the results with paired-associate learning, this implies that any positive relationship will be weak at best, and that the factor of imagery cueing is unlikely to be important in it. Moreover, because free recall, particularly if immediate, is a positive function of verbal rehearsal (e.g. Greene, 1987; Jacoby & Bartz, 1972; Rundus & Atkinson, 1970), it may not be related to imagery differences if this occurs.

From six studies (Tasks 67 to 83, Table 12; Dwyan, 1988; Hishitani, 1985a, 1985b; Lorenz & Neisser, 1985; McKelvie & Demers, 1979; McLeod, 1986, Experiment 2), 18 estimates gave a confidence interval of .138 to .448, with a midpoint of .301. The acceptable and very good standards are met and minimally met, respectively. This contrasts with the non-significant relationships with single-picture and paired-associate studies. The present conclusion also applies to "visual" free recall, because the stimuli were usually either pictures or concrete (high imagery) words and, in the case of abstract (low imagery) words, were accompanied by instructions to form images in six out of eight cases.

In searching for a factor to predict size of effect for these studies of free recall, I considered group definition, intentionality of learning, cueing, and delay. Although McKelvie and Demers (1979) and Hishitani (1985b) both obtained significant positive results with extreme groups, Hishitani (1985a) did not, and Dwyan (1988) obtained nonsignificant or negative ones with a median split. Because 13 of the 18 estimates giving an overall significant relationship came from studies with intentional instructions, this factor did not prohibit the VVIQ/recall relationship, as it did with paired-associate learning.

For cueing, the confidence intervals were -.227 to .468, centered on .137 for picture stimuli, .027 to .507, centered on .320 for concrete verbal stimuli, and a mean of two values of .222 for instructed imagery and of .344 for abstract verbal stimuli. These intervals overlap, showing that imagery cueing (particularly pictures) is not an important factor in the relationship. Because it was significant only with concrete words, the results provide some support for the overlap hypothesis, but this conclusion is tempered by the r of .344 for abstract words. Given that there are only two estimates in each case, further work should be done with instructed imagery and with abstract words. Although the mean correlations for the nine short-term (.283) and nine long-term (.284) memory estimates were very similar, the

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confidence intervals of $-.034$ to $.548$ and $.159$ to $.400$, respectively, indicated that only the latter reliably exceeded 0. This weakly supports the prediction, derived from Marks' (1989b) position that imagination plays an increasing role in long-term memory, that the VVIQ/free recall relationship would be stronger over shorter than longer delays.

Recognition memory for a number of stimuli. In these tasks (Tasks 87 to 98, Table 12; weights for Task 88, Table A), pictures or words are presented, then shown mixed up with new ones. Subjects judge each stimulus as "old" or "new" (single-stimulus test) or choose which one is old (forced-choice test).

In comparing the literal, associative, and free recall tasks, I argued that detailed visual imagery may be more important for the first two than for the last. Greater imagery vividness should also offer an advantage for recognition memory of pictorial stimuli, but this will depend on the nature of the alternatives on the recognition test. If the discrimination between new and old items is very clear, vividness may be redundant; on the other hand, if it is extremely difficult, so that imagination comes into play as in the single-stimulus studies (Heuer et al., 1986; Reisberg et al., 1986; Reisberg & Leak, 1987), the relationship between vividness and performance may be negative. For verbal material, there may be no relationship between imagery vividness and accuracy, because any additional detailed visual coding may be redundant for a task that does not require retrieval. Therefore, I predicted no VVIQ relationship for word recognition or for easy picture recognition; however it might be positive for a moderately difficult and negative for an extremely difficult picture test.

On the other hand, Marks (1977) hypothesizes that good visualizers should perform better on recognition memory for pictures than for words, and that the opposite should occur for poor visualizers. This also implies that good visualizers should correctly recognize more pictures than poor visualizers (Marks, 1983b, p.113) and, although he does not state it, that the opposite should hold for words. Surprisingly, Marks does not distinguish between concrete and abstract words, but his reasoning on words seem most applicable to the latter. If concrete words invoke some imagery coding, they should be recognized better by good than poor visualizers, although perhaps not to the same extent as pictures.

From the recognition memory studies (Chaguiboff & Denis, 1981; Childers et al., 1985; Denis, 1982; Devane, 1988; Marks, 1977, 1983b; McKelvie, 1993a; McKelvie & Demers, 1979; Mou, Anderson, Vaughan, & Rouse, 1989; Phillips, 1978; Swann & Miller, 1982; Wallace, 1990a), 26 estimates yielded a VVIQ/accuracy confidence interval including 0 (Table 12). As with single pictures and paired-associate learning, the VVIQ does

not generally predict recognition accuracy.

Seven of these estimates were obtained with faces which, because they are difficult to encode verbally, constitute an interesting class of stimuli for present purposes. Although Marks does not discuss them specifically, his argument suggests that good visualizers should recognize them better than poor visualizers. On the other hand, because faces vary in subtle ways along common dimensions, they may provide an example of a fine discrimination task, particularly if they are matched on the recognition test. This implies that the VVIQ/accuracy relationship might be negative. The validity coefficient confidence interval for the VVIQ/face recognition accuracy estimates contained 0 (Table 12), indicating no association.

In reviewing the variables that might affect the VVIQ/recognition memory relationship, I considered intentionality of learning, because it was a factor with paired-associate learning. However, it was not important. Learning was intentional with many nonsignificant results (McKelvie, 1993a; McKelvie & Demers, 1979; Mou et al., 1989; Phillips, 1978; Wallace, 1990a) and in 5 of the 10 significant positive ones (Denis, 1982, 1987). To investigate whether cueing and delay predicted a significant relationship, I calculated the confidence intervals for picture stimuli (-.038 to .254, centered on .111) and concrete verbal stimuli (-.107 to .407, centered on .162). There was only one case each for abstract stimuli (.14) and for instructed imagery (-.09). These data show that cueing was not important, and that there was no VVIQ/memory association for either pictures or words, contradicting both Marks' prediction of a positive and negative relationship for pictures and words, respectively, and my suggestion (based on Marks' reasoning) of a positive and negative relationship for concrete and abstract words, respectively. However, these calculations were conducted with the assumption that Marks' (1977, 1983b) experiment employed concrete words, which is not clear. If they were abstract, the concrete word confidence interval became .024 to .459, centered on .266 (exceeding zero), and the abstract mean of two estimates (.14, -.31) became -.09. This supports the possibility of a differential relationship between the VVIQ and recognition accuracy for concrete and abstract words. However, the VVIQ/recognition memory relationship appeared with immediate testing, where the confidence interval was .040 to .258, centered on .151, but not with delayed testing, where it was -.399 to .156, centered on -.132.

The data on stimuli warrant a closer look, because this analysis combined four kinds of pictures (scenes, faces, art, and objects). There was no relationship between imagery vividness and recognition memory accuracy for photographs of faces (McKelvie, 1992b; McKelvie & Demers, 1979; Phillips, 1978), pictures of objects/concrete verbal material (Chaguiboff &

Denis, 1981; McKelvie & Demers, 1979), abstract drawings or art (Marks, 1983b; Mou et al., 1989), concrete words (Chaguiboff & Denis, 1981; Marks, 1983b; McKelvie & Demers, 1979; Wallace, 1990a), abstract words (Marks, 1983b; McKelvie & Demers, 1979; Wallace, 1990a), abstract text (Denis, 1982), or advertisements (Childers et al., 1985). Indeed, one association for words (nature unspecified) was negative (Marks, 1977, 1983b). However, positive relationships appeared with photographs of scenes (Marks, 1977, 1983b), social information if visually cued (Swann & Miller, 1982), and information from concrete text (Denis, 1982), some of which was social (Denis, 1987). This suggests that the VVIQ/recognition memory relationship occurs with complex meaningful stimuli, which may be more likely to induce visual images. Photographs of faces are an exception, but this relationship could have been inhibited by the target-distractor similarity. This idea could be tested by manipulating distractor/target similarity with scenes, text information and faces, with the expectation that the relationship would appear with faces and an easier test and would disappear in the other two cases with a more difficult test.

Commentary on Memory Studies

General. For the four memory tasks, there were 88 numerical estimates, giving a confidence interval for the the VVIQ/accuracy relationship of .064 to .209, centered on .137 (Table 12). This meets the .10 marginal standard.

Although weak, could this relationship reflect a subject or experimenter effect? Of the 88 estimates, 50 were obtained with concurrent testing and, in all of them, order of test/task administration was not balanced or unclear. As before, I conducted a 3×2 chi square test on the 6 significance/design frequencies: 10 (significant positive/concurrent), 13 (significant positive/delayed), 33 not significant/concurrent, 18 (not significant/delayed), 7 (significant negative/concurrent), and 1 (significant negative/delayed). Here, the association was almost significant, $X^2(2, N = 82) = 5.61, p < .10$, but it indicates that, whereas insignificant and significant negative relationships were more likely with concurrent than with delayed testing, positive significant ones were slightly more likely with delayed. When z_p s were compared, the delayed mean value ($M = 0.290, SD = 0.408, n = 30$) was significantly higher than the concurrent one ($M = 0.054, SD = 0.291, n = 52$), $t(80) = 3.06, p < .01$. Thus, in contrast to the self-report and cognitive/perceptual VVIQ/task relationships, which were unrelated to the delay between VVIQ and criterion task, those for memory were higher with a delay, even more strongly undermining a subject expectancy effect. This contrast was confirmed by a 3×2 (Task $\times$ Delay) analysis of variance, in which the interaction was significant, $F(2, 218) =$

4.48, $p < .05$. I further examined the cases where testing was delayed and the VVIQ was given first. Of the 24, 21 were obtained with extreme groups. However, 10 were positively significant and 11 were not significant, indicating that significance was not more likely with this design. Since the vast majority of memory studies also employed group testing, I conclude that the VVIQ/task data were, like those from self-report and other objective performance studies, not likely to be contaminated by either subject or experimenter effects.

When the factor of cueing was investigated with four separate estimates, only the confidence interval for concrete words exceeded 0 (see Table 12). It included both the marginal and acceptable standards. For delay, the interval for shorter (.075 to .241) and for longer (-.141 to .194) overlapped, but only the former exceeds 0 (minimally acceptable).

These analyses show a marginal association between the VVIQ and visual memory, particularly over shorter delays and with concrete word stimuli. This is consistent with Marks' (1989b) view that the VVIQ will be more successful in predicting short- than long-term memory, but not his claim of greater success with pictures than with words. The data also do not support the suggestion that instructed visual imagery generation will enhance the VVIQ/memory accuracy relationship.

Of course, this approach neglects distinctions among the four main tasks. The following conclusions are tentatively offered: (a) With single-picture stimuli, where memory for detail is required, the overall VVIQ/accuracy relationship is absent ($r = -.005$), but a positive association may appear if instructions or stimuli encourage generation of visual images and discourage verbal processing; however, if the memory test is difficult and involves fine discrimination, the relationship may be negative. The VVIQ also positively predicts speed of response. (b) For paired-associate learning, the overall VVIQ/accuracy relationship is also absent ($r = .141$), but it may occur with extreme groups on the VVIQ, interacting picture stimuli, incidental testing and, possibly, immediate testing. (c) With free recall, the relationship between the VVIQ and accuracy was acceptable ($r = .301$), and was more likely to occur with concrete word stimuli and with delayed testing. (d) For recognition memory, there was no overall VVIQ-accuracy relationship ($r = .097$), but it may occur with immediate testing and for complex meaningful stimuli such as photographs of scenes, social information, and narrative text.

Evaluation of the overlap hypothesis. For the 82 out of the 88 estimates that could be classified as concurrent or delayed, the mean obtained z_r for the VVIQ/memory task relationship was 0.159 ($SD = 0.358$), which gave a validity coefficient confidence interval of .081 to .222 (midpoint = .158).

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This exceeds 0, but the mean z_r was lower than predicted ($M = 0.466$, $SD = 0.057$), $t(81) = 48.73$, $p < .01$. The memory estimates were then considered according to delay, cueing, and type of task.

Although Marks predicted that the VVIQ would be more successful in predicting visual memory performance over short than long intervals, the VVIQ taps the processes of long-term memory and imagination, which Marks indicates are involved over longer intervals. Thus, the VVIQ may not be related more strongly to short- than to long-term tests. In fact, from the summed weights, the VVIQ/task relationship was not predicted to be different for the 28 short-term ($M = 18.8$, $SD = 2.4$) and 12 long-term memory tasks ($M = 20.7$, $SD = 5.0$), $F(1, 38) = 2.55$, $p < .12$. Similarly, the predicted mean z_r s (corrected for concurrent or delayed testing and unclear cases considered as concurrent) were not significantly higher for the 67 short- than for the 15 long-term individual cases ($M_s = 0.461, 0.489$, Table 9), $F(1, 80) = 3.00$, $p < .10$. Consistent with these predictions, the obtained difference was not significant ($M_s = 0.168, 0.113$, Table 9), $F(1, 80) = 0.28$, $p > .20$. This supports the overlap hypothesis, which assumed that greater veridicality of imagery over shorter than longer periods of time (leading to a positive VVIQ/task relationship) would be counteracted by an effect for imagination over longer periods.

For cueing (instructed imagery, picture stimulus, concrete word stimulus, abstract word stimulus), Table 9 displays the values predicted and obtained with z_r scores. The predicted effect was significant, $F(3, 73) = 3.55$, $p < .05$, with a higher and lower VVIQ/task relationship expected for instructed imagery and abstract words respectively ($M_s = 0.483, 0.409$), $t(23) = 2.45$, $p < .05$. However, the obtained results were not significant, $F(3, 73) = 0.51$, $p > .20$. Finally, although predicted differences among the four kinds of task were almost significant, $F(3, 78) = 2.58$, $p < .06$, with the strongest and weakest relationships expected from the single-stimulus and free recall studies, respectively ($M_s = 0.486, 0.441$), the obtained effect was not significant, $F(3, 78) = 1.46$, $p > .20$, and the single-stimulus and free recall relationships were the weakest and strongest, respectively ($M_s = 0.072, 0.311$).

As before, the success of the overlap hypothesis in predicting memory performance was also evaluated by correlating predicted and obtained z_r scores (corrected for concurrent and delayed testing with unclear cases considered as concurrent). Contrary to expectation, r was significant ($-.444$, $p < .01$, 76 pairs), but negative! This shows that greater overlap between VVIQ and memory task processes was associated with a *weaker* VVIQ/performance relationship. One factor that may contribute to this finding is that the obtained VVIQ/memory performance relationship was *stronger* when

the administration time between the VVIQ and criterion task was *delayed* rather than concurrent; z_r scores reflect this variable but were predicted to be *higher* for *concurrent* than delayed testing. To investigate this possibility further, I calculated correlations between obtained z_r s and predicted summed scores, without correction for delay (82 pairs). With $r = -.326$, $p < .01$, the value was slightly lower than that for the corrected z_r score (-.444) but it remained negative and significant. I conclude that the inverse association between the overlap model and the VVIQ/memory performance relationship is not due simply to the difference between concurrent and delayed testing.

Another possibility is that it reflects the significant negative VVIQ/accuracy relationships found in studies where fine discriminations were required. Of the nine largest negative relationships, five were obtained in Reisberg et al.'s tasks (Heuer et al., 1986; Reisberg & Leak, 1987; Reisberg et al., 1986) and two were obtained in Mou et al.'s (1989) with abstract art. Furthermore, all seven relationships expected from the overlap model were higher than the mean expected z_r for that task. Of course, this argument does not ameliorate the criticism that the overlap model is inadequate; it did not take the task difficulty variable into account. However, to examine the contribution of these seven relationships to the negative r for the overlap hypothesis, I calculated a new correlation between predicted and obtained z_r s with these seven cases removed (reducing the pairs to 69). The result was $r = -.269$, $p < .05$. Although this is higher than the previous value of -.444, it indicates that the overlap model continued to negatively predict the VVIQ/memory relationship without studies requiring fine test discriminations.

Implications of memory studies for construct validity. Marks predicted positive VVIQ/accuracy relationships for single-stimulus, paired-associate, and recognition memory tasks, but made no specific mention of free recall. Taking the results for each task individually, only those for the free recall gave a significant relationship, disconfirming Marks' predictions. In addition, if it is assumed that Marks contrasted pictures with abstract words, he predicted that the VVIQ/recognition memory relationships would be positive and negative, respectively. This was also not the case. However, if this assumption implies a positive relationship for concrete words, Marks' prediction is supported, although the relationship was not weaker than that for pictures. Marks' prediction of a stronger VVIQ/task relationship for shorter than longer delays was only partly upheld. Although the short-term but not the long-term relationship was significant, they overlapped with each other. On the other hand, Marks' (1983b, p. 111) prediction that good visualizers would respond faster than poor visualizers in the successive picture dis-

crimination task was upheld ($r = .523$). Moreover, if Marks' theorizing is treated as making the general prediction that there would be a positive VVIQ/task relationship for "visual" memory, this was confirmed, albeit marginally.

The generally significant VVIQ/memory relationship also supported the overlap hypothesis, although its size was again overpredicted. In addition, of the six predictions for the effects of variables, four (and possibly five) were confirmed. However, the significant negative r between predicted and obtained relationships violated the overlap hypothesis.

These memory findings provide patchy evidence for Marks' theoretical construct and the validity of the VVIQ. At best, the overall positive, but marginal, VVIQ/task relationship is supportive but the overlap hypothesis, which was fairly successful with self-report and cognitive/perceptual studies, was disconfirmed by the negative r . This may imply that there are theoretical problems with the construct, that the VVIQ fails to adequately measure it, or that the studies are flawed. It is also possible that the weights were incorrectly assigned to the memory tasks or that the memory tasks did not depend much on vividness. Indeed, Ahsen (1985b, pp. 33-34) has argued for the latter possibility on the grounds that imagination is a component of the VVIQ but not of recall tasks, implying that the VVIQ is unlikely to predict such performance. If this accounts for the overall weak VVIQ/task relationship, the data do not undermine the validity of the VVIQ as a measure of vividness.

Inductively, the contrast between the VVIQ/reaction time task relationships in Gur and Hilgard's (1975) simultaneous (not significant $r = .134$) and successive (significant $r = .421$) picture discrimination conditions provides further discriminant validity evidence that imagery is involved. Marks' prevention of verbal rehearsal also demonstrates that the imagery in his task was visual, counteracting the suggestion from self-report studies that visual processing may not be central to the VVIQ.

General Discussion of the Overlap Hypothesis on the Basis of All Criterion Task Studies

Size of task relationships and effects of variables. For each of the self-report, cognitive/perceptual, and memory tasks, the overlap hypothesis overpredicted the obtained VVIQ/task relationship, leading to a general significant difference between predicted ($M = 0.487$, $SD = 0.097$) and obtained ($M = 0.254$, $SD = 0.319$) mean z 's for all 71 tasks combined with the task as the unit of analysis, $t(70) = 19.83$, $p < .01$ (single-sample t -test using standard error based on the predicted SD). Thus, although the validity coefficient of .253 (confidence interval .184 to .231) corresponding to

the mean z_r over all 71 tasks is significant, $t(70) = 6.70$, $p < .01$, and indicates an acceptable VVIQ/criterion task relationship, it falls short of the value predicted by the overlap hypothesis. With each of the 263 individual z_r s as the unit, the confidence interval for the validity coefficient was .252 to .310 (midpoint = .269). However, I worked here with the 257 (6 missing from memory) for which predictions were made. The obtained mean z_r ($M = 0.289$, $SD = 0.265$, 257 cases) was greater than 0, $t(256) = 18.06$, $p < .01$, and the validity coefficient exceeded the acceptable standard. Again, however, it was lower than the predicted value ($M = 0.522$, $SD = 0.086$, 257 cases), $t(256) = 43.15$, $p < .01$.

On the other hand, there was some success in predicting the effects of certain variables. For self-report tasks, the predicted significant effect of content occurred, and the predicted ordering of the five types of content tasks was obtained fairly well. The exact predicted ordering for control, absorption, and creativity did not emerge, but cognitive task order was predicted perfectly. In addition, the effect of short- versus long-term memory tasks was not expected or obtained. However, for cueing and for type of memory task, no effect was obtained, although it was expected in the first case and almost in the second.

Finally, predictions were derived for mean z s across the self-report, cognitive/perception, and memory tasks. For individual z_r scores (corrected as before), the VVIQ/task relationship for self-report was expected to be higher than that for cognitive/perception, which was in turn expected to be higher than that for memory (M s = 0.576, 0.507, 0.466, Table 9), $F(2, 250) = 38.63$, $p < .01$. The obtained relationships conformed to this pattern (M s = 0.387, 0.279, 0.159, Table 9), $F(2, 250) = 10.88$, $p < .01$.

Correlations between predicted and obtained z_r scores. Over all 71 tasks, the r of -.023 between predicted and obtained z s was not significant. This failure of the overlap model to predict all VVIQ/task relationships considered together contrasts unfavorably to Kosslyn et al.'s (1984) finding of a r of .56 between obtained task similarity and that predicted from their imagery module model. This suggests that the present attempt to characterize the processes of the VVIQ is even less successful than that of Kosslyn et al., who described their own VVIQ model as inadequate. On the other hand, it may not be entirely appropriate to compare the present overlap r with Kosslyn et al.'s, because they derived it from data over tasks that were all completed by the same subjects.

Conclusion. Although the overlap hypothesis was not generally successful in terms of the overall r between predicted and obtained relationships, outcomes were different among the three kinds of task. For memory, the hypothesis was contradicted, whereas for self-report and cognitive/percep-

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tion tasks, it was supported. In addition, six out of eight predicted effects of variables were clearly obtained (self-report imagery content versus no content; ordering for cognitive tasks; short- versus long-term memory; self-report vs. cognitive/perception vs. memory tasks) or partly obtained (five kinds of self-report content; four types of memory task), and only two were not (ordering for self-report tasks of control, absorption, and creativity; effects of cueing in memory). Finally, although the overlap hypothesis overpredicted the size of the obtained VVIQ/task relationships, the weights assigned to criterion tasks were somewhat speculative. In addition, the point predictions from the overlap hypothesis were based on estimated maximum values for the validity coefficients. The gap between obtained and predicted effect sizes may mean that VVIQ/task relationships are not determined solely by common processes. Overall, I conclude that the overlap hypothesis based on the weighted-processes model provides a useful start to understanding what the VVIQ measures.

General Discussion of Construct Validity of the VVIQ

During a critical exchange with Marks about imagery questionnaires, particularly the VVIQ (Marks, 1983a; Kaufmann, 1981, 1983), Kaufmann pointed out various methodological defects in the validation studies and summarized how they should be rectified. He then threw down the following gauntlet:

"I challenge Marks to show that when such a designis implemented, the essential features of the nomological net structure that has been established for the imagery construct can be replicated using imagery questionnaires as a measure of imagery ability. Nothing remotely approaching this state of affairs has been achieved as yet" (Kaufmann, 1983, p. 249).

Kaufmann's "imagery construct" is probably Paivio's (1971) *I* (imagery, or imageability), which is defined differently from vividness (J.T.E. Richardson, 1980, p. 85) as ease of image generation rather than similarity to perceptual experience. Horne (1993) has also argued that imagery as a sensuous experience is not simply reducible to imagery as a representation. Therefore, it is unreasonable to expect that VVIQ/task relationships will replicate those of *I*. However, we can consider Kaufmann's challenge by asking whether the present body of research provides a parallel network of relationships that establishes the construct validity of the VVIQ as a measure of vividness rather than of *I*.

The VVIQ and Kaufmann's Challenge

Two major criticisms referred to the construct of visual imagery vividness rather than to the VVIQ: Marks' conception of experienced vividness does not capture the functional essence of imagery, and vividness is not an ability: it is not a trait, it does not have maximum performance properties, and it is not stable. These three specific criticisms were addressed by the effects of context, social desirability and training, by alternate-form and test-retest reliability, and by cases of imagery fluctuation and a logical analysis of randomizing influences in the brain. I concluded that imagery vividness was not an unstable state or simply a stable maximum performance trait, but possessed some features of a typical performance trait.

Five other difficulties (ambiguity in the operational definition of vividness, demand characteristics, blocking of items, unanchored ratings, the possibility of scale reversal) referred to the test itself, and threatened to induce lenient responding and contamination by extraneous variables. VVIQ scores were positively skewed and covered only a restricted range of the rating scale, but their standard deviation was not judged to be lower than expected from a normal distribution. They were also correlated with social desirability and confidence, but the level of contamination in the first case was judged to be acceptable, and the second relationship was only a problem if the objective criterion task was taken first. In addition, VVIQ scores were not contaminated by the eyes instruction or choice of parent at the beginning of the test. Finally, internal consistency was acceptable, but alternate-form and test-retest reliabilities were, respectively, unacceptable and lower than desirable, although these judgments may be harsh for a trait that is somewhat dynamic.

However, the best answer to all the objections would be that the VVIQ successfully predicts criterion relationships.

VVIQ/Criterion Task Relationships

Deductive evidence. From Marks' discussions of the VVIQ, I extracted 17 specific predictions (1, Marks, 1972, p. 96; 1, Marks, 1977; 14, Marks, 1983b, pp. 109-123; 1, Marks, 1989b) covering the three kinds of criterion task. Of these, 11 were supported by research evidence in the form of a significant relationship between the VVIQ and task performance. In addition, for the 13 variables whose effect was predicted by the overlap hypothesis, 10 were confirmed. The overlap hypothesis was also supported by the correlational evidence for self-report and, to a lesser extent, for cognitive/perceptual tasks. These results provide positive evidence for a theoretical VVIQ construct with maximum performance properties and for the validity of the VVIQ as a test. Furthermore, if the failure of the weighted-process-

es model with memory tasks is due to the absence of vividness as an important dimension in task performance or to inadequate criterion task weights, then this negative result cannot be counted against the construct or the test.

Apart from the specific predictions from Marks and from the overlap hypothesis, the overall significant VVIQ validity coefficient of .269 (263 cases) shows an acceptable relationship between the VVIQ and criterion task performance, adding support to the vividness construct and to test validity. However, the strongest relationship (.369) occurred with self-report tasks, particularly for those in which imagery content was judged (.459), followed by cognitive/perceptual ones (.273) and, finally, by memory ones (.137). Although I concluded that the high VVIQ/self-report relationship was unlikely to be contaminated by positive cueing, the argument was based on the lack of effect of VVIQ/task delay on significance and size of the validity coefficient. Because the self-report tasks in which imagery content was judged directly shared many common processes with the VVIQ, it is possible that, even with a delay, subjects would notice the overlap and be induced to respond in a consistent manner. If so, more pertinent evidence rests on the non-imagery content self-report and objective tasks. Furthermore, the latter are unlikely to be contaminated by social desirability, which may be a factor in the higher VVIQ/self-report relationship.

The validity coefficients for the non-imagery content, cognitive/perception, and memory tasks were .253, .273, and .137 respectively, demonstrating acceptable or marginally acceptable relationships. However, they may have been suppressed by VVIQ and methodological factors that can be improved: ambiguity in the operational definition of vividness, the subjective unanchored task, the possibility of rating scale reversal, a truncated range of scores, and questionable reliability over content and time. In addition, most studies did not employ the optimum extreme group split and often did not report whether there were controls for subject and experimenter effects. Finally, most of the objective performance studies did not contain either subject variable or experimental variable controls for the possibility of non-visual, particularly verbal, processing. If these factors were properly dealt with, stronger associations with the VVIQ may emerge. Indeed, in studies with objective theoretically-relevant tasks and good experimental control, validity coefficients were very good (.409 in Finke's perception substitution tasks and .536 in Marks' single-stimulus memory task).

Inductive evidence. Results with self-report studies suggested that the most important VVIQ process was vividness, followed by vision, then control, affect, and creativity, and that imaginative involvement also played a role. Cognitive/perception studies confirmed imaginative involvement as a

factor, provided evidence that VVIQ/task relationships did not simply reflect demand characteristics, and demonstrated the important role of imagery, but they also showed that creativity and, particularly, control were functionally weaker. Memory evidence added to the discriminant validity of the imagery process.

Together, these findings point to general vividness as the dominant VVIQ process, particularly if imaginative involvement is conceived of as a contributor. Indeed, this may justify my 5-point weighting scale, in which weights up to 3 reflected increased richness in the form of detail and weights of 4 and 5 reflected more global aspects of experience, which might be interpreted as involvement or absorption. Imagery is also important (weight of 4), but creativity (imagination) and manipulation (control) are less so (weights of 2 and 1, respectively). However, affect should perhaps be added as a VVIQ process.

Kaufmann's Challenge: A Response

Although the methodological flaws recognized by Kaufmann (1983) have not disappeared, I think that the evidence is sufficient to judge his skepticism about what the VVIQ measures. Many individual results were not significant but, when aggregated, they revealed VVIQ/task relationships that were acceptable or marginally acceptable. Although improved methodology may clarify these effects, I think that their size will be limited by endemic aspects of the construct and test: vividness is a somewhat dynamic trait with some typical performance features and the necessary self-report task requires a judgment that is difficult to anchor. Overall, my interpretation of the relationships is that they constitute a network that is favorable to the construct validity of the VVIQ.

Future Research with the VVIQ: A Plea for Direction

Research on Construct Validity

Many previous studies with the VVIQ were designed to investigate general imagery theory. Although this attests to the value that researchers place on the test, its validity often seems to have been assumed. I think that future studies should more clearly separate the issues of test validity and of the role of imagery vividness in cognition, which is controversial (Begg, 1988; Neisser, 1970). There is experimental research on this issue (Anderson & Hidde, 1972; Bower, 1972, pp. 68-69; Einstein, McDaniel, & Lackey, 1989; Gerler & Nelson, 1982; Hishitani, 1991; Horne, 1993; Iaccino & Byrne, 1989; Kerr & Neisser, 1983; Paivio, 1968; Rowland & Franken, 1976; Standing, 1973), but more is needed. As the function of vividness is

better understood, weights given to processes in the VVIQ and in other tasks can be revised, so that better predictions can be formulated for VVIQ/task relationships, allowing a better assessment of test validity.

Modifications to the VVIQ

1. To reduce ambiguity in the operational definition of vividness, I suggest that its components be more clearly identified and be made consistent with Marks' published descriptions of vividness. In particular, the term "vivid" should be replaced as a scale descriptor. If Marks' "vague and dim" (Rating 4, Table 1) corresponds to "clear and vivid" (Ratings 1, 2, 3), "vivid" might become "bright."

2. The implicit demand that a visual image will be experienced should be removed, perhaps by replacing the statement before each block of items "Consider carefully the picture that comes before your mind's eye" (Table 1) by "Consider whether a picture comes before your mind's eye."

3. To broaden the range of scores, increase the number of scale points to 6 or 7, with a majority referring to vividness that is satisfactory or better.

4. Reverse the rating scale so that a high rating indicates greater vividness.

5. Develop objective exemplars for the rating scale points, perhaps utilizing Chara and Verplanck's (1986) approach with methodological improvements (see McKelvie, 1990a).

6. To improve reliability, increase the number of items. Because there was evidence that test-retest reliability was raised under Ahsen's filters, instructions might be altered to encourage subjects to better focus their attention on the rating task.

7. Drop the practice of rating items twice with eyes open or closed. However, consider matching these instructions with the criterion task.

The following recommendations assume that Marks' VVIQ will be used. However, they can also be conducted with the revised version, for which complete reliability data will first be required.

Ideal VVIQ Validity Study

To reduce the likelihood that subjects (and experimenters) make a connection between the VVIQ and criterion task, the design should be double-blind (Berger & Gaunitz, 1977; Kaufmann, 1983). Although the finding of no difference between concurrent and delayed administration for the significance or size of VVIQ/task relationships suggested no subject expectancy effect, few studies have included specific controls to avoid it. Indeed, there was a significant and unacceptably high correlation between the VVIQ and task confidence if the criterion task preceded the VVIQ. I rec-

ommend as standard practice that the VVIQ and criterion tasks be administered at different times by different experimenters and that their order be counterbalanced. In addition, if the extreme groups method is employed with the VVIQ given first, all subjects should be tested on both occasions, to avoid the possibility that good and poor visualizers (and experimenters) realize why they have been invited back. Under these circumstances, the 25%-30% split, rather than the median split, should be used. This approach will also permit the calculation of r using all subjects. If there are outliers, the value with and without them should be reported. Finally, researchers may suspect a serious problem of restriction in range if the SD of the VVIQ distribution is less than 0.64.

Using Cohen's (1977, p. 102) Table 3.4.1 with linear interpolation, it is also possible to recommend sample sizes, based on the following assumptions: $\alpha = .05$, power ($1 - \beta$) = .70, and effect sizes of $r = .369$ (self-report tasks), .273 (cognitive/perception tasks), and .137 (memory tasks). The results may be surprising: $n = 46$ (self-report), 89 (cognitive/perception), and, for memory, a mammoth $n = 444$! If the effect sizes are defined by the standards established here, the recommended sample sizes are $n = 109$ (acceptable $r = .25$), 384 (marginal $r = .15$), and 616 (marginal $r = .10$). If extreme group research is planned, the number of subjects to be tested is even greater. Using Cohen's (1977, p. 22) Table 2.2.1, the six correlations just considered become $d = 0.79, 0.57, 0.28, 0.50, 0.30$, and 0.20 , respectively. From Table 2.4.1 (Cohen, 1977, p. 55), the required number in each group of good and poor visualizers is 21, 40, 172, 50, 138, and 310 respectively. It can be seen that these numbers are approximately 50% of those required for the correlations. Thus, if the extreme groups are chosen using a median split, the total number to be tested is essentially the same. However, if the present recommended split of 25%-30% is adopted, the required totals will be considerably higher. For 30%, they are 70, 133, 573, 166, 460, and 1032. The two largest numbers here raise serious questions about the feasibility of conducting extreme group research to detect effect sizes of the order .10 to .14. Of course, if a number of studies are conducted and the results are aggregated to provide confidence intervals, the sample sizes in each case can be smaller.

Investigators should also include better experimental control conditions to exclude alternative explanations (e.g., in terms verbal processing) and better subject variable controls in the form of a social desirability test for self-report tasks and, as suggested by Kaufmann (1983), a test of general mental ability for objective criterion tasks. Such cautionary measures should permit a more definitive assessment of discriminant validity for the VVIQ.

Extending Previous Studies to Assess VVIQ Validity

1. Gather additional delayed alternate-form reliability data.
2. Compare the blocked and random formats of the VVIQ. First, investigate whether the strongest relationships between the blocked VVIQ and criterion performance (e.g., Finke's perception, Marks' single-picture recall, or Denis' prose recognition tasks) are replicated with the random format. If this is successful, investigate other tasks to find out if a stronger relationship with the random than blocked VVIQ will emerge. Reliability data for the random format will also be required.
3. To assess the degree to which imaginative involvement is a component of the VVIQ, conduct further work in which a measure of hypnotic susceptibility and the VVIQ are investigated as predictors (e.g., Wallace, 1990b). To investigate the possibility that responses to both tests are mediated by demand characteristics, compare the reactions of good and poor imagers to hypnosis and simulating instructions. If the VVIQ/susceptibility relationship reflects demand characteristics, good and poor imagers would differ from each other in both conditions.
4. Replicate Marks' (1973a) single-stimulus memory study with procedures to avoid subjects' perceiving the connection between the VVIQ and recall task, with a control for general mental ability, and with a non-visual control task.
5. Replicate Reisberg et al.'s (1986) color memory study with manipulation of delay (shorter/longer) and of carefully operationally-defined task difficulty (moderate/difficult alternatives). Extend the analysis to other stimuli (e.g. faces, scenes).
6. Investigate the relationship between the VVIQ and recognition memory for material that varies in complexity and meaningfulness.
7. Conduct a paired-associate learning study with concrete and abstract word pairs only in an independent groups design. Include conditions with and without imagery generation instructions and with interactive pictures accompanying the words. Manipulate delay and incidental/intentional instructions.
8. To investigate the hypothesis that the strongest VVIQ/task relationship occurs with free recall, compare the performance of good and poor visualizers on each of the four kinds of memory task (single-stimulus picture, paired-associate recall, free recall, recognition memory).
9. Paivio (1972, p. 262) suggested that more work on the differential effects of vividness and of imageability (*I*) is required. To further investigate the discriminative validity of the VVIQ, its success as a predictor should be compared directly with measures of individual differences in *I*. I suggest studies with different memory tasks, starting with prose, where both

imagery/spatial ability (Ernest, 1991) and imagery vividness (Denis' work) predict performance.

10. One reason for the generally weak VVIQ/memory relationship may be that performance on memory tasks is evaluated solely in terms of accuracy or veridicality. However, real-life memorizing, like the VVIQ, involves imagination so that, as Ahsen (1985b, p. 34) puts it, memory tasks are "overdefined." One way to explore the relationship between the VVIQ and memory tasks which allow for this factor would be to investigate pseudomemory, where subjects incorporate into memory implied, suggested, or imagined information from leading questions (Loftus & Loftus, 1980; Loftus & Palmer, 1974) or from meaningful distractors (Leibrich & White, 1983). Would the imagination of good visualizers render them more susceptible than poor visualizers to such cues? Another possibility is to utilize the size distortion effect in which a drawing from memory of the central object in a photograph is smaller than the original (Legault & Standing, 1992). If this frame extension effect is produced in imagination by "a perception schema which inappropriately incorporates background information into the memory image" (Legault & Standing, 1992, p. 121), it might be greater for good than poor visualizers. Furthermore, if imagination contributes more to performance with delay, the VVIQ/size distortion effect should also increase with delay.

New Directions for VVIQ Research

Given the present evidence of construct validity, for what purposes can the VVIQ be used?

VVIQ as an Independent Variable

Marks (1987, p. 3) has stated that the VVIQ was designed for "experimental, laboratory research," not clinical applications. However, it was based on Sheehan's (1967) QMI, which he argued was useful in both clinical and experimental situations. It follows that the VVIQ may be useful in applied settings where imagery is evoked. For example, imagery has been used as a therapeutic technique (e.g., Sheikh & Jordan, 1983; Strosahl & Ascoug, 1981), as a mental rehearsal device, often in sports settings (Suinn, 1983), and to enhance sexual arousal (Kelley & Byrne, 1978; Smith & Over, 1990). Because VVIQ scores were marginally and acceptably related to physiological and motor responses, respectively, they might also predict the effect of visual imagery in these settings, with greater benefit for good than poor imagers.

On the other hand, even if the VVIQ is a valid measure of vividness, it does not follow that it can be used to identify individuals who would and

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would not benefit from such imagery instructions. The reliability and validity coefficients may be sufficient for research purposes, but they do not justify the VVIQ as a diagnostic instrument for personal assessment. Moreover, norms are inadequate for score interpretation.

Because Murphy (1990) warns that imagery vividness may not adequately represent the variety of dimensions that are tapped by mental rehearsal, I think that future research should investigate the functional component of imagery in applied settings (e.g., generation versus vividness). Here, the VVIQ can serve as an independent variable.

VVIQ as a Dependent Variable: The Work of Ahsen

Because vividness is more dynamic than most abilities and skills, the VVIQ can serve as a dependent variable. Indeed, if scores fluctuate over various contexts, as Marks (1989b) acknowledges, the questionable test-retest and alternate-form reliabilities will be less of a problem here than for other purposes. In particular, by manipulating filters and other mental sets, adapted versions of the VVIQ may help to understand the vividness construct (Ahsen, 1987, 1990, 1991; Hilgard, 1987; Marks, 1987, 1993; Sheehan, 1987).

So far, I have only examined Ahsen's work for its bearing on the VVIQ as a psychometric test. However, he makes three other major claims for his results: they show how data can be examined in creative ways to reveal new statistical distinctions; they demonstrate that unvividness is functional; and they show that imagery is a dynamic process governed by developmental factors.

Statistical distinctions. Ahsen (1993b, p. 31; 1993d, p. 154) states that the usual "averaging" method for scoring and evaluating psychometric tests conceals distinctions that might be revealed by other forms of statistical treatment. Authors accept this general point (Hilgard, 1987, p. 1; McKelvie, 1993b, p. 359; Reisberg & Heuer, 1988, pp. 97-98), but doubt that Ahsen's duplication method of counting scores is important for judging psychometric reliability (see earlier discussion and Marks, 1993). However, Ahsen (1990, 1991, 1993d) argues that his treatment of the data reveals other facts: (a) a general tendency to report dimmer imagery under the father than the mother filter, but also (b) consistent individual differences between responses to father and mother filters, with some people reporting dimmer imagery under father, some reporting dimmer imagery under mother, and some reporting similar imagery in both cases.

Although Ahsen (1990, 1991) and Sussman (1993) report substantial percentage differences between father- and mother-filtered imagery, which Ahsen (1990, p. 10) characterizes as statistically significant for both males

and females, they does not provide inferential statistics. Thus, it is not clear that the percentage counting method confers any advantage over the usual approach of comparing mean scores. Indeed, my analysis of Ahsen's data showed that father-dimming was significant only for women.

More interesting is Ahsen's second claim of consistent within-subject differences between the two filters for certain individuals. However, even here, I think that a standard statistical treatment of the data would have illuminated his point better than a count of the number of items rated as dimmer under father or mother for each subject (Ahsen, 1990, 1991). To identify the people who clearly show a difference in one direction or the other, we could compute a repeated-measures *t*-test on the paired ratings across the 16 items for each subject. A significant value would demonstrate that the ratings were consistently different. A stricter approach would be to compute a confidence interval for each person's total score under each filter, using the standard error of measurement for the VVIQ. For example, the empirical test-retest reliability for the VVIQ was estimated to be .735 and the mean *SD* was 0.692. These give $SE_{\text{meas}} = 0.692 (1 - .735)^{0.5} = 0.356$, and a 95% confidence interval of $\pm 0.698 (1.96 \times 0.356)$ around a score. If the intervals did not overlap, the parental ratings would be considered to be different. My general objection to counting duplications is that the method demands that "similarity" of responding requires perfect consistency and that any deviation represents a meaningful difference. However, these judgments of similarity and difference should allow for the element of random error in test responses.

Although psychometric reliability was not Ahsen's concern, no less a person than Buros (1978, p. 1026) has criticized the standard approach, pointing out that "no correlational technique can provide information about the repeatability of precision of scores generally...". Although I question exact duplications as a measure of consistency, Ahsen's and Buros' concern about the blind use of standard statistical procedures prompts me to ask whether duplication rates across parental filter are higher or lower than expected. Although Ahsen (1990) set a standard of 75% agreement, it is essentially arbitrary. My suggestion is that, just as the standard for content validity over a variable should be similar to that for reliability over one source of error, so the duplication rate over a variable should be similar to the duplication rate for reliability. As an illustration, I obtained the number of duplications for McKelvie's (1986) study of VVIQ reliability. In the 3-week delayed test-retest condition, the *Ms* and *SDs* for the number of duplicated responses were $M = 6.62$, $SD = 3.11$ (20 males) and $M = 7.52$, $SD = 3.07$ (20 females). Returning to Ahsen's (1990, 1991) parental filter raw data, the following values were obtained: $M = 9.63$, $SD = 4.70$ 16 Bent

Hall males), $M = 8.80$, $SD = 4.74$ (44 Bent Hall females), and $M = 6.10$, $SD = 3.66$ (48 Pakistani women). Taking the test-retest duplications as the standard, and using planned contrasts with a $MSE = 16.0$, male duplications were higher for parental filter than for male test-retest, $t(143) = 2.24$, $p < .05$, but neither of the female comparisons were significant, $t(143) = 1.19, 1.33$, respectively. This suggests that response consistency across filter was greater for males than for females, which is consistent with the three corresponding r s of .91 (males), .59 and .68 (females).

However, there is another sense in which duplication is appropriate. Ahlsen's general concern about response repetition took the form of a worry that choice of parent at the beginning of the test might influence subsequent responses to the extent that the person's *classification as a good or poor visualizer* would be altered (Ahlsen, 1985a, p. 5; 1990, p. 32). Here, duplication refers to the extreme group to which a subject is assigned. In fact, when developing correlation coefficient standards for content validity and reliability, I included Ebel's (1965, p. 334) equivalence between r s and percentage of cases remaining the same, and I expressed Ahlsen's r s across parental filters in these terms (McKelvie, 1993b). Corresponding to the combined Bent Hall (Ahlsen, 1990) and Pakistani Women (Ahlsen, 1991), parental coefficients of .705 and .678, the percentage of cases falling into the same extreme groups (33% split) was 65.9 and 71.9, respectively. These estimates of the degree of duplication of extreme categories alert us to the practical consequences of parental filters for research practice.

The Unvividness Paradox. Although imagery investigators generally assume that vivid imagery is better than dim imagery (the "ability" theory of vividness, Ahlsen, 1987, p. 14), Ahlsen (1985a, p.3; 1986, p.3; 1987, p. 16; 1988; 1990, p. 7; 1991, p. 23) argues that his effects with parental filters show that unvivid imagery has its own unique effects. Because this implies that both poles of the vividness dimension are functional, Ahlsen (1985a) terms this phenomenon the "Unvividness Paradox." To support his claim, Ahlsen (1986, p. 4; 1987, p. 17) cites the case of a mother who cannot form an image of her dead baby. However, Hilgard (1987) and Marks (1993, p. 344) agree that dimmer imagery may be an inhibitory effect and may, therefore, not be paradoxical, and A. Richardson (1988, p. 111) states that Ahlsen's criteria for discriminating expressive from repressive vague imagery are unclear. Ahlsen (1987, p. 17; 1993d, pp. 157, 182) recognizes that unvivid imagery may reflect personal trauma, which implies that vivid imagery is suppressed, but he also argues that "an image may be vague simply because it is supposed to be vague to create the effect it is designed for" (e.g., in advertising, Ahlsen, 1988). Ahlsen (1986, p. 4; 1987, pp. 16-17) also asks why images of hope are often vague, and he points to the use of

vague imagery in art, music, and poetry to communicate haunting effects. To settle this matter, Reisberg and Heuer (1988) state that the unvividness hypothesis will have to survive the same empirical validation process as the vividness trait hypothesis. I agree, and suggest that Ahsen should specify tasks in which unvivid imagery will be advantageous, a point that he seems to recognize: "The imagery tests need to involve the paradoxical nature of unvividness in defining imagery tasks and related performance effects" Ahsen (1985a, p. 4). This practice would be consistent with Marks' (1972, p. 83; 1973a, 1983a, p. 24; 1989b, p. 461; 1993, p. 344) appeal to evidence and to his implied acknowledgment that vivid imagery may not always predict better performance, and would provide further evidence for the construct validity of the VVIQ. The finding that recognition memory performance is better for poor than good visualizers when the test is extremely difficult (Heuer et al., 1986; Reisberg et al., 1986; Reisberg & Leak, 1987) provides a useful starting point. Indeed, if conditions can be established under which dimmer imagery is functional, it provides another reason to consider imagery vividness as a typical performance trait. Notably, Horne (1993) suggests a factor that determines the value of vivid imagery: it is advantageous when voluntary but disadvantageous when autonomous.

Imagery as a dynamic, developmental process. Critics of Ahsen's work with the AA-VVIQ recognize that his approach may be valuable for demonstrating the dynamics of imagery as it varies under controlled conditions (Hilgard, 1987, p. 2; Marks, 1987, p. 3; A. Richardson, 1988, p. 112; J.T.E. Richardson, 1988, p. 117; Sheehan, 1987, p. 11), and Ahsen (1990, pp. 49-52) has proposed a number of directions that his filter studies might take.

In the spirit of Sheehan's (1987, p. 12) comment that we should understand the methodological and theoretical factors that create filter effects, I think that the first step should be to carefully examine reported vividness under the two parental filters, with proper controls for demand characteristics, order, and carry-over effects. To provide a baseline against which any filter differences could be compared, it would also be useful to include a control group in which subjects rate VVIQ items twice with no mention of parents. I also agree with Marks (1987) and Sheehan (1987) that subject sampling should be extended beyond clients in therapy and undergraduates, and should include different age groups. For example, although no details are given, Ahsen (1993a, pp. 22) claims that actors report less vivid imagery with mother than with father in mind. Particular attention should be given to establishing consistent individual differences in response to the mother and father filters

and to replicating the finding of general imagery dimming under father for females.

Another feature of Ahsen's methodology that should be studied is his instruction to "Think of your father/mother in your mind" before the second to fourth blocks of items, which does not clearly tell subjects what to do (Sheehan, 1987, p. 10). In contrast, for the first block, visual images of each parent are rated directly. In reply, Ahsen (1987, p. 41) states that he initially used this instruction so that people answering the QMI would not be biased in favor of the visual modality and also because the QMI and VVIQ both preface each block of items with the term "think." In fact, Ahsen (1987, pp. 41-43) reports that some subjects answer the AA-VVIQ while also holding a visual image of their parents, whereas others only keep their parents abstractly in mind. Moreover, he indicates that this choice can alter the rating of vividness of the VVIQ item depending on the parent's attitude towards that item. There is also another logical possibility for Ahsen's instruction: subjects may *include* a visual image of their parents in the general image, so that they are part of the scene (for example, see Ahsen, 1993d, p. 173). To study if these three interpretations produce different effects, they should be experimentally manipulated. In particular, does father-dimming for females occur because the idea or image of father (outside the imaged scene) dominates, thereby suppressing the vividness of the scene, or because the visual image of the father *in* the scene is also vague?

In view of Ahsen's (1987, p. 12) initial goal to investigate "the roots of personal emotional history," and his hypothesis that parental filter effects have a developmental basis (Ahsen, 1991, p. 23; 1993d, p. 158), the second step should be to design studies to locate the source of any effects. For example, dimmer visual imagery with father in mind for females may have a psychoanalytic basis, with the father perceived as threatening. Indeed, Sheehan (1987, p. 11) has noted that imagery vividness may reflect affect and Ahsen (1993a, p. 17) has speculated that "the father figure in contemporary society is in big trouble" and is creating feelings of insecurity. Indeed, when students formed visual images of their parents standing in front of them, 25% reported that father was remote (distant) and only 5% reported that mother was remote (Ahsen, 1977, p. 31). As part of this endeavor, we should ensure that the effect is related to father and not just males in general. Needed here are conditions with filters for brother/sister, grandfather/grandmother, uncle/aunt, and close male friend/close female friend. These conditions might also throw light on another explanation for filter effects: they may be generally related to the emotional relationship with the person in the filter (Marks, 1987, p. 8). For example, Sussman (1993) describes an interesting case where the person received all her love

and nurturance from her grandmother. With filters, imagery dimming was greater for father compared to grandmother than for father compared to mother, and was even greater for her mother compared to her grandmother. This proposal could be tested by identifying individuals who show consistently different responses to different filters, and by relating these differences to reports of personal relationships with the people concerned. We might also investigate if differences are stronger with good visualizers, who might be more threatened by the vividness of their images than poor visualizers. A third possibility is that any filter effect for females, in which the VVIQ ratings were higher under father than mother, may reflect *more* vivid imagery for the *females in general*. This would be consistent with an own-gender effect in recognition memory for photographs of faces (Shapiro & Penrod, 1986), particularly for females (McKelvie, 1981). If the finding of a relatively high performance of females on female faces is due differential processes of attention and judgment of the two genders by males and females (McKelvie, 1981), Ahsen's and McKelvie's results on imagery may have a similar basis. Again, imagery ratings under male and female filters other than parents would provide evidence for this hypothesis.

Ahsen (1993d, 1993e) has recently emphasized other findings on parental imagery that were not obtained with the VVIQ. When students formed visual images of their parents standing in front of them, 67% reported seeing their father to the left of mother (i.e., to the subject's left) (Ahsen, 1977), and Ahsen claims that reversal of these positions is associated with recent trauma (Ahsen, 1993e, pp. 205, 208). Indeed, he cites the case of a person whose physiological pain was apparently alleviated after the father was restored to his left position in the visual image (Ahsen, 1993c, p. 134). Molteno (1993) did not find a significant overall difference in left-right positions with another group test of students, but she did replicate it with individual testing on some of these people, and she found more psychopathic symptoms among those who reported father on the right (50%) than on the left (25%). Although further replication is required, particularly in view of Ahsen's (1977) finding that the left-right father/mother incidence of 70%/30% for a group of emotionally disturbed adults was similar to the original normal student group, these studies suggest that it may be more natural to see father on the left and mother on the right, and that the latter positions are associated with mental difficulties.

Like filter effects, Ahsen thinks that the left-right preferences have a developmental basis (Ahsen, 1993e, p. 200), which he claims is related to the fact that a baby is usually placed with the right cheek on the left breast of the mother for nursing, so that the natural position for father and mother becomes the left and the right, respectively Ahsen (1993c, pp. 133-134).

Of more importance for the present review, Ahsen links his findings on parental filters to those on parental position, which he refers to as a "hemispheric effect" (Ahsen, 1993e). Specifically, Ahsen (1993d, p. 179) asserts that the changed imagery under mother and father filters "was based on an underlying hemispheric function." Ahsen's use of the term "hemisphere" is somewhat confusing, because he refers to work on the differential functions of the two hemispheres in the brain (Ahsen, 1977; 1993e, pp. 199-200), which implies that images to the left and right of subjects are projected on to the right and left hemispheres, respectively, but he also states that "The notion as to which hemisphere they are located in is not addressed by us; instead the terms 'visual field' or 'visual space in front' are used to label the positions" Ahsen, 1993e, p. 200; see also p. 223). Apart from this ambiguity, it is not clear to me what Ahsen (1993e) means precisely when he argues that *parental filter effects* are based on a hemispheric function. However, Ahsen's claim that father on the left and mother on the right is the preferred placement in imagery for most people may help to account for the possible order effect in the AA-VVIQ, which contains the column for rating father to the left of the column for mother. Subjects may rate items by first of all imagining their parents in front of them, and the placement of columns may reinforce them to imagine father on the left and mother on the right. If the image of father was less vivid than that of mother, then the VVIQ item rating for father may have been higher than that of mother (at least for females). The importance of this account is that any "order effect" is not simply a methodological artifact, but reflects a natural positioning of parents in imagery. Rigorous testing of it would require a number of conditions, including data on rated vividness of parents in each position.

Final Statement

The VVIQ validity coefficient across all criterion tasks of .269 (95% confidence interval .252 to .310) was deemed acceptable as evidence of a relationship. In addition, although the VVIQ/criterion task *r*s varied from .369 (self-report) through .273 (cognitive/perception) to .137 (memory), they may be limited by a construct that is somewhat dynamic. Furthermore, visual vividness may have been only a minor component on some criterion tasks. On the other hand, the operation of potentially confounding variables (e.g., social desirability with self-report tests, general mental ability with objective tests) on the validity coefficients could not be entirely ruled out. On balance, I conclude that the evidence favors the construct validity of the VVIQ, with a more definitive conclusion awaiting further research with a modified test and improved methodology.

In a discussion of construct validity, Cronbach (1989, p. 165) writes:

In an occasional instance, the test or construct will be found totally devoid of merit, but the usual report will not come out unequivocally for or against the test interpretation. More likely, it will suggest modifications in the technique or the interpretation...

From this point of view, it is not surprising that my conclusion concerning the VVIQ is not more categorical. However, it is not based simply on a simple "box score" (Meehl, 1990, p. 198) of significant and nonsignificant findings, but rather on a critical quantitative analysis whose results are judged against proscribed standards. Consequently, I trust that it is more substantial than the outcome of literature reviews in "soft" psychology that often consist of a mass of "uninterpretable" relationships (Meehl, 1990). If so, it may lead to the future advances implied by Cronbach's expectation in the remainder of his passage:

The investigation should aim to illuminate the test and the related construction so that persons making decisions see more clearly how to use the test, and so that those pursuing research know where the greatest perplexities lie.

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APPENDIX

*Assignment of Weights to VVIQ and Criterion Task Processes**VVIQ Processes*

I began by counting the number of items on which each process seemed to be required. Nine processes (verbal, cueing, generation, holding, complexity, vision, imagery, vividness, and self-report) appeared on all 16 items and were each given a weight of 3. Holding is not mentioned explicitly but is logically implied: the image must be held at least briefly in short-term or immediate memory in order to be judged. However, since the VVIQ instructions emphasize vision, imagery experience, and judgment via the self-report rating scale, these processes were given an extra point, that is, a weight of 4. Vividness was deemed to justify a further point for a weight of 5, because it is the main focus of the test. Memory seemed to be involved explicitly on 12 items (which referred to familiar scenes), and imagination was implied in at least 8. Memory and imagination were given a weight of 2. Control was given 1 because manipulation of images seemed to be involved on 6 items.

Because the VVIQ instructions indicate that subjects should try to generate visual images, it has been argued that the most appropriate criterion tasks should also activate imagery (Palmer, 1980; Reisberg et al., 1986; Swann & Miller, 1982). This issue has been discussed in most detail by Katz (1983, p. 46-57; 1987, 177-181), who claims that most convergent validity criterion tasks can be performed in various ways, so that subjects are not obligated to utilize visual imagery. Moreover, subjects may vary not only in imagery skill, but in the likelihood that they will employ it, so that they will only do so if they decide that it is necessary. It follows that relationships between the VVIQ and criterion tasks may be weak because imagery was not employed in the latter case. Therefore, Katz suggests that individual differences in imagery vividness, as measured by the VVIQ, are only likely to operate if subjects are also required to utilize imagery on the criterion test. Of course, if the VVIQ measures the extent to which subjects spontaneously employ vivid imagery, relationships might appear in the absence of cues to utilize imagery on the criterion. However, Cohen and Saslona (1990) do not think that the VVIQ reflects habitual imagery use. I agree that the VVIQ was not designed to measure the degree to which imagery is spontaneously used, but rather its quality when aroused. Although a high score indicates weak imagery or even none at all, implying that the person would be unlikely to spontaneously utilize imagery, the instruction to "carefully consider the picture that comes before your mind's eye" (see Table 1), clearly implies that imagery should be generated. For this reason, cueing (intention, instruction) was given an extra point (i.e., 4 rather than 3).

Criterion Task Processes

To provide specific predictions for the VVIQ/task relationships, and to permit an evaluation of the summed weight model for the VVIQ, I also assigned weights to each criterion task on each of the 12 VVIQ processes. This endeavor was even more speculative than the assignment of weights to the VVIQ itself, because it was often difficult to logically evaluate the importance of the processes on many tasks. The major guideline was that, if a process was judged to have the same importance in the criterion task as in the VVIQ, the criterion task weight for that process was the same as in the VVIQ. Weights for each process were assigned as follows. For verbal stimulus, they were either 3 or 0 for a verbal or nonverbal (usually pictorial) stimulus, respectively. Cueing was given a weight of 4 if subjects were initially instructed to generate visual images, and 2, 1, or 0 if there were no such instructions but the stimuli were pictures, concrete (high imagery) or abstract (low imagery) words respectively. This practice was based on Paivio's (1971, p. 179) judgment of the availability of imagery for the different kinds of stimuli. Generation was interpreted as the difficulty of arousing a visual imagery experience. Because images are generated from concrete words on the VVIQ, this kind of stimulus material was also given a generation weight of 3. It was assumed that pictures would arouse imagery more easily and abstract words with more difficulty, giving respective weights of 1 and 5. Again, if the stimuli were concrete words, memory and imagination were given weights of 2 to reflect the sources of imagery. However, with pictures, they were only weighted at 1, because the stimulus itself provides visual information directly. For abstract words (with imagery instructions), memory and imagination were weighted at 1 and 3, to reflect the fact that some creativity would be involved. However, at the

testing stage of an immediate memory test, it was assumed that, for any stimuli, initially-generated images would be retrieved, giving weights of 3 and 1 for memory and imagination, respectively. For a delayed memory test, Marks' (1989b) argument that long-term memory images are probably somewhat imaginative was recognized, and the immediate memory and imagination weights were lowered and raised, respectively, to 2 and 2 for delayed memory and imagination. Since holding of the visual image in short-term memory was given 3 on the VVIQ, this weight was assigned for tasks in which generated imagery was likely to be "inspected" when producing a response. If this was judged to be less likely, holding was given 1 and if subjects were specifically instructed to hold images, it was given 5. Manipulation of imagery was judged to be absent on many tasks (weight = 0 for control), but was given 5 when clearly important. Complexity was weighted as 3 (as on the VVIQ) if the visual images preserved spatial arrangements (e.g., scenes, interactive images combining elements) or were detailed (e.g., artwork, faces), 2 if they were simpler (e.g., of single objects or concrete word referents), or 1 if they represented simple figures. Weights for vision and imagery were 4, 3, 2, or 0 and were assigned on the assumption that the degree of activation would depend on two sources: the combination of cueing and generation difficulty, followed by specific task demands. The cueing/generation guidelines were as follows: weights of 4 for instructed imagery and a pictorial stimulus, 3 for instructed imagery and concrete words or just pictures, 2 for instructed imagery and abstract words or just concrete words, and 0 for just abstract words. These weights were then modified (if necessary) by task demands. Vividness was weighted from 5 to 1 as follows. If the task clearly required that details of the image were needed for response, the weight was 3; if some detail would be helpful (e.g., in paired-associate recall), the weight was 2; if little detail, but only schematic imagery was necessary (e.g., for free recall of words or identity of objects), the weight was 1. Weights of 4 or 5 were given if a more global judgment of "closeness to reality" was required (in addition to presence of detail). In recognition memory studies, vividness was weighted as 3 if the alternatives were very similar (e.g., faces), 2 if they were less similar (e.g., scenes), and 1 if they represented different objects. The process of self-report received either 4 or 0, depending on whether the task was subjective or objective. In some studies, usually involving memory, there were two stages (e.g., presentation and later testing). To assign weights to the total task, separate ones were given for each stage, and then an average was taken. In addition, in memory studies with repeated-measures designs, the weight for cueing was assigned by modifying the weaker ones upwards to reflect the more powerful ones. For example, if abstract words were presented with pictures, the initial cue of 0 was raised to 1 (mean of 0 and 2), and the weights on the other processes were assigned on that basis.

From these weights, I derived a summed score that reflected how similar each task was to the VVIQ. However, some tasks were assigned weights higher than those on the VVIQ (e.g., the Gordon Test of Visual Image Control [TVIC] was given 5 for control, whereas the VVIQ was given only 1). To obtain a score that represented similarity between the task and the VVIQ, the weight included in the sum was obtained by calculating the absolute size of the difference between VVIQ process weight and task process weight and subtracting that difference from the VVIQ weight. For example, in the case of the TVIC, control was initially weighted at 5. This gave a difference of 4 from the VVIQ weight of 1. In the all weights summed score for the TVIC, the weight was therefore $1 - 4 = -3$.

Table A
Judged Weights for Processes in VVIQ and in Selected Criterion Tasks

Process	VVIQ	Task Number																
		1	6	8	10	11	12	13	22	23	24	30	33	36	45	52	68	88
Verbal	3	3	3	3	3	0	3	3	0	3	3	3	0	0	0	3	3	0
Cueing	4	4	1	4	4	2	1	1	2	4	4	0	4	4	0	2.5	1	2
Generation	3	3	3	3	3	1	3	3	1	3	3	0	1	1	0	3	3	1
Memory	2	2	2	2	2	1	2	2	1	2	2	0	0	1	0	2.5	2.5	2
Imagination	2	2	2	2	2	1	2	2	1	2	2	0	5	1	0	1.5	1.5	1
Holding	3	3	3	3	3	5	1	1	3	3	3	0	3	5	0	2	1	1
Control	1	0	0	0	5	0	0	0	5	5	1	0	0	0	0	0	0	0
Complexity	3	3	3	2	3	1	3	3	1	2	3	0	1	3	0	3	2	3
Vision	4	4	4	1	4	4	2.5	2.5	4	4	4	4	5	5	4	2.5	2	3
Imagery	4	4	3	4	4	4	2.5	2.5	4	4	4	0	5	5	0	2.5	2	3
Vividness	5	5	3	5	1	4	2.5	0.5	2	1	2	0	3	3	0	2	1.5	2
Self-report	4	4	4	4	4	4	4	4	0	0	0	0	0	0	0	0	0	0
Total	38	37	31	33	30	23	26.5	24.5	16	25	31	7	17	20	4	23.5	18.5	18

Note. Tasks were: 1. Visual Vividness rated from a Question, 6. VES (Visual Elaboration Scale), 8. QMI (Questionnaire Upon Mental Imagery), 10. TVIC (Test of Visual Imagery Control), 11. EI (Eidetic Imagery), 12. Absorption, 13. Habitual Use, 22. Spatial Control, 23. Physiological/Motor Control, 24. Hypnotic Susceptibility, 30. Proofreading, 33. Perceptual Aftereffects, 36. Single-stimulus Marks Task, 45. Gestalt Closure, 52. Paired-associate Learning/Short-term Memory/Instructions Given/Concrete Words, 68. Free Recall/Short-term Memory/No Instructions/Concrete Words, 88. Recognition Memory/Short-term Memory/No Instructions/Scene Pictures.

VVIQ Items (Marks, 1973)

For items 1-4, think of some relative or friend whom you frequently see (but who is not with you at present) and consider carefully the picture that comes before your mind's eye.

Item

1. The exact contour of face, head, shoulders and body.
2. Characteristic poses of head, attitudes of body, etc.
3. The precise carriage, length of step etc., in walking.
4. The different colors worn in some familiar clothes.

Visualize a rising sun. Consider carefully the picture that comes before your mind's eye.

Item

5. The sun is rising above the horizon into a hazy sky.
6. The sky clears and surrounds the sun with blueness.
7. Clouds. A storm blows up, with flashes of lightning.
8. A rainbow appears.

Think of the front of a shop which you often go to. Consider the picture that comes before your mind's eye.

Item

9. The overall appearance of the shop from the other side of the road.
10. A window display including colors, shapes and details of individual items for sale.
11. You are near the entrance. The colour, shape and details of the door.
12. You enter the shop and go to the counter. The counter assistant serves you. Money changes hands.

Finally, think of a country scene which involves trees, mountains and a lake. Consider the picture that comes before your mind's eye.

Item

13. The contours of the landscape.
14. The colour and shape of the trees.
15. The colour and shape of the lake.
16. A strong wind blows on the trees and on the lake causing waves.

